

Florida Institute of Technology

**Protective Environmental Recovery Spacecraft for Expended
Upper-Stages (PERSEUS)**



SPC 5082 Fall 2025 Capstone

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Table of Contents

List of Figures and Tables.....	6
1.0 Mission Overview.....	8
1.1 Proposal.....	8
1.2 Background.....	8
1.3 Overview.....	9
1.4 Space Environment.....	11
1.5 Astrodynamics.....	11
1.6 Launch Vehicle Selection.....	11
1.7 Guidance, Navigation, and Control.....	12
1.8 Propulsion.....	12
1.9 Spacecraft Configuration and Structure.....	12
1.10 Spacecraft Mechanisms.....	12
1.11 Thermal Control.....	12
1.12 Power Systems.....	13
1.13 Communications and Data Systems.....	13
1.14 Reliability and Risk.....	13
1.15 Launch and Mission Operations.....	13
1.16 Cost.....	13
2.0 Target Object Payload.....	14
3.0 Space Environment.....	15
3.1 The Vacuum Environment.....	15
3.2 The Neutral Environment.....	16
3.3 The Radiation Environment.....	18
4.0 Astrodynamics.....	19
4.1 Trajectory Description.....	19
4.1.1 Ascent from Earth.....	20
4.1.2 Parking orbit.....	20
4.1.3 Hohmann Transfer.....	20
4.1.4 Characterization.....	21
4.1.5 Rendezvous.....	21
4.1.6 Docking.....	22
4.1.7 Deorbit.....	22
4.2 Trajectory Computations.....	23
4.3 DV Summary.....	30
4.4 Summary and Recommendation.....	33
5.0 Launch Vehicle Selection.....	34
5.1 Launch Vehicle Trade Study.....	34
5.2 Launch Vehicle.....	37
6.0 Guidance, Navigation & Control.....	37

6.1 Top level GNC Description.....	37
6.1.1 Mission Phase.....	37
6.1.1.1 Ascent.....	38
6.1.1.2 Parking Orbit.....	38
6.1.1.3 Hohmann Transfer.....	38
6.1.1.4 Characterization.....	38
6.1.1.5 Rendezvous.....	39
6.1.1.6 Docking.....	39
6.1.1.7 Deorbit.....	39
6.1.2 GNC Toolbox.....	39
6.1.2.1 GN Algorithms and Hardware.....	40
6.1.2.2 GNC System Equipment and Block Diagram.....	41
6.1.2.3 GN Subsystem Summary.....	42
6.2 ADCS Introduction.....	42
6.2.1 Spacecraft Pointing Requirements.....	43
6.2.2 Attitude Determination.....	43
6.2.2.1 Attitude Determination Sensor Selection.....	44
6.2.2.2 Attitude Determination Algorithms.....	44
6.2.2.3 Spacecraft Physical Characteristics.....	45
6.2.2.4 Disturbance Torques.....	46
6.2.2.4.1 Atmospheric Torque.....	46
6.2.2.4.2 Magnetic Field Torque.....	46
6.2.2.4.3 Gravity Gradient Torque.....	47
6.2.2.4.4 Solar Radiation Pressure Torque.....	47
6.2.3 Attitude Control.....	47
6.2.3.1 Attitude Control Law Algorithms.....	48
6.2.3.2 Attitude Control Actuators.....	48
6.2.3.3 Attitude Control Actuator Sizing.....	48
6.2.4 ADCS Hardware Components.....	49
6.2.4.1 ADCS System Equipment and Block Diagram.....	50
6.2.5 ADCS Summary.....	50
6.3 GNC Summary.....	51
7.0 Propulsion.....	51
7.1 Main Spacecraft Engine Trade Study.....	51
7.2 Main Spacecraft Engine Specifications.....	53
7.3 Main Spacecraft Engine Tank Sizing.....	53
7.4 Main Spacecraft Engine Schematic.....	55
7.5 RCS Thrusters Trade Study.....	56
7.6 RCS Thrusters Specifications.....	58
7.7 RCS Thrusters Tank Sizing.....	58
7.8 RCS Thrusters Schematic.....	60

7.9 Summary of Propulsion System.....	61
8.0 Spacecraft Configuration & Structure.....	61
8.1 Spacecraft Configuration and Structure Overview.....	61
8.2 Spacecraft Configuration and Structure Description.....	63
9.0 Spacecraft Mechanical.....	67
9.1 Robotic Arm Manipulator.....	67
9.2 Capture Mechanism.....	73
9.3 Power and Data Architecture.....	75
10.0 Spacecraft Thermal Control.....	77
10.1 Thermal Control System Overview.....	77
10.2 Thermal Control System Requirements.....	78
10.3 Thermal Control System Components.....	79
10.3.1 Passive thermal control.....	80
10.3.2 Active Thermal Control.....	81
10.3.3 Thermal Control Cooling System.....	81
10.3.4 Thermal Control Heating System.....	82
11.0 Power Systems.....	83
11.1 Power Subsystem Overview.....	83
11.2 Power Generation.....	85
11.3 Solar Panel Selection and Sizing.....	86
11.4 Batteries.....	86
11.5 Power Summary.....	87
12.0 Communications & Data Systems.....	88
12.1 Communications Subsystem Overview.....	88
12.2 Link Budget.....	90
12.3 Communications Summary.....	93
13.0 Reliability & Risk.....	93
13.1 Low Mission Impact Power System Failure.....	95
14.0 Launch & Mission Operations.....	96
15.0 Cost.....	98
15.1 Cost Estimation Method.....	98
16.0 Summary & Conclusion.....	100
Appendix A: References.....	101
Appendix B: Navigation.....	106
Appendix C: Propulsion Calculations.....	108
Appendix D: Astrodynamics Calculations.....	112
Appendix E: Spacecraft Thermal Calculations and Radiator Sizing.....	118
Appendix F: Power and Communication.....	119
Appendix G: Mechanical Subsystem Calculations.....	122
Appendix H: Ground Operations Antennas.....	129

List of Figures and Tables

Figure 1. The Top 500 Space Debris Objects that Pose the Greatest Risk.....	9
Figure 2. Timeline of Mission.....	10
Figure 3. CAD Drawing of PERSEUS in its Deployed Configuration.....	11
Table 1. Detailed Information for the S/L-16 Zenit-2 Upper Stage (SATNO 28353).....	14
Figure 4. The Nozzle -View of the Zenit-2 Upper Stage.....	15
Figure 5. Density of Atmospheric Constituents Versus Altitude.....	17
Figure 6. AO Flux Versus Altitude for Solar Minimum and Solar Maximum.....	17
Figure 7. PERSEUS in a Parking Orbit with the SL-16 in Higher Orbit.....	23
Figure 8. SL-16 Rocket Body 1.866 degrees ahead of PERSEUS.....	24
Figure 9. PERSEUS Entering a Hohmann Transfer.....	24
Figure 10. PERSEUS Establishing New Circular Orbit.....	25
Figure 11. PERSEUS Enters an NMC Around the SL-16 Rocket Body.....	26
Figure 12. PERSEUS Completes the First Quarter of the NMC.....	26
Figure 13. PERSEUS Completes the First Half of the NMC.....	27
Figure 14. PERSEUS Completes Three Quarters of an NMC Around the Target.....	27
Figure 15. PERSEUS Completes the Full NMC Around the SL-16 Rocket Body.....	28
Figure 16. PERSEUS Enters a Hohmann Transfer to Deorbit the Target.....	29
Figure 17. PERSEUS Crosses 120 km Altitude with the Target.....	29
Table 2. Events in Hohmann Trajectory to Target Orbit.....	31
Table 3. Key Events from Closing with the Target to Deorbiting.....	32
Table 4. Events in Edelbaum Trajectory to Target Orbit.....	33
Table 5. Launch Vehicle Selection.....	34
Figure 18. PERSEUS in SpaceX Fairing (Space Exploration Technologies Corp., 2020)....	35
Figure 19. ULA's Vulcan 9 Fairing Options (United Launch Alliance, 2023).....	36
Table 6. GN Algorithms and Maneuvers by Mission Phase.....	40
Table 7. Selected Navigation Hardware.....	41
Figure 20. GNC System Block Diagram.....	42
Table 8. Attitude Pointing Requirements.....	43
Table 9. Selected Attitude Determination Sensors.....	44
Table 10. Moments of Inertia of Simplified Spacecraft Geometries.....	45
Table 11. Disturbance Torques Affecting PERSEUS.....	47
Table 12. Sizing Criteria for the RWUs and RCS Thrusters.....	49

Table 13. Maximum 90 degree Slew Rate Capability of Each ADCS Actuator.....	49
Figure 21. ADCS Subsystem Block Diagram.....	50
Table 14. Main Spacecraft Engine Trade Study.....	52
Figure 22. Main Spacecraft Engine Schematic.....	55
Table 15. RCS Thrusters Trade Study.....	57
Figure 23. RCS Thrusters Schematic.....	60
Figure 24. PERSEUS in the Undeployed Configuration.....	62
Figure 25. PERSEUS in the Deployed Configuration.....	62
Figure 26. PERSEUS and the Target Debris Post-Capture.....	63
Figure 27. PERSEUS' Structural Layout and Dimensions.....	64
Table 16. Bill of Materials (BOM).....	64
Figure 28. PERSEUS' Component Layout (Excluding the Capture Mechanism).....	65
Figure 30. PERSEUS System Block Diagram.....	66
Figure 31. Proto-Flight Manipulator Arm (Britton, 1977).....	68
Figure 32. A Block Diagram of the Robotic Arm.....	69
Table 17. Mechanical Sub-System Component Breakdown by Vendor/Model/Quantity.....	70
Figure 33. A Technical Illustration of the REACCH	74
Table 17. Capture Timeline of the Mechanical Subsystem.....	76
Table 18. Operating Temperatures of Each Component.....	79
Figure 34. Thermal Control Panel with Heat Pipes.....	80
Figure 35. Space Rated Minco HK 6907 Thermofoil Heaters.....	82
Figure 36. Power Block Diagram.....	84
Table 19. Solar Array.....	85
Table 20. Battery Information.....	87
Figure 37. Communications and Data Block Diagram.....	88
Figure 38. Communications Black Out Periods Before Capture.....	89
Figure 39. Communications Black Out Periods After Capture.....	89
Table 21. Uplink Budget.....	91
Table 22. Downlink Budget.....	92
Table 23. Qualitative Risk Analysis.....	94
Figure 40. Fever Chart Showing Concerns and Their Risk Levels.....	95
Table 24. NASA QuickCost Estimation Variables for PERSEUS Mission Cost.....	99
Table 25. Additional Adjustments to the QuickCost Estimate.....	99

1.0 Mission Overview

1.1 Proposal

The proposal requests for the team to design a Low Earth Orbit (LEO) mission to remove large orbital debris greater than 5000 kg. The mission shall be able to demonstrate the ability to safely rendezvous, capture, and dispose of an unprepared and uncooperative derelict rocket body by means of controlled atmospheric re-entry. The spacecraft shall be capable of rendezvous and proximity operations and must include a reliable capturing mechanism capable of securing and transporting large debris. With all these requirements in mind, the team has decided to target a Zenit-2 upper stage in LEO.

1.2 Background

A far too common result of space exploration is space debris left in orbit. Space debris refers to derelict, non-functional objects, including spent rocket stages, defunct satellites, and fragments generated from collisions or the impact of objects. With an ever-growing spacecraft population in LEO comes an increasing accumulation of space debris, and with it a rising concern for a Kessler Syndrome scenario (Dremman et al, 2025). The Kessler Syndrome refers to a cascade of orbital collisions that could potentially jeopardize near-Earth space activities and render space unusable (Nishida et al, 2018; Stadnyk et al, 2021). To address the threat of orbital debris congestion, active debris removal (ADR) has become a critical area of focus for preserving the safety and sustainability of future space activities.

The most effective approach of ADR is to target derelict space debris objects that pose the greatest risk of generating further debris fragmentation, determined by mass times probability of collision (NASA Orbital Debris Program Office, 2011). In 2011, a NASA study was

completed identifying the top 500 objects of space debris that pose the highest risk of collision and fragmentation (Figure 1). Another study was completed in 2021, narrowing down the top 50 most statistically concerning objects (McKnight et al, 2021). In April of 2025, an update of this list was published showing 36 derelict rocket boosters, of which, 17 were S/L-16 Zenit-2 upper stages (McKnight et al, 2025).

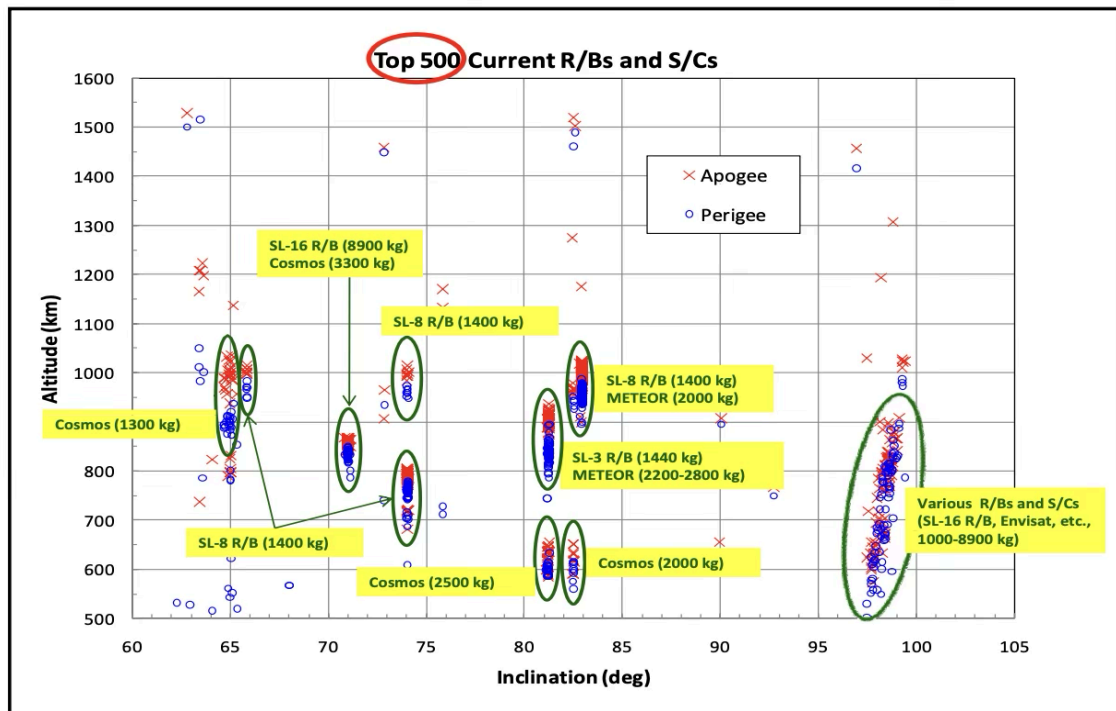


Figure 1. The Top 500 Space Debris Objects that Pose the Greatest Risk of Collision and Fragmentation (NASA Orbital Debris Program Office, 2011)

1.3 Overview

The spacecraft will be launched into a parking orbit in Low Earth Orbit (Figure 2). A Hohmann Transfer will be completed followed by Natural Motion Circumnavigation (NMC) and rendezvous to allow for close proximity operations to commence. To secure the Zenit-2 upper stage, a capturing mechanism will grab the rocket booster using eight tentacle-like arms. To wrap up the mission, a deorbit burn will take place to bring both the PERSEUS spacecraft (Figure 3)

and the Zenit-2 upper stage into Earth's atmosphere to burn up. Any larger pieces of debris that do not burn up shall splash down in the Pacific Ocean.

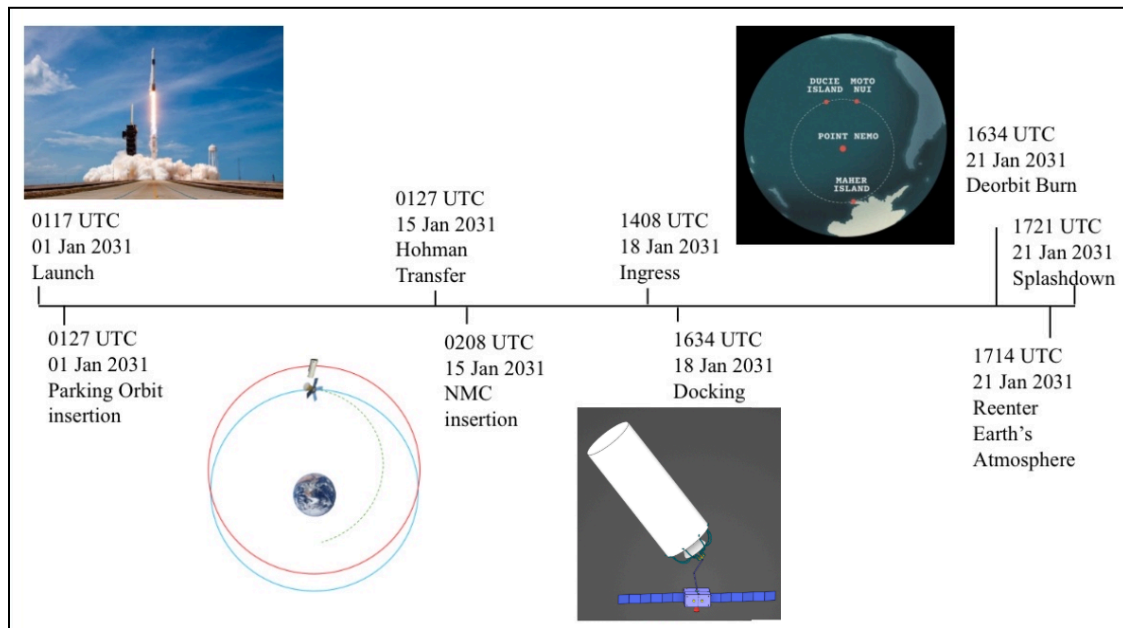


Figure 2. Timeline of Mission

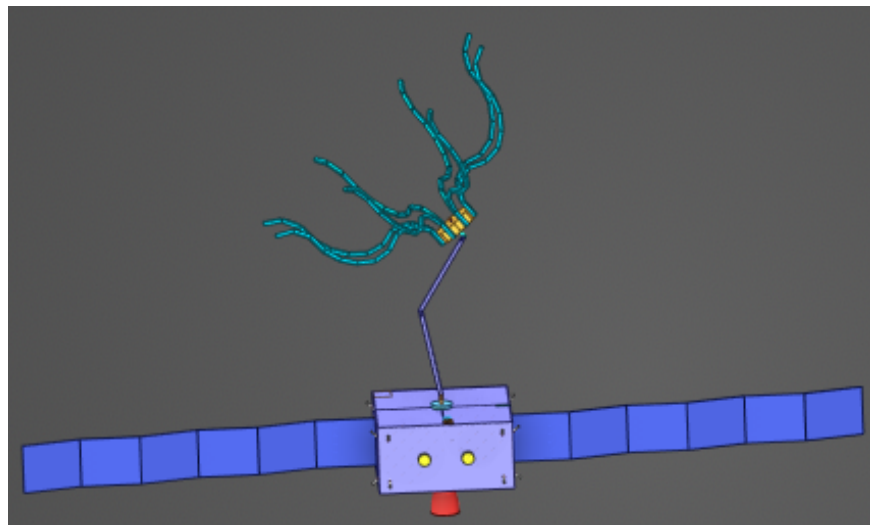


Figure 3. CAD Drawing of PERSEUS in its Deployed Configuration

1.4 Space Environment

The Space Environment section will discuss the hazards the spacecraft will encounter within the vacuum, neutral, and radiation environments. Mitigation techniques and methods will also be addressed for this mission.

1.5 Astrodynamics

The Astrodynamics section will describe the trajectories and maneuvers required to complete the mission. Delta-V requirements are also discussed.

1.6 Launch Vehicle Selection

The Launch Vehicle Selection will explain the rocket selected to transfer the spacecraft to its required orbit and reasons for selection.

1.7 Guidance, Navigation, and Control

The Guidance, Navigation, and Control (GNC) section will include the GNC algorithms and hardware that have been chosen for this mission. Additionally, this section will cover the Attitude Determination and Control (ADCS) algorithms and hardware chosen for the mission. A block diagram of the GNC and ADCS subsystems will also be included.

1.8 Propulsion

The Propulsion section will cover the Reaction Control System (RCS) thruster and main engine selection, the schematics of the RCS system and main engine system, the type and amount of fuel required, and the required size of the tanks.

1.9 Spacecraft Configuration and Structure

The Spacecraft Configuration and Structure section will discuss the spacecraft configuration, frame, material, size, and weight.

1.10 Spacecraft Mechanisms

The Spacecraft Mechanisms section will detail the capture mechanism used by the spacecraft to capture the upper stage. This section will describe in detail the robotic arm and end-effector design, as well as its associated power and data architecture.

1.11 Thermal Control

The Thermal Control section will explain the thermal requirements and limitations expected throughout the mission lifecycle. Conduction and radiation analysis will be discussed, as well as thermal hardware selection.

1.12 Power Systems

The Power Systems section will discuss the mission power requirements, power system design, and will include a block diagram for the system. A description of different operating modes will be given based on the battery charge status.

1.13 Communications and Data Systems

The Communications and Data Systems section will detail the communication system design, antenna selection, link budget, and associated avionics. A link budget is shown which details the flow of communications from the surface and data from the subsystems to the flight computer.

1.14 Reliability and Risk

The Reliability and Risk section will explain the risks involved with this mission as well as the failure modes and redundancies considered for each subsystem.

1.15 Launch and Mission Operations

The Launch and Mission Operations section will discuss the launch site, ground stations, and required support personnel to carry out the mission.

1.16 Cost

The Cost section will cover the cost estimation method used as well as the total estimated cost of the mission.

2.0 Target Object Payload

The S/L-16 Zenit-2 upper stage (SATNO 28353) was chosen as the target object for this mission because it was identified as the first object on the list of ‘The Top 50 Statistically-Most-Concerning Derelict Objects in LEO’ published by McKnight et al, 2025 (Table 1). Zenit-2 upper stages in LEO are critical targets of interest due to their large size of 3.9 meters in diameter by 11.047 meters in length, and high mass of 9000 kg (Figure 4) (Bylard et al, 2017; NASA Orbital Debris Program Office, 2011).

Table 1. Detailed Information for the S/L-16 Zenit-2 Upper Stage (SATNO 28353)

NORAD ID	28353
Int'l Code	2004-021B
Perigee (km)	846.8
Apogee (km)	855.0
Period (mins)	101.8
Semi Major Axis (km)	7221
RCS (m^2)	8.3334
Launch Date	June 10, 2004
Source	Commonwealth of Independent States (former USSR) (CIS)
Launch Site	Tyuratam Missile and Space Complex (TTMTR)



Figure 4. The Nozzle -View of the Zenit-2 Upper Stage

After launching their payload into orbit, the Zenit-2 upper stages inherently develop a state of spin, which is enhanced by subsequent outgassing. However, over time this spin state decreases as a result of internal damping and external disturbances caused by Earth's magnetic field and atmosphere, leaving a final spin that is aligned with the principal axis of inertia (Bylard et al, 2017). Additionally, studies have shown consistent engine-side-to-zenith orientations of expended upper stages suggesting predictable target attitudes and enabling predictable, simplified capture operations (Nishida et al, 2018; Okamoto et al, 2025).

3.0 Space Environment

3.1 The Vacuum Environment

Operating a spacecraft in a vacuum environment imposes constraints on structural integrity, material selection, mechanism design, and thermal control due to the absence of an atmosphere and the resulting lack of atmospheric pressure. Structures must be able to withstand pressure differentials experienced in orbit. Additionally, structures cannot be cooled by convective heat transfer; instead, they must rely on methods of conduction and/or radiation.

The very low-pressure environment may also induce the off-gassing, or outgassing, of organic materials, a process in which volatile chemicals escape from the surface of the material into the surrounding environment, resulting in mass loss. Another concern with the outgassing of materials is the deposition of a thin layer of contamination on sensitive surfaces of the spacecraft, altering thermo-optical surface properties (Tribble, 2003). Outgassing rate is temperature dependent, and deposition rate is affected by solar UV, AO, and surface temperature. NASA requires outgassing to be limited to less than 1% total mass loss (TML) in accordance with the ASTM E595 material test guidelines (Scialdone et al, 2000).

3.2 The Neutral Environment

Within the neutral environment, spacecraft are subjected to mechanical interactions, including aerodynamic drag and physical sputtering, as well as chemical interactions, most notably atomic oxygen attack. Atomic oxygen (AO) is produced by the photodissociation of diatomic oxygen (O_2) into free oxygen atoms due to the absorption of short-wavelength (200-400 nm range) solar UV radiation (Dever et al, 2005; Dooling & Finckenor, 1999). AO fluence is most prevalent in the ionosphere (Figure 5) (Silverman, 1995) and varies as a result of solar activity consistent with the 11-year solar cycle (Figure 6) (Dooling & Finckenor, 1999).

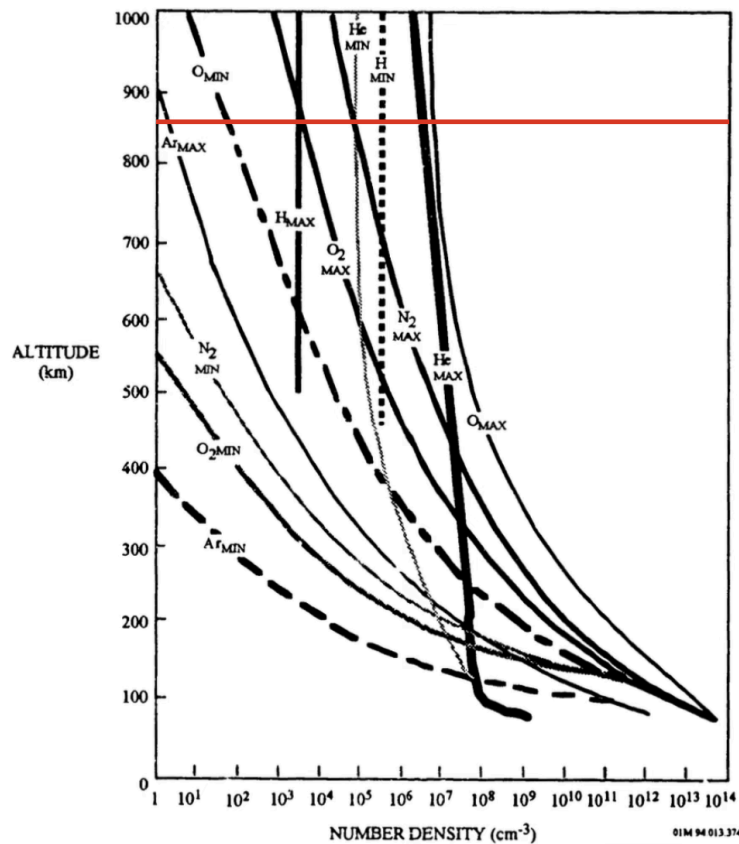


Figure 5. Density of Atmospheric Constituents Versus Altitude (Silverman, 1995). Red line depicts the altitude of the target object.

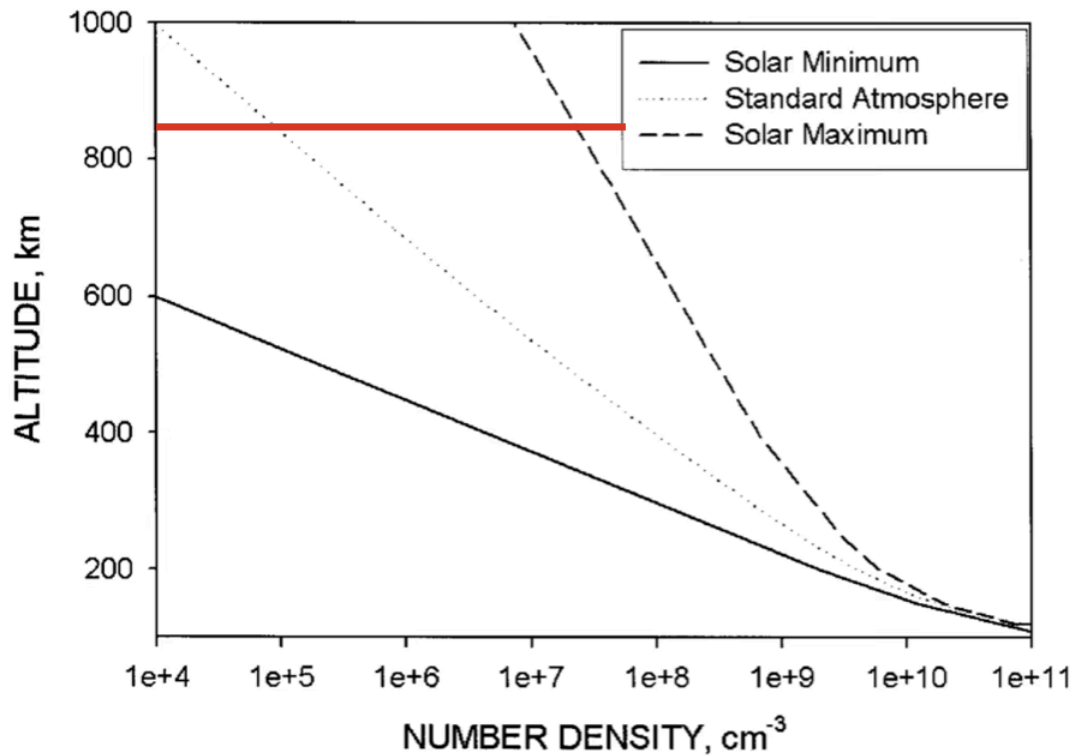


Figure 6. AO Flux Versus Altitude for Solar Minimum, Nominal (Standard Atmosphere), and Solar Maximum Conditions (Dooling & Finckenor, 1999). Red line depicts the altitude of the target object.

As space debris travels at orbital speeds of 7.8 km/s, the ram-facing surfaces are exposed to a stream of AO in the atmosphere at an energy of approximately 5.2 eV (Finckenor et al., 2020). Since oxygen atoms are highly corrosive, prolonged exposure to AO significantly changes the mass and surface properties of materials due to oxidative erosion. This mass loss can affect thermal, optical (solar absorptivity and thermal emissivity), and mechanical properties, and increases overall contaminant deposition (Dooling & Finckenor, 1999).

While PERSEUS will remain in Low Earth Orbit (LEO) for a relatively short amount of time, considerations must be given to protect the structure and components against AO.

Moreover, since the target object has been in LEO orbit for over 20 years, the effects of AO play a significant role in determining the best capture method and mechanism to account for the deterioration of the surface materials.

3.3 The Radiation Environment

Radiation presents a significant challenge for the design and operation of spacecraft in LEO. The primary radiation sources that will affect this mission include solar electromagnetic radiation, and charged-particle radiation generated by galactic cosmic rays and solar particle events (SPEs) (Tribble, 2003). Solar ultraviolet (UV) radiation produces non-ionizing effects that degrade spacecraft surfaces, altering mechanical, thermal, and electrical properties over time. Meanwhile, charged-particle radiation introduces ionizing effects, where cumulative exposure can damage semiconductors, optical coatings, and solar arrays, resulting in reduced performance, material embrittlement, and optical darkening of sensors and windows (Anderson, 1994).

In addition to these gradual effects, radiation also causes short-term, potentially mission-critical disruptions. High-energy particles penetrating the spacecraft can induce single-event effects (SEE), such as bit flips in memory, latch-ups in circuits, or even permanent component failures (Perez, 2008). These transient malfunctions can cascade into broader system issues if not properly mitigated. To address these risks, spacecraft must integrate measures such as shielding, radiation-hardened components, fault-tolerant architectures, and error correction techniques to ensure long-term survivability and reliable operation throughout the spacecraft's lifespan.

4.0 Astrodynamics

The PERSEUS mission is designed to remove hazardous orbital debris in LEO by deorbiting it and bringing it back down into Earth's atmosphere. As a proof-of-concept, the first PERSEUS vehicle will launch in 2031. This mission will target one of the most dangerous pieces of orbital debris in LEO, the Zenit-2 second stage SL-16 rocket body (SATNO 28353). This defunct rocket body resides in a nearly circular orbit at an altitude of approximately 844 km. PERSEUS will rendezvous with this target, attach itself to the rocket body, and perform a series of maneuvers to deorbit the space debris back into Earth's atmosphere.

4.1 Trajectory Description

PERSEUS will separate from the launch vehicle at an altitude of 743.8 km above Earth's surface. From the parking orbit, it will perform a series of burns to execute a Hohmann transfer to arrive at a circular orbit 843.8 km above Earth's surface. In order to ensure PERSEUS arrives at the same true anomaly as the target, PERSEUS will wait until the target is 1.8661 degrees ahead of PERSEUS in its orbit before PERSEUS executes the 50.37 minute Hohmann transfer. Once PERSEUS arrives at the rendezvous point, it will have the same semi major axis as the target while matching true anomalies. Since the target has a slight eccentricity, and PERSEUS will be in a circular orbit, PERSEUS will naturally move in an ellipse relative to the target inertial frame. This ellipse will have a semi major axis of 10.42 km and a semi minor axis of 5.21 km, providing PERSEUS with a naturally safe standoff distance while it characterizes the target. Once the target is characterized, PERSEUS will close the relative distance and capture the target. With the target captured, PERSEUS will execute a Hohmann transfer to arrive at an altitude 75 km above Earth's surface. As PERSEUS arrives at 120 km above Earth's surface, it will exhibit a

flight path angle of -1.5094 degrees to prevent skipping off of Earth's atmosphere. This altitude and flight angle combination ensures PERSEUS and the target reenter Earth's atmosphere and ultimately splash down in the Pacific Ocean. Below are the phases of the PERSEUS mission and a brief description of the actions in each phase.

4.1.1 Ascent from Earth

In this phase, PERSEUS will launch on a Falcon 9 rocket out of Vandenberg, CA. To ensure PERSEUS is inserted into the correct plane, the launch must execute at an azimuth of 156.66 degrees on 01 Jan 2031 at 0117 UTC or 1126 UTC.

4.1.2 Parking orbit

In this phase PERSEUS will remain in a circular orbit 743.8 km above Earth's surface while it does a complete system checkout. This checkout will commence over a 2-week period. Once all systems are mission capable, PERSEUS will wait until the target is 1.8661 degrees ahead in its orbit before executing its first burn.

4.1.3 Hohmann Transfer

In this phase, PERSEUS will execute a series of burns to conduct a Hohmann transfer to arrive at the rendezvous orbit. The first burn to enter the Hohmann transfer will be 26.03 m/s. This will place PERSEUS on a 50.37 min transfer orbit trajectory. Once PERSEUS arrives at the target altitude, it will conduct a 25.94 m/s burn to circularize the orbit at an altitude of 843.8 km.

4.1.4 Characterization

Once PERSEUS is established in a circular orbit at the same semi-major axis as the target, it will start to move in a natural elliptical path around the target. This ellipse will have a semi-major axis of 10.42 km and a semi-minor axis of 5.21 km. The period of the natural ellipse around the target will be the same as the period of PERSUS' orbit around Earth. This natural motion, also referred to as Natural Motion Circumnavigation (NMC), is made possible by the difference in eccentricity of the orbits. PERSEUS' circular orbit results in a constant velocity and altitude as it orbits Earth. On the other hand, the target has an eccentric orbit, which results in it having a lower altitude and faster velocity at perigee, and a higher altitude and slower velocity at apogee. While the two space objects have the same period, but varying altitudes and velocities, the result is a natural motion that appears to be an ellipse relative to the target. This NMC will be crucial for the characterization phase because it allows PERSEUS to use its sensors to identify the target and ensure it has selected the correct target. Once the target has been identified to be the correct target (SATNO 28353), PERSEUS will use its sensors to determine the spin rate of the target. Once the spin rate has been determined, ground operators will plan a safe trajectory for PERSEUS to follow for its rendezvous in the next phase.

4.1.5 Rendezvous

In this phase, PERSEUS will use a series of small maneuvers to rendezvous with the target. This will bring PERSEUS out of its 10.42 x 5.21 km NMC and will close the relative distance down to 0 km. In total, it is expected that the PERSEUS vehicle will use 10.73 m/s to close with the target and grapple the Zenith side of the target.

4.1.6 Docking

Once PERSEUS is within reach of the target, it will use its Electro-Adhesive/Gecko (EA/G) capture mechanism to attach itself to the target. PERSEUS will do another system checkout over a period of 72 hours. The intent of this system checkout is to ensure PERSEUS is in good condition and ready to conduct maneuvers with a significantly different moment of inertia. With the target captured and all systems checked, PERSEUS will enter the next phase for deorbiting.

4.1.7 Deorbit

In this phase, PERSEUS will stay attached to the target while performing a series of maneuvers to deorbit the debris. Starting at the target's perigee of 838.6 km altitude, PERSEUS will perform a 213.24 m/s maneuver to enter into a Hohmann transfer. The target altitude for the Hohmann transfer will be 75 km above Earth's surface. After 39.94 minutes, PERSEUS will reach an altitude of 120 km. At that altitude, PERSEUS will move to the final phase of the mission.

4.1.8 Reentry

In this phase PERSEUS will, for the first time in several months, start to experience significant effects from Earth's atmosphere. Since the Hohmann transfer targeted an altitude of 75 km, PERSEUS will have a flight path angle of -1.509 degrees as it passes through an altitude of 120 km. This flight path angle prevents PERSEUS from skipping off of Earth's atmosphere. The drag caused by Earth's atmosphere will reduce the kinetic energy of PERSEUS and the captured target. This will result in the PERSEUS/target object pair being unable to maintain an

orbital velocity and establishing a new trajectory toward Earth's surface at point NEMO ($48^{\circ} 52.6' \text{ S}$, $123^{\circ} 23.6' \text{ W}$), as this is the spot in the South Pacific Ocean furthest from any land. This phase, as well as the entirety of the mission, will be complete once PERSEUS and the target object splash down in the Pacific Ocean.

4.2 Trajectory Computations

This mission will use a Hohmann transfer trajectory with a launch on a Falcon 9 rocket inserting the PERSEUS spacecraft at an inclination of 71 degrees and an altitude of 743.8 km above Earth's surface. This will be a parking orbit for two weeks so that PERSEUS can conduct systems checkout. The target for this mission is the SL-16 rocket body, SATNO 28353, which orbits Earth with a semi-major axis of 7221.9 km and an inclination of 71 degrees. The orbits are depicted in Figure 7.

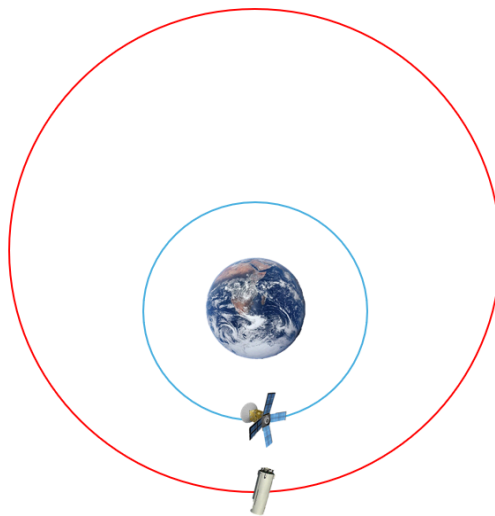


Figure 7. PERSEUS in a Parking Orbit with the SL-16 in Higher Orbit

PERSEUS will conduct a two week system check to make sure everything is operational after the launch. Once the system check is complete the target will be 1.866 degrees ahead of PERSEUS in its orbit as shown in Figure 8.

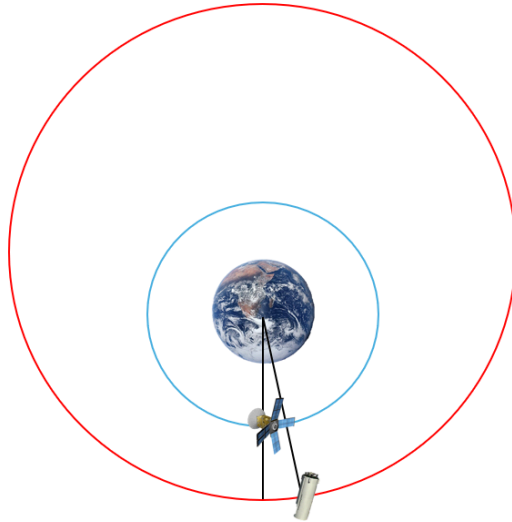


Figure 8. SL-16 Rocket Body 1.866 degrees ahead of PERSEUS

PERSEUS will then execute a 26.03m/s burn to enter the Hohmann transfer depicted in Figure 9.

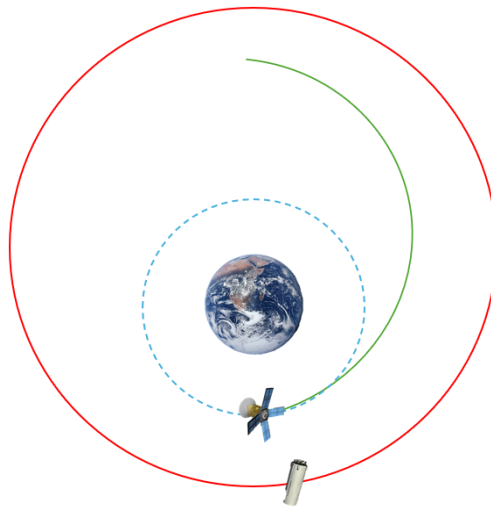


Figure 9. PERSEUS Entering a Hohmann Transfer

After 50.37 minutes of flight, PERSEUS will execute a 25.94 m/s burn to establish a circular orbit with a semi-major axis of 7221.9 km. This is the same semi-major axis as the target, SL-16 rocket body, so that the two objects have the same orbital period. The newly established orbit is depicted in Figure 10.

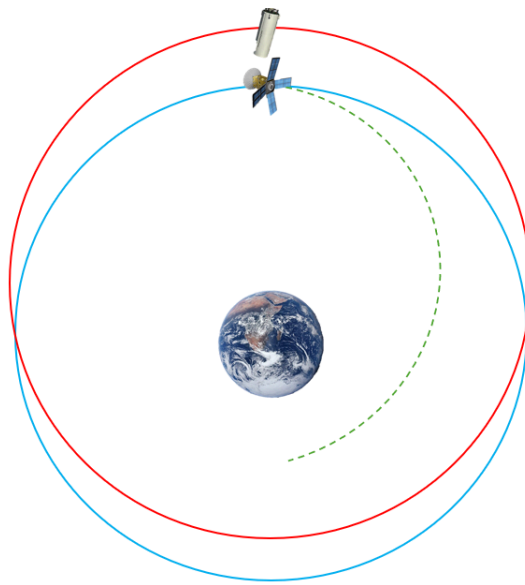


Figure 10. PERSEUS Establishing New Circular Orbit

PERSEUS' new orbit will match the semi-major axis, inclination, and RAAN of the SL-16 rocket body. The key difference between the orbits will be the eccentricity. Since the two objects have different eccentricities, but the same orbital period, they will appear to naturally maneuver around each other in a 10.42×5.21 km ellipse. This relative motion is called Natural Motion Circumnavigation (NMC). Figure 11 depicts the two orbits in an Earth Centered Inertial frame on the left and in a relative frame, centered on the target, on the right. For the relative frame, the radial direction, +R, originates at the center of mass and points away from the Earth (Zenith), and the in-track direction, +I, points in the direction of motion.

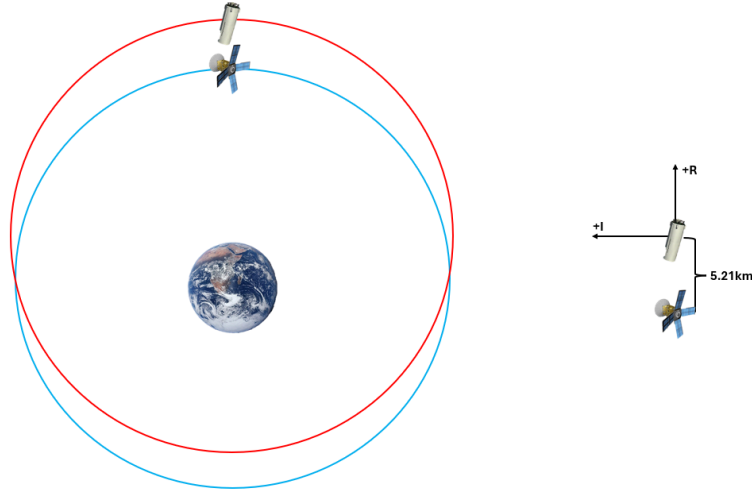


Figure 11. PERSEUS Enters an NMC Around the SL-16 Rocket Body

In the figure above, PERSEUS is shown to arrive 5.21 km below, or negative radial, of the target. At this moment, the target is at a higher altitude than PERSEUS and therefore has less velocity. PERSEUS will move through its orbit faster and appear to move in front of the target as shown in Figure 12.

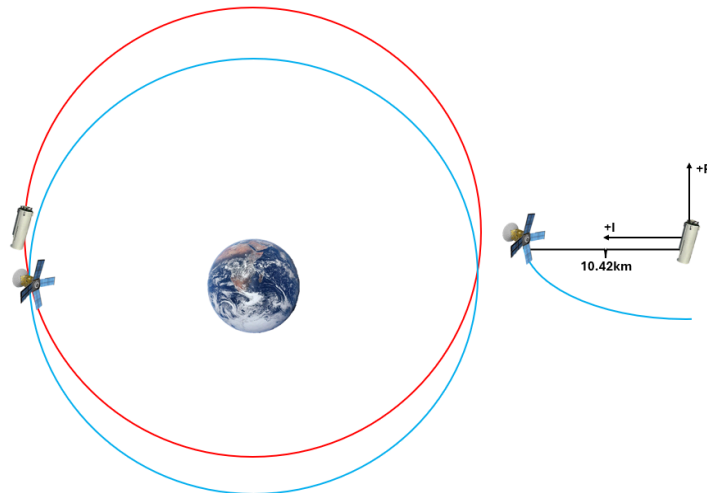


Figure 12. PERSEUS Completes the First Quarter of the NMC

At this moment, the two objects are co-altitude, but the target will soon become lower in altitude and PERSEUS will appear to slow down in its orbit relative to the target. This will result in the relative motion depicted in Figure 13.

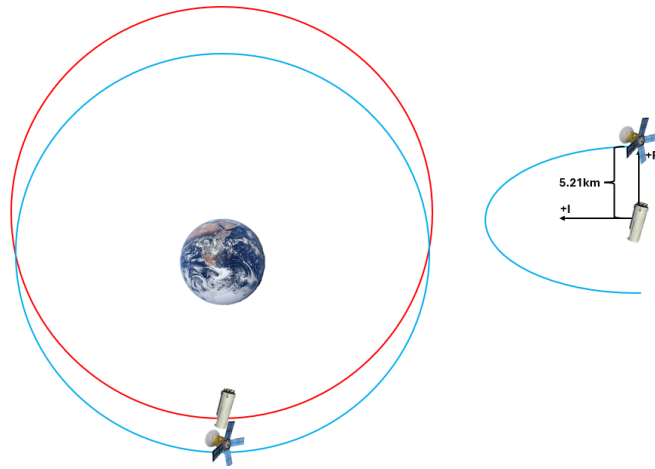


Figure 13. PERSEUS Completes the First Half of the NMC

At this point, PERSEUS will have a lower velocity than the target and will appear to move through its orbit slower. Although PERSEUS will appear to move slower and drift behind the object, its orbit will naturally bring them back to the same altitude. This will result in the relative motion depicted in Figure 14.

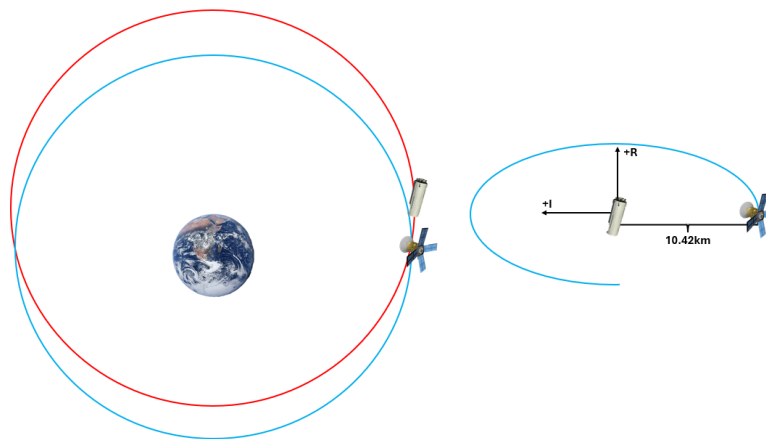


Figure 14. PERSEUS Completes Three Quarters of an NMC Around the Target

Now that the objects are coaltitude again, PERSEUS will start to increase its velocity and lower its altitude relative to the target. This will result in PERSEUS completing the NMC as depicted in Figure 15.

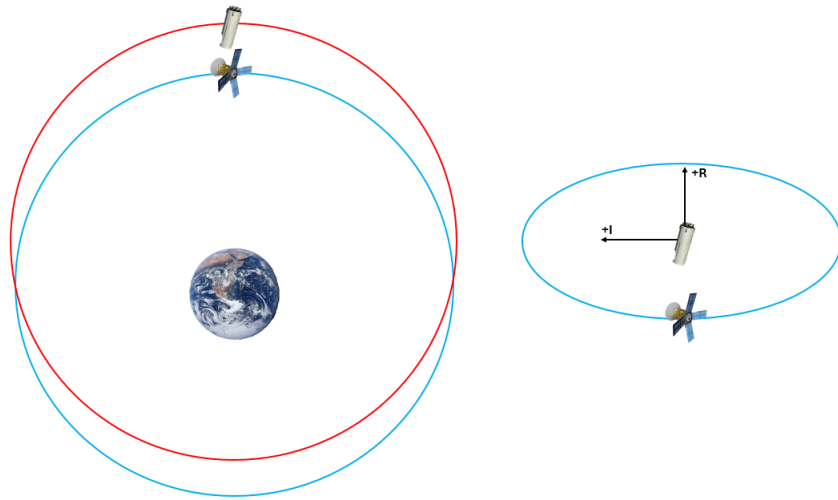


Figure 15. PERSEUS Completes the Full NMC Around the SL-16 Rocket Body

The NMC uses no fuel while allowing PERSEUS to characterize all six faces of the target. Characterizing the target will be crucial for the mission because it ensures PERSEUS' rendezvous with the correct object, and it allows PERSEUS to determine the spin rate of the target. With the spin rate known, PERSEUS operators can map a safe flight path for PERSEUS to close in with the target and bring it within reach of its payload. PERSEUS will remain in this NMC for three days to allow for adequate observations of the target and planning by the mission operations center.

Once PERSEUS is attached to the target, it will do a 72-hour systems checkout. This will ensure all subsystems are in good condition and are ready to conduct the deorbit burn with the captured object. Once all systems are checked, PERSEUS will conduct a 213.24 m/s burn to enter a Hohmann transfer trajectory toward Earth to deorbit the target. This Hohmann transfer will target an altitude of 75 km above Earth's surface as shown in Figure 16.

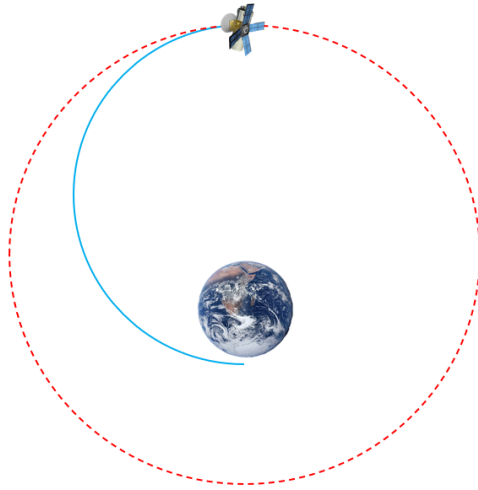


Figure 16. PERSEUS Enters a Hohmann Transfer to Deorbit the Target

After 39.94 minutes into the Hohmann transfer, PERSEUS will cross an altitude of 120 km above Earth's surface as depicted in Figure 17.

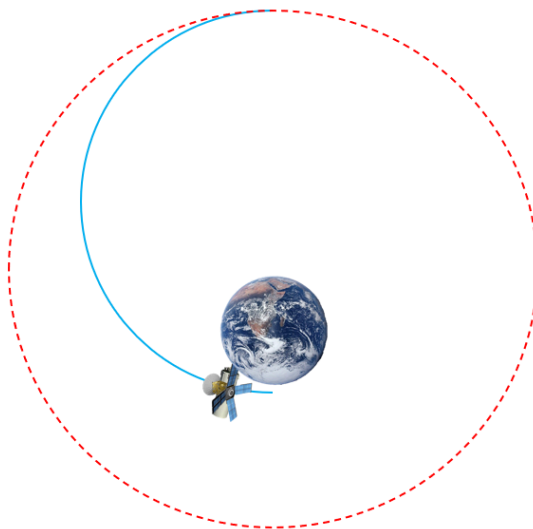


Figure 17. PERSEUS Crosses 120 km Altitude with the Target

PERSEUS will experience drag from Earth's atmosphere which will reduce the velocity of its orbit and start to change the spacecraft's trajectory to remain within Earth's atmosphere.

Since the original deorbiting Hohmann transfer targeted an altitude of 75 km, PERSEUS will have a flight path angle of -1.509 degrees as it crosses an altitude of 120 km. This flight path angle ensures PERSEUS does not skip off of Earth's atmosphere. With PERSEUS and the target in tow experiencing significant drag, they will shortly splash down in the Pacific Ocean.

The Edelbaum trajectory was also analyzed when planning this mission. The trajectory was calculated and considered when going from the parking orbit to the target orbit. The Edelbaum trajectory is a continuous thrust using a low acceleration thruster. Since the PERSEUS mission will not use electric propulsion, ideal for Edelbaum trajectories, this analysis was done using acceleration from the reaction engine assembly used mainly for attitude control. The acceleration used to calculate the Edelbaum trajectory was 0.014667 m/s^2 . The burn time to get from the parking orbit to the target circular orbit was 59 minutes and 3 seconds. This burn would use 51.97 m/s. Since this trajectory would take longer than the Hohman transfer, the Hohman transfer was the trajectory selected for this mission. The calculations for the Hohman transfer and Edelbaum trajectory can be found in Appendix D.

4.3 DV Summary

The Hohmann transfer trajectory was selected for the mission. The velocity values at key points in the Hohmann transfer trajectory is outlined in Table 2 below.

Table 2. Events in Hohmann Trajectory to Target Orbit

Event	Velocity Value (km/s)
Initial orbit velocity in parking orbit	7.481171
Velocity of Hohmann transfer at parking orbit altitude	7.507203
Delta-V to enter Hohmann transfer	0.026032
Velocity of Hohmann transfer at target orbit altitude	7.403254
Velocity of circular target orbit	7.429196
Delta-V to establish circular target orbit	0.025942
Total Hohmann transfer Delta-V	0.051974

The Hohmann transfer trajectory starts at the parking orbit altitude. In the parking orbit, PERSEUS has a velocity of 7.481171 km/s. After performing a 26.032 m/s burn, PERSEUS will establish itself in the Hohmann transfer trajectory toward the target orbit. At the parking orbit altitude, the velocity of PERSEUS will be 7.507203 km/s. Once PERSEUS arrives at the target orbit altitude, it will have a velocity of 7.403254 km/s. At that altitude, PERSEUS will execute a 25.942 m/s maneuver to establish a circular orbit at the target orbit altitude. The velocity of PERSEUS in the target circular orbit will be 7.429196 km/s. The total Delta-V used in the Hohmann transfer trajectory will be 51.974 m/s.

Once PERSEUS arrives at the target circular orbit, it must close with the target in order to achieve nearly zero meters of relative distance. After closing with the target, it must deorbit the target back into Earth's atmosphere. The velocity values at key events throughout that process are identified in Table 3.

Table 3. Key Events from Closing with the Target to Deorbiting

Key Event	Velocity Value (km/s)
Delta-V to close with target from NMC orbit	0.010726
Orbital velocity after closing with target	7.434560
Orbital velocity of deorbit trajectory at target's altitude	7.221324
Delta-V to deorbit	0.213236
Total Delta-V (closing and deorbit)	0.223962
Total Mission Delta-V (Hohmann transfer, closing, and deorbit)	0.275937

Once PERSEUS is established in an NMC relative to the target, it must execute a maneuver to close its relative distance with the target. The velocity of the closing maneuver will be 10.726 m/s. Once the relative distance is closed, PERSEUS will grapple the target and execute a deorbiting maneuver. The orbital velocity of PERSEUS and the target after the grappling is complete will be 7.43456 km/s. The desired orbital velocity to deorbit both spacecraft is 7.221324 km/s. Therefore, a Delta-V of 213.236 m/s is required to perform the deorbit maneuver and put the spacecraft on a trajectory to enter Earth's atmosphere. The total Delta-V to close with the target and perform the deorbit maneuver will be 223.962 m/s, thus the total mission Delta-V to include the Hohmann transfer, closing maneuver, and deorbit burn will be 275.937 m/s.

The other trajectory considered in the mission analysis was an Edelbaum trajectory to arrive at the target orbit from the parking orbit. The velocity values at key points in the Edelbaum trajectory are shown in Table 4 below.

Table 4. Events in Edelbaum Trajectory to Target Orbit

Key Event	Associated Value
Initial orbit velocity in parking orbit	7.481171 km/s
Velocity of circular target orbit	7.429196 km/s
Total Hohmann transfer Delta-V	0.051975 km/s
Acceleration of low thrust engine	0.014667 m/s ²
Time of Flight	59 minutes 3 seconds

The Edelbaum trajectory starts at the parking orbit which has a velocity of 7.481171 km/s. PERSEUS will need to achieve a velocity of 7.429196 km/s at the desired target orbit altitude, thus requiring a Delta-V of 51.975 m/s. With PERSEUS' low thrust engine's acceleration of 0.014667 m/s², it will take a time of flight of 59 minutes and 3 seconds to achieve the desired orbit altitude and velocity.

4.4 Summary and Recommendation

The Hohmann transfer was the trajectory selected for this mission. Both the Edelbaum transfer and Hohmann transfer used approximately the same Delta-V, however the Hohmann transfer had a shorter time of flight by 8 minutes and 41 seconds. The shorter time of flight and simplicity of the Hohmann transfer, made the Hohmann transfer the desirable choice when compared to the Edelbaum transfer.

5.0 Launch Vehicle Selection

5.1 Launch Vehicle Trade Study

With the mission requirement to launch into LEO from the western range using rockets that are flight proven, the options were limited. The two rockets that were considered were United Launch Alliance's (ULA) Vulcan and SpaceX's Falcon 9. Table 5 below highlights what the team compared between the two rockets.

Table 5. Launch Vehicle Selection

	Falcon 9	Vulcan
Cost	\$70 Million	\$110 Million
Fairing Option 1	CPAF: 1575 mm	CPAF: 1575 mm
Total Mass Allowed (kg)	10885	9000
Dimensions (in)	460 x 177	612 (or 840) x 182
Fairing Option 2	CPAF: 2624 mm	LVFA: 4991 mm
Total Mass Allowed (kg)	19050	27000
Dimensions (in)	460 x 177	612 (or 840) x 182
Number of Launches	538	2

The two smaller Circular Payload Attachment Fittings (CPAF) from SpaceX are listed without consideration for the largest CPAF due to limited information widely available (Space Exploration Technologies Corp., 2025). The Launch Vehicle Forward Adapter (LVFA) and other CPAF from Vulcan are the only two options from ULA (United Launch Alliance, 2023). The dimensions of each fairing are approximate. Figure 18 shows that PERSEUS fits within the SpaceX fairing. SpaceX provides a CAD file of their fairing envelope (Space Exploration Technologies Corp., 2020).



Figure 18. PERSEUS in SpaceX Fairing (Space Exploration Technologies Corp., 2020)

On the contrary, ULA only provides two figures to account for their two fairing options. They have a standard one as well as an extended one as shown in Figure 19 below.

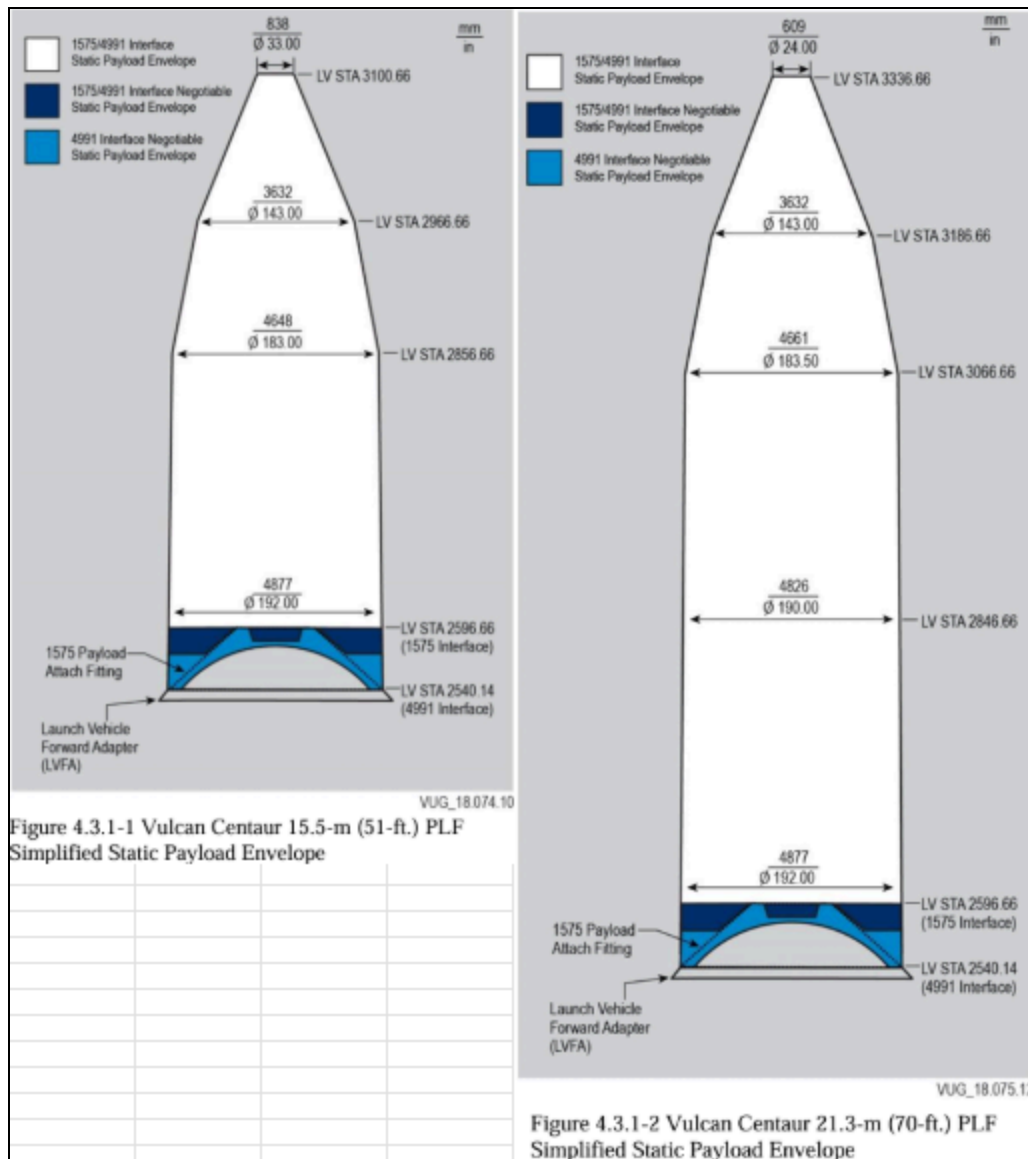


Figure 19. ULA’s Vulcan 9 Fairing Options (United Launch Alliance, 2023)

With these options in mind for fairing size, the team ultimately decided to go with the Falcon 9 due to its cost and reliability. The continuous success of Falcon 9 cannot be overlooked for a large fairing size. With the mission plan, the largest available fairing options were not needed.

5.2 Launch Vehicle

With 538 successful launches and a price of around \$70 million, the team decided to select SpaceX's Falcon 9 (Space Exploration Technologies Corp., 2020). The CPAF of 2624 mm was also selected. Since this was one of the first trade studies conducted, the team wanted to be sure that our spacecraft was not heavily constrained by overall spacecraft mass, which is why the smallest option was not selected. Now that the design is complete, it is evident that the mass of the spacecraft is low compared to the allowable mass. Since PERSEUS needs to launch from Vandenberg, other smaller rockets cannot be considered.

6.0 Guidance, Navigation & Control

6.1 Top level GNC Description

Guidance, Navigation, and Control (GNC) are critical to mission success. This system ensures accurate guidance, navigation state determination, and attitude control throughout each mission phase. For the PERSEUS mission, GNC algorithms and hardware have been chosen to ensure the mission is accomplished accurately, effectively, and safely.

6.1.1 Mission Phase

Below is a description of the PERSEUS mission based on each of the mission phases. Each phase includes a description of how the GNC subsystem will operate throughout its duration, as well as what each phase will accomplish for the mission.

6.1.1.1 Ascent

During the ascent phase, a SpaceX Falcon 9 launch vehicle will leave Vandenberg Space Force Base to carry PERSEUS to a circular parking orbit 100 km below the target's average altitude. PERSEUS will perform no GNC operations during the ascent phase.

6.1.1.2 Parking Orbit

Once PERSEUS decouples from the Falcon 9 Circular Payload Attachment Fitting (CPAF) and enters its parking orbit, on-orbit checkout will begin. All GNC components will power on, and PERSEUS will begin receiving navigation information. Reaction wheels will hold the desired attitude while a star tracker and IMU provide pointing information. Reaction Control System (RCS) thrusters are available for reaction wheel momentum dumping.

6.1.1.3 Hohmann Transfer

After a thorough check of onboard systems, PERSEUS will perform a Hohmann transfer maneuver to an elliptical NMC orbit around the target using the main engine. Powered Explicit Guidance (PEG) guidance will be used during this maneuver, while the Global Positioning System (GPS) component updates the navigation state.

6.1.1.4 Characterization

Once in the NMC orbit, PERSEUS will study the motion of the target using the flash LiDAR and optical camera components in order to verify acquisition of the correct target, study its motion, and choose a rendezvous tactic.

6.1.1.5 Rendezvous

To approach the target, the Clohessy-Wiltshire guidance equations will be utilized with the flash LiDAR and optical camera components updating target-relative navigation. PERSEUS will approach from the target's zenith direction, as its main thruster is always pointed in its zenith direction due to gravity-gradient torque (Okamoto, 2025), and this is the direction that PERSEUS' capture mechanism was designed to capture from.

6.1.1.6 Docking

Once PERSEUS is within range of the capture mechanism, flash LiDAR, optical camera, and Forward Looking Infrared (FLIR) components will continue to be used during the docking procedure to ensure precise relative maneuvering. Once the target is captured, the onboard RCS thrusters will be used to nullify its spin.

6.1.1.7 Deorbit

To begin the deorbit phase, PERSEUS will slew the target 90 degrees to align PERSEUS' main engine in the retrograde direction in order to begin the deorbit burn. PEG guidance will be used throughout the burn. The PERSEUS/target pair is placed along a ballistic entry trajectory towards a splashdown in the Pacific Ocean.

6.1.2 GNC Toolbox

A combination of GNC hardware and software have been selected to meet the needs of the PERSEUS mission. This section will describe the components and capabilities available in the "toolbox" for the GNC subsystem of PERSEUS.

6.1.2.1 GN Algorithms and Hardware

PERSEUS will have to navigate from its parking orbit to a rendezvous with an uncooperative target with enough precision to capture it as it spins. This challenge requires various guidance algorithms, coordinate systems, navigation algorithms, navigation sensors, and navigation measurement filtering techniques. Tables 6 and 7 below list the guidance algorithms, navigation algorithms, and navigation hardware used during the mission.

Table 6. GN Algorithms and Maneuvers by Mission Phase

	Ascent	Parking Orbit	Orbit Adjustment	Target Rendezvous	Deorbit
Guidance algorithms	N/A	Vgo, PEG	Vgo, PEG	Vgo, PEG	Vgo, PEG, Ballistic entry
Maneuvers supported	N/A	Momentum Dumping	Hohmann Transfer	Lambert targeting CW relative motion	Hohmann Transfer
Navigation algorithms	N/A	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method	Orbit Determination: State vector and COE determination Coasting: Cowell's method Powered: Super G method

Table 7. Selected Navigation Hardware

Instrument	Function
Star tracker (Terma T3 Star Tracker)	Provide accurate pointing and navigation information to correct IMU drift and add redundancy to the GPS system
Inertial Measurement Unit (HBK 3DM-CX5-IMU)	Provide data on angular velocity changes and linear accelerations on all axes
Flash LiDAR (ASC GSFL-16KS)	Provide target-relative navigation during the characterization and docking phases
Optical camera (Infinity Avionics ORION12MP-550)	Provide target-relative navigation and visualization during the characterization and docking phases
Forward Looking Infrared (FLIR) (Teledyne Blackfly S BFS-U3-13Y3)	Provide target-relative navigation from the center of the REACCH end-effector
GPS Receiver (L3Harris AS-48917 GPS Band Omnidirectional Antenna)	Provide navigation data throughout the mission

PERSEUS will use the Geodetic, ECEF, J2000 ECI, and LVLH coordinate systems throughout its operation. The only gravity model necessary will be Earth's, since PERSEUS will never leave LEO. PERSEUS will filter the measurements from its navigation sensors using the Kalman filtering method. The full GNC design tree is shown in Appendix B.

6.1.2.2 GNC System Equipment and Block Diagram

Figure 20 below shows a block diagram that describes the software and hardware used by the PERSEUS ADCS and GNC subsystems.

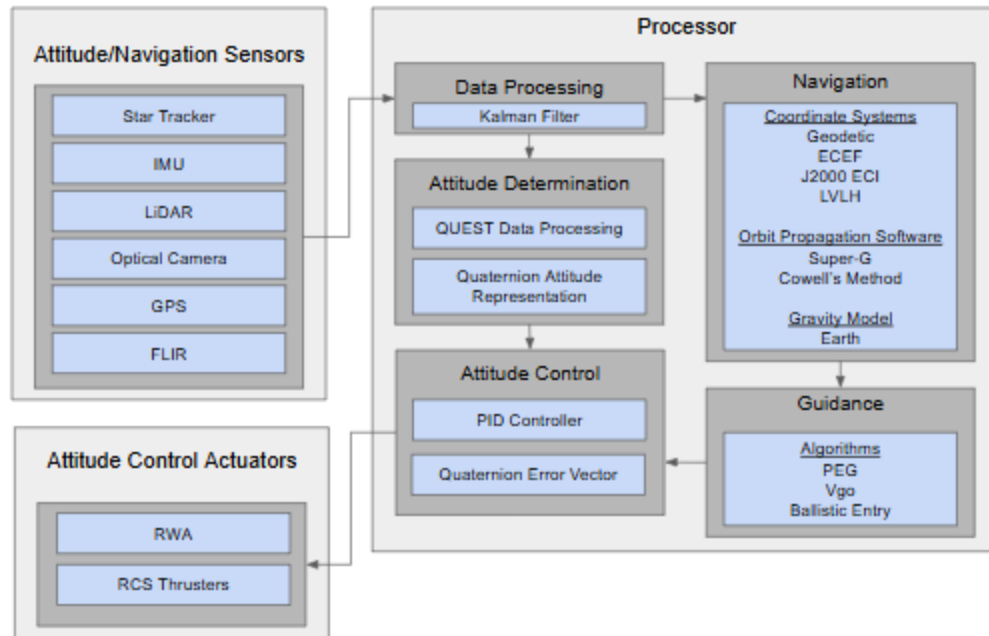


Figure 20. GNC System Block Diagram

6.1.2.3 GN Subsystem Summary

The design of the GNC subsystem for PERSEUS has been selected based on analysis obtained in the iterative process of mission design. This design will be adequate for each phase of the PERSEUS mission, providing accurate guidance and navigation through the deorbit of the target debris.

6.2 ADCS Introduction

During each phase of the mission, PERSEUS will need to be pointed for various operations, including orienting the antenna towards ground stations, pointing solar panels towards the sun, and aligning the spacecraft for target capture. To meet these pointing requirements, hardware, algorithms, and requirements have been defined for each mission phase to ensure each objective is met. The selected sensors will collect data on PERSEUS' attitude in relation to the necessary coordinate frames defined previously in Figure 20. Navigation sensor

data will be refined using Kalman filtering techniques to create a dataset for the navigation algorithms to determine PERSEUS' attitude. This data will feed into the control algorithms and hardware to correct errors between the desired and actual attitude.

6.2.1 Spacecraft Pointing Requirements

The attitude pointing requirements are shown in Table 8 (Shao et al, 2011). These requirements were driven by the data sheets of the listed contributing components.

Table 8. Attitude Pointing Requirements

Spacecraft Requirement Item	Requirement Data	Units	Contributing Component
GPS Receiver Targeting Direction	Zenith		GPS Receiver
Spiral Antenna Targeting Direction	Nadir		Spiral Antenna
Star Tracker Targeting Direction	Space, 37.5 deg from sun (full moon accepted)		Star Tracker
Solar Array Targeting Direction	Sun		Solar Array
Capture Mechanism Targeting Direction	Target		Capture Mechanism
Optical Camera Targeting Direction	Target		Optical Camera
Flash LiDAR Targeting Direction	Target		Flash LiDAR
Pointing Control Error	10	Arcsec	RWUs, RCS
Pointing Knowledge Error	10	Arcsec	Star Tracker, IMU
Pointing Jitter	0.1/60	Degrees/sec	Flash LiDAR, Optical Camera
Pointing Knowledge Error Temperature Stability	0.04% for IMU 0.1 for Star Tracker	%/Kelvin, Arcsec/Kelvin	Star Tracker, IMU
Slew Rate	<0.3 full performance <1.5 reduced performance	Degrees/sec	Star Tracker
Slew Rate Resolution	<0.003	Degrees/sec	IMU
Independent Antenna Pointing DOF	0	Degrees	Spiral Antenna
Independent Solar Array Pointing DOF	360	Degrees	Solar Array

6.2.2 Attitude Determination

Attitude determination is a process in which sensors are used to collect data and then algorithms are applied to determine the orientation of the spacecraft in a reference frame. PERSEUS will collect data from multiple sensors and that data will be fed to a processor that will apply the attitude determination algorithms.

6.2.2.1 Attitude Determination Sensor Selection

Based on the design needs of the PERSEUS mission, the selected sensors are shown in Table 9.

Table 9. Selected Attitude Determination Sensors

Instrument	Function
Star tracker (Terma T3 Star Tracker)	Provide accurate pointing and navigation information and correct IMU drift
Inertial Measurement Unit (HBK 3DM-CX5-IMU)	Provide data on angular velocity changes and linear accelerations on all axes

A star tracker uses a database of stars and an optical camera to sense the spacecraft's attitude. It uses the pattern of stars seen by the camera in combination with the known star positions in its database to provide highly accurate attitude data.

An IMU is an attitude determination sensor that is made up of a combination of accelerometers and gyroscopes. The accelerometers are used to sense changes in linear acceleration across all axes, while the gyroscopes sense changes in angular velocities. In tandem, these sensors paint a picture of the linear and rotational direction and motion of PERSEUS.

6.2.2.2 Attitude Determination Algorithms

The QUEST (QUaternion ESTimator) algorithm will be used with quaternion attitude representation to process attitude determination data. QUEST was chosen because of its accuracy and heritage. To use QUEST, quaternions must be used for attitude representation, so Direction Cosine Matrices (DCMs) and Euler angles were not considered. Additionally, Quaternions are not affected by the gimbal lock problem, as they are in other representations.

6.2.2.3 Spacecraft Physical Characteristics

In order to determine the actuated and environmental torques affecting PERSEUS, the moments of inertia about each axis must be estimated. Considering the PERSEUS only configuration and the PERSEUS/target object paired configuration, the MOIs of each are shown in Table 10. The PERSEUS only configuration was calculated as a solid rectangular prism. The PERSEUS/target object paired configuration was calculated with the PERSEUS bus as a solid rectangular prism, the capture arm as a solid cylinder, and the debris as a hollow cylinder. A more detailed calculation was performed using the ANSYS Discovery CAD software, but considering the preliminary nature of the configuration, the simplified geometry values were found to be a more realistic estimate of similar spacecraft. Calculations of the MOIs are explained in Appendix B.

Table 10. Moments of Inertia of Simplified Spacecraft Geometries

Variable	PERSEUS (Solid Rectangular Prism)	Arm (Solid Cylinder)	Target (Hollow Cylinder)	Combined
Length (m)	1.92			
Width (m)	1.92			
Height (m)	0.98			
Diameter (m)				
Thickness (assumed) (m)				
Mass (kg)	1,723.26	90.00	9,000.00	10,813.26
I _x (kg*m ²)	667.30	14.86	108,420.44	175,350.51
I _y (kg*m ²)	667.30	14.86	108,420.44	175,350.51
I _z (kg*m ²)	1,058.77	4.18	33,786.56	34,849.56

6.2.2.4 Disturbance Torques

Over the course of the PERSEUS mission, various environmental torques will be experienced. This necessitates an attitude control system to nullify these torques as they will inherently affect spacecraft pointing. After separating from the Falcon 9 launch vehicle, PERSEUS will manage its own pointing control until capturing the debris, at which point it will begin handling attitude control for the combined system until splashdown. The calculation of these disturbance torques is shown in Appendix B.

6.2.2.4.1 Atmospheric Torque

Atmospheric drag is significantly stronger at lower orbital altitudes, so it is expected that PERSEUS will experience the greatest amount in its parking orbit (Shao et al, 2011). However, since PERSEUS will be nadir-pointing while in its parking orbit with a symmetrical ram direction, there won't be any induced atmospheric torque. After capturing the debris in a higher orbit, the ram direction will no longer be symmetric, leading to an increase in torque, but it will still be insignificant compared to the gravity gradient torque. All disturbance torques are shown in Table 11.

6.2.2.4.2 Magnetic Field Torque

The magnetic field torque is induced by the Earth's magnetic field on the spacecraft, increasing in strength in lower orbital altitudes and higher latitudes (Shao et al, 2011). The PERSEUS only configuration and the PERSEUS/target object pair will experience insignificant magnetic field torques (Table 11).

6.2.2.4.3 Gravity Gradient Torque

Gravity gradient torque is induced by separated areas with significantly different masses attached to the same system (Shao et al, 2011). The PERSEUS only configuration does not have areas of significantly different masses, so it therefore experiences an insignificant gravity gradient torque. However, once attached to the target debris, the system will experience a significantly larger gravity gradient torque (Table 11).

6.2.2.4.4 Solar Radiation Pressure Torque

Solar radiation pressure will push against surfaces exposed to the sun (Shao et al, 2011). The effects of solar radiation pressure-related torque are insignificant on the PERSEUS only system due to the coincidence of the center of solar pressure and center of mass. The effects are greater once attached to the debris, but overall the torque remains insignificant (Table 11).

Table 11. Disturbance Torques Affecting PERSEUS

Disturbance Torque	PERSEUS	Combined Configuration
Solar Radiation Pressure (Nm)	0	1.02E-03
Aerodynamic (Nm)	0	8.70E-04
Magnetic (Nm)	3.97E-05	7.93E-05
Gravity Gradient (Nm)	1.08E-04	3.89E-02

6.2.3 Attitude Control

Using the sensed/calculated actual attitude of PERSEUS and the desired attitude in an inertial reference frame, PERSEUS can be commanded to re-orient itself to correct the error between these two pieces of data. PERSEUS accomplishes this using its available attitude actuators. By continually applying this process, PERSEUS actively maintains closed-loop control of its attitude.

6.2.3.1 Attitude Control Law Algorithms

Using reaction wheels, PERSEUS will be three-axis stabilized, requiring active control. The selected attitude control law will use a Proportional Integral Derivative (PID) controller. PID was chosen because the Integrator removes steady state errors, a problem not solved using a Proportional Derivative (PD) controller. The error vector will be expressed to the PID controller as a quaternion, since attitude information will already be converted to quaternions for compatibility with the QUEST algorithm.

6.2.3.2 Attitude Control Actuators

To accomplish the active control of PERSEUS, attitude actuators will be used. These actuators will keep PERSEUS stable during each mission phase. The actuators chosen for PERSEUS will be a Reaction Wheel Assembly (RWA), and RCS thrusters. The RWA will conduct the majority of the spacecraft pointing in order to save propellant. RCS thrusters are necessary for the more torque-intensive requirements of the PERSEUS/target object paired configuration's pointing, orbit maintenance, the Clohessy-Wiltshire rendezvous burn, and to perform the occasional momentum dump.

6.2.3.3 Attitude Control Actuator Sizing

To determine the necessary sizes of the ADCS actuators, the MOIs were calculated using simplified geometries of the largest, heaviest configurations encountered during the mission. This means that all other configurations will have adequate attitude control available. The sizing criteria are shown in Table 12 below. The RCS thrusters were sized for their use during the Clohessy-Wiltshire rendezvous burn, so they far exceed the necessary torques for attitude control. The slew time is unconstrained due to the nature of the mission, so the Reaction Wheel

Units (RWUs) were chosen based on the magnitudes of expected disturbance torques. The slew rate capabilities of each of the ADCS actuators are shown in Table 13. Calculations are shown in Appendix B.

Table 12. Sizing Criteria for the RWUs and RCS Thrusters

Sizing Calculation	Value	Quantity (units)	Configuration
Reaction Wheels			
Torque to match disturbance	3.89E-02	T (N*m)	PERSEUS
Max stored momentum	5.00E-01	N*m*s	PERSEUS
Reaction Control System Thrusters			
Force to match disturbance torque	7.01E-03	F (Newtons)	Combined
Force for momentum dump (1-sec burn)	9.00E-02	F (Newtons)	Combined

Table 13. Maximum 90 degree Slew Rate Capability of Each ADCS Actuator

Maneuver	Value (s)	Configuration
Slew 90 deg with RWA	365.1550067	PERSEUS
Slew 90 deg with Thrusters	4.387801622	Combined

6.2.4 ADCS Hardware Components

The selected ADCS hardware components were chosen based on the requirements of each mission phase. The selected RWUs will be four of Space Inventor's WHL-500mNms. There will be four to ensure the mission will continue if one RWU fails. The selected RCS thrusters will be 16 of the Aerojet Rocketdyne MR-106L 22N hydrazine thrusters. They were chosen for their low mass, Isp, and thrust force characteristics. There will be 16 thrusters in order to cover any potential rotations or translations of the spacecraft, having 4 thrusters on each side not facing nadir or zenith.

6.2.4.1 ADCS System Equipment and Block Diagram

Figure 21 below shows a block diagram that describes the software and hardware used by the PERSEUS ADCS subsystem.

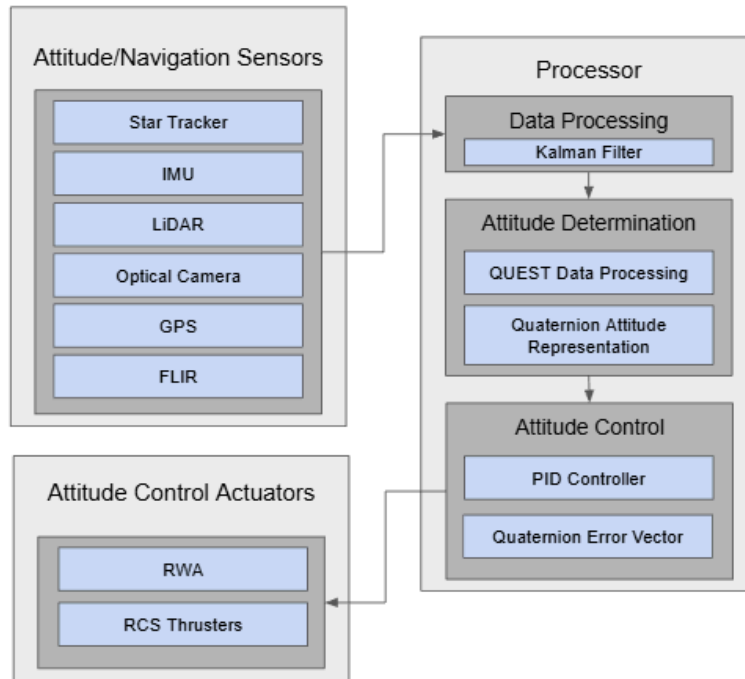


Figure 21. ADCS Subsystem Block Diagram

6.2.5 ADCS Summary

The chosen ADCS design, software, and hardware has been determined based on many iterations of mission design, and is ideal for attitude determination and control of PERSEUS. Analysis has been performed on each component as Moment of Inertia (MOI) and ADCS pointing requirements evolved throughout the design, validating the algorithm and hardware choices for PERSEUS.

6.3 GNC Summary

The PERSEUS mission involves unique challenges, requiring a robust GNC toolbox to ensure mission success. The selected tools allow PERSEUS to maintain a high degree of accuracy with a low cost by using flight-proven algorithms and hardware. PERSEUS will be able to accomplish all of the GNC mission objectives throughout each phase of the mission with the current GNC subsystem design.

7.0 Propulsion

7.1 Main Spacecraft Engine Trade Study

Initially, different types of engines were considered. Due to the mission design, it was found that it is ideal to have a liquid propellant rocket engine that does not utilize cryogenic fuel. This is due to the cooling requirements to prevent boil-off (Sutton & Biblarz, 2011). Engines that use solid rocket propellant were not considered due to their inability to restart. The team also wanted to consider only flight proven hardware, and, therefore, did not consider other rocket engine types.

The three engines considered were the R-40B, MR-80B, and the R-42 from L3Harris (formerly Aerojet Rocketdyne). Table 14 below shows the comparison of mass, specific impulse, calculated propellant mass, thrust, and calculated burn time between the three considered engines (Aerojet Rocketdyne, 2020; L3Harris Technologies, 2023).

Table 14. Main Spacecraft Engine Trade Study

	Maneuver 1 - Hohmann Transfer	Maneuver 2 - Hohmann Transfer	Maneuver 4 - Deorbit	Totals
Delta-V (m/s)	26.03	25.94	213.23	265.20
R-40B				
Mass (kg)	10.5			
Isp (sec)	293			
Prop Mass (kg)	13.54	13.49	751.48	778.51
Thrust (N)	4000			
Burn Time (sec)	9.76	9.73	559.73	579.22
MR-80B				
Mass (kg)	168			
Isp (sec)	215			
Prop Mass (kg)	18.42	18.36	1018.11	1054.88
Thrust (N)	2500			
Burn Time (sec)	15.62	15.57	895.57	926.75
R-42				
Mass (kg)	4.53			
Isp (sec)	305			
Prop Mass (kg)	13.01	12.96	722.96	748.93
Thrust (N)	890			
Burn Time (sec)	43.88	43.72	2515.63	2603.23

With this comparison, the initial dry mass was estimated to be 1500 kg with the information the team has at the early stage in the design. While the dry mass was found to be around 748 kilograms, the decisions remain the same. While the R-40B has a lower specific

impulse than the R-42, the R-42 has the highest burn time. Due to the R-42 engine's low thrust and high burn time, it was determined that it would not be ideal for the mission (L3Harris, 2023). The MR-80B is a throtttable engine, which makes it desirable (Aerojet Rocketdyne, 2020). It is a monopropellant engine with the lowest specific impulse of all three engines. Due to this fact, its weight, and required propellant mass, was deemed unsuitable (Aerojet Rocketdyne, 2020). Therefore, the R-40B was selected as the main spacecraft engine.

7.2 Main Spacecraft Engine Specifications

The flight-proven R-40B engine uses MMH/NTO (MON-3) (Aerojet Rocketdyne, 2020). This bipropellant consists of Monomethylhydrazine as the fuel and Nitrogen Tetraoxide with approximately 3% Nitric Oxide as the oxidizer. The engine length is 56.6 centimeters and the exit diameter of the nozzle is 40 centimeters (Aerojet Rocketdyne, 2020). It utilizes an Aerojet Rocketdyne Single Seat valve (Aerojet Rocketdyne, 2020). For this mission, only one main spacecraft engine is needed.

7.3 Main Spacecraft Engine Tank Sizing

To size the propellant tanks for the main engine, the sizing process published by Feizal B Muhalim and Krishnan (n.d.) was used as a reference guide. The initial dry mass estimate for the mass of the propellant was 1500 kg, with a final mass of 748 kg. To factor in a safety margin, 750 kg was used to find the mass of propellant. The mass of the propellant came to be 712 kg. In the first iteration of calculations where a dry mass estimation of 1500 kg was used, the mass of the propellant was 778 kg and increased to a mass of 1000 kg for safety assurance. After changing the dry mass to 750 kg, the required propellant did not change enough to warrant another configuration design iteration. Therefore the mass of the propellant remained 1000 kg.

The density of the MMH is 870 kilograms per meter cubed, and the density of NTO (MON-3) is around 1433 kilograms per meter cubed. The mass flow rate of the propellant came from the Aerojet Rocketdyne data sheet for this engine and it is 140 kg/sec. The oxidizer to fuel ratio is 1.6 for this fuel (Feizal B Muhalim & Krishnan, n.d.). The mass flow of the fuel and oxidizer were calculated to be 54 kg/sec and 86 kg/sec, respectively. The next values found were the mass of the fuel and the mass of the oxidizer, which was calculated to be 384.6 kg for the fuel and 615.4 kg for the oxidizer. Additionally, the volumes for each were calculated to be 0.46 meters cubed for the fuel and 0.45 meters cubed for the oxidizer. The initial pressure in the pressurant gas tank is 3 MPa and the final pressure is 0.9 MPa (Feizal B Muhalim & Krishnan, n.d.). From that, the volumes of pressurant gas required for fuel and oxidizer are 0.46 and 0.45 meters cubed respectively. Since only one pressurant tank is being used, the required volume overall is 0.915 meters cubed. The radius for the fuel tank was calculated to be 0.47 meters and the radius for the oxidizer tank is 0.46 meters. The factor of safety was set to 1.3, and 7075 Aluminum Alloy was selected, which has a yield strength of 505 MPa. (Feizal B Muhalim & Krishnan, n.d.). 7075 Aluminum Alloy was selected due to its compatibility with the fuel and oxidizer and high strength to weight ratio (Feizal B Muhalim & Krishnan, n.d.). With that information, the thickness of each was found, but the required thickness was very small and around 0.0005 meters. Therefore, it was decided that the thickness will be 1 millimeter. The mass of the fuel tank was found to be 7.89 kilograms and the oxidizer tank is 7.73 kilograms.

For the pressurant tanks, the mass of Helium required was calculated to be 4.45 kilograms. The storage pressure was set to 30 MPa, as this is what is most commonly used in industry. With that, the volume of the Helium tank came out to be 0.09 meters cubed. The thickness was calculated to 0.01 meters, and the mass is 29.78 kilograms.

In conclusion, the radius of the MMH tank is 0.47 meters, and the thickness is 0.001 meters. The mass is 7.89 kilograms and the volume is 0.46 meters cubed. The radius of the NTO tank is 0.468 meters, and the thickness is 0.001 meters. The mass is 7.73 kilograms, and the volume is 0.45 meters cubed. The Helium tank has a radius of 0.28 meters, and the thickness is also 0.001 meters. The volume is 0.092 meters cubed, and the volume is 0.092 meters cubed.

7.4 Main Spacecraft Engine Schematic

A main spacecraft engine schematic was created. This is shown in Figure 22 below:

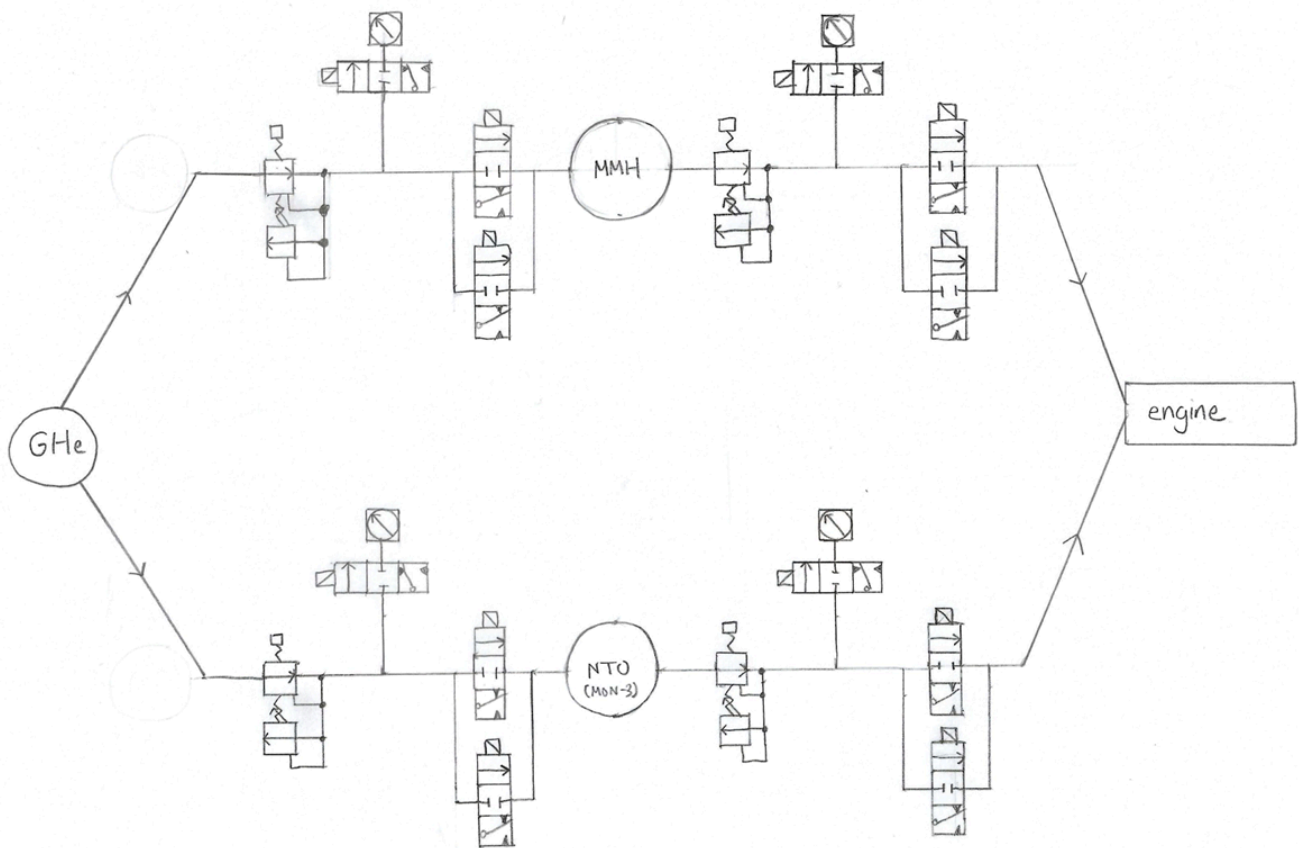


Figure 22. Main Spacecraft Engine Schematic

The flow starts at the Helium tank and flows through a pressure regulator. After the regulator, the flow goes through the transducer for a pressure reading. Next is the solenoid valve.

There are two present so that there is a redundancy. This pressurant gas then flows into the fuel and oxidizer tanks. The fuel is then pushed into the next pressure regulator on the fuel line, and the oxidizer is pushed into its pressure regulator. After the regulator is the pressure transducer for another pressure reading. Next are the solenoid valves, where each line has a redundancy valve as maintaining flow control is very important. Finally, the fuel and oxidizer meet up at the engine.

7.5 RCS Thrusters Trade Study

To begin the thruster trade study, a rough estimation of a Delta-V requirement of 40 m/s was used (Shao et al, 2011). Due to limited information on spacecraft size, the thrusters were compared based on thrust, specific impulse, and propellant mass. These comparisons are shown in Table 15 below.

Table 15. RCS Thrusters Trade Study

Dry Mass (kg)	1500	10500
MR-111G		
Isp (sec)	225	
Propellant Mass (kg)	26.97	188.76
Thrust (N)	4	
Burn Time (sec)	15000	105000
MR-106L		
Isp (sec)	230	
Propellant Mass (kg)	26.38	184.69
Thrust (N)	22	
Burn Time (sec)	2727.27	19090.91
AJ10-220		
Isp (sec)	268	
Propellant Mass (kg)	22.67	158.70
Thrust (N)	62	
Burn Time (sec)	967.74	6774.19

It is important to note that the spacecraft will capture a 9000 kg target object, so while it may make sense in most scenarios to opt for a 4 Newton thruster, the team is aware that more thrust will be required to control the spacecraft. On the other hand, the team wants to avoid so much thrust that more debris is created from the target object as the condition and integrity of the debris is unknown. The MR-106L appears to be the ideal option to meet both requirements. The specific impulse and propellant mass of the AJ10-220 thruster was attractive, but due to it having the highest thrust of all three, it was not selected.

7.6 RCS Thrusters Specifications

The characteristics of the MR-106L thruster include using hydrazine as propellant (Aerojet Rocketdyne, 2020). The status of this thruster is that it is flight proven. From nozzle end to mounting flange, the length is 98 millimeters, and the nozzle exit diameter is 34 millimeters (Aerojet Rocketdyne, 2020). This thruster uses a dual seat valve (Aerojet Rocketdyne, 2020). A configuration of 16 RCS thrusters is the plan as of now based on the success of the Apollo missions (National Aeronautics and Space Administration, n.d).

7.7 RCS Thrusters Tank Sizing

To size the propellant tank for the RCS thrusters, Feizal B Muhalim and Krishnan (n.d.) was referenced as a guide for the process, again. Same as with the main spacecraft engine tanks sizing, the initial dry mass situation is the same. For safety, 40 kg was used to find the mass of propellant. The mass of the propellant came to be 3.6 kg. A large amount of fuel was added to account for ACDS requirements and preferences.

The density of the Hydrazine is 794.52 kilograms per meter cubed. The mass flow rate of the propellant came from Aerojet Rocketdyne for this engine, and it is 0.009 kg/sec. The volume for the hydrazine tanks were calculated to be 0.013 meters cubed. The initial pressure in the pressurant gas tank is 3 MPa and the final pressure is 0.9 MPa. These numbers were pulled from Feizal B Muhalim and Krishnan (n.d.). From that, the volumes of pressurant gas required came out to be 0.021 meters cubed. The radius for the fuel tank was calculated to be 0.043 meters. The factor of safety was set to 1.3, and 7075 Aluminum was selected, which has a yield strength of 505 MPa. (Feizal B Muhalim & Krishnan, n.d.). With that information, the thickness was found, but the required thickness was very small and around 0.0002 meters. Therefore, it was decided

that the thickness will be 0.001 meters. The mass of the four tanks was found to be 0.26 kilograms.

For the pressurant tanks, the mass of Helium required was calculated to be 0.103 kilograms. The Helium tanks are to be broken up into four tanks. The storage pressure was set to 30 MPa, as this is what is most commonly used in industry. With that, the volume of the Helium tanks came out to be 0.0005 meters cubed. The thickness was calculated to 0.003 meters, and the mass is 0.17 kilograms.

In conclusion, the radius of the hydrazine tank is 0.043 meters, and the thickness is 0.001 meters. The mass is 0.26 kilograms, and the volume is 0.01 meters cubed. The radius of the Helium tank is 0.02 meters and the thickness is 0.003 meters. The mass of the Helium tank is 0.17 kilograms, and the volume is 0.0005 meters cubed. Since there are 16 RCS thrusters, the system was broken up into four clusters of thrusters for an ideal spacecraft configuration, thus the smaller numbers.

7.8 RCS Thrusters Schematic

A schematic was created for the RCS thrusters. It is shown in Figure 23 below:

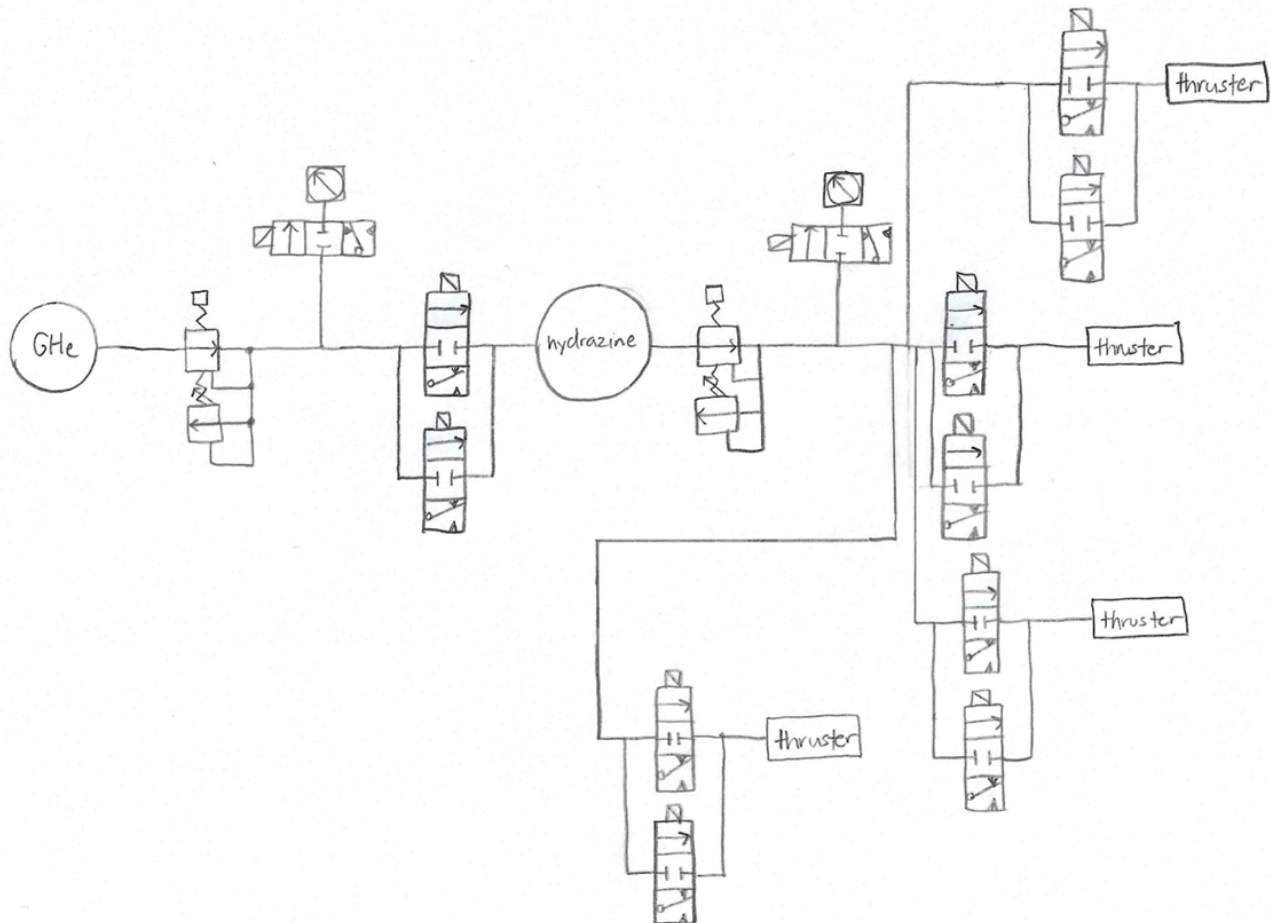


Figure 23. RCS Thrusters Schematic

The flow starts with the pressurant tank where the Helium flows from and to the pressure regulator. The next item is the pressure transducer, so that a pressure reading may be had. Next comes the solenoid valves, where a second one exists for redundancy. The Helium hits the Hydrazine tank and pushes the hydrazine to the pressure regulator. After the pressure regulator is the pressure transducer. Then, the line breaks up into four different thrusters. Each thruster has its

own solenoid valve and a redundant solenoid valve. There exist four of these systems in the spacecraft to account for the 16 thrusters needed for the mission.

7.9 Summary of Propulsion System

To summarize, the engine selected for this mission is the L3Harris R-40B engine due to its thrust, weight, and specific impulse. The RCS thruster selected is the MR-106L in a configuration of 16 thrusters. The propellant mass for the main engine is 1000 kilograms, and the propellant mass for the thrusters is 40 kilograms. The volume for the MMH tank is 0.46 meters cubed, and the volume for NTO is 0.45 meters cubed. The volume for the Helium tank for the main engine is 0.0915 meters cubed. The volume for the hydrazine tank is 0.013 meters cubed and the Helium tank for this system has a volume of 0.0005 meters cubed. There are four hydrazine tanks with four Helium tanks. The schematics for each showcase a blowdown pressurization system.

8.0 Spacecraft Configuration & Structure

8.1 Spacecraft Configuration and Structure Overview

The PERSEUS structure was designed to have four separate open quadrants with a denser center rod to handle axial loads from the main engine and capture mechanism. The structure is made up of wrought aluminum alloy 6061 T6. This quadrant structure houses the components that make up PERSEUS's bus. Figure 24 below depicts PERSEUS in the undeployed configuration, while Figure 25 below depicts PERSEUS in the deployed configuration. Figure 26 shows the joint PERSEUS-Debris configuration.

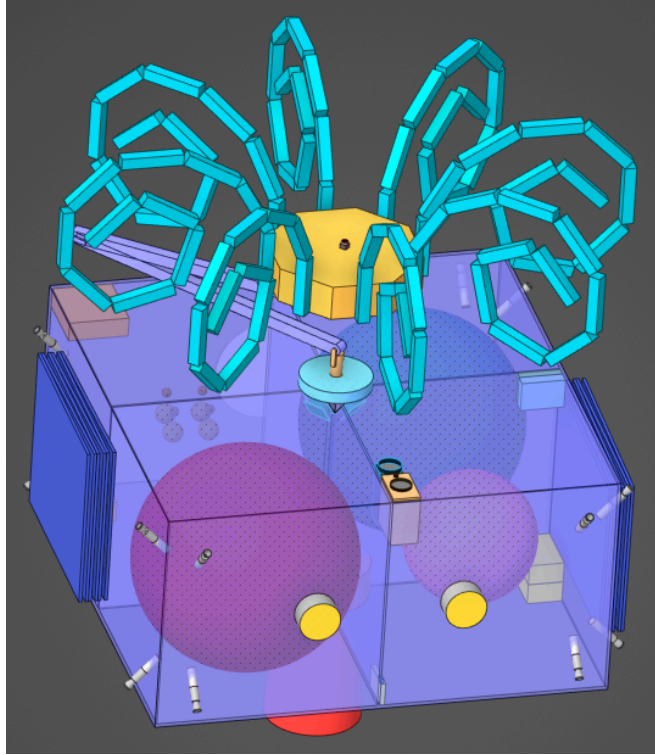


Figure 24. PERSEUS in the Undeployed Configuration

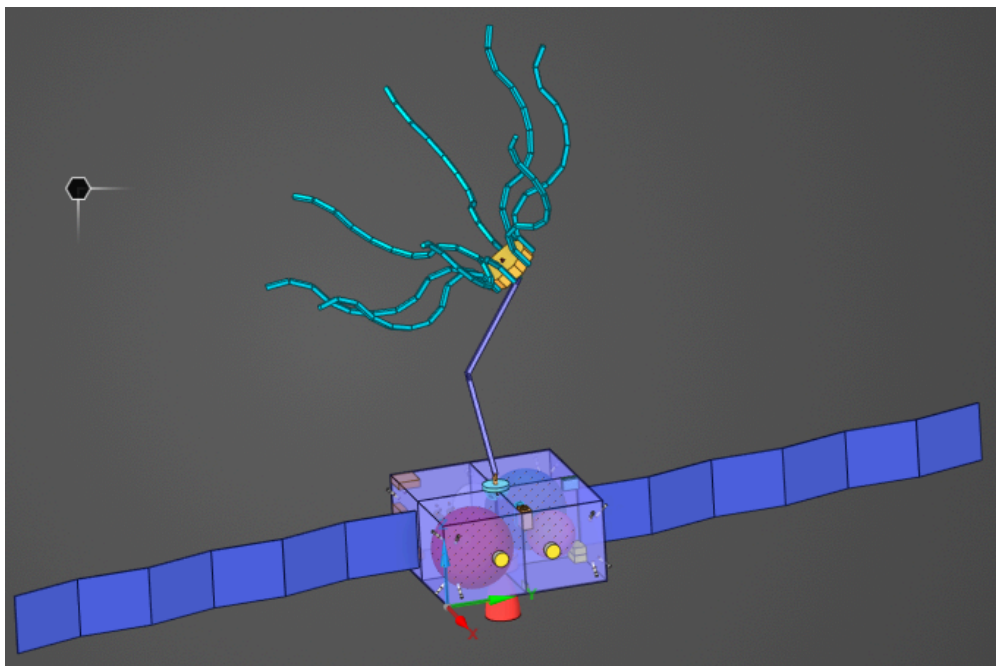


Figure 25. PERSEUS in the Deployed Configuration

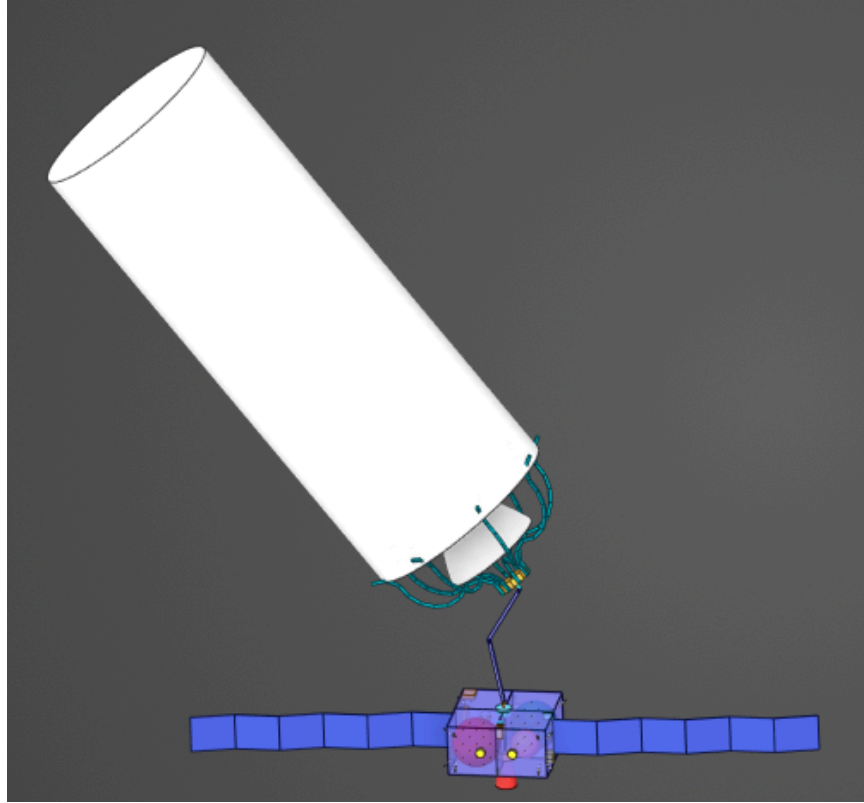


Figure 26. PERSEUS and the Target Debris Post-Capture

8.2 Spacecraft Configuration and Structure Description

Wrought aluminum alloy 6061 T6 was chosen for its structural integrity under high forces, low density, and preferable thermal properties. This material is a common choice in spacecraft design, so it has a high TRL and is cheaper to source. The quadrant design was chosen to naturally separate the four groups of components. The width and height of the quadrants were driven by the size of the main propulsion tanks, leading to dimensions of 1.92 x 1.92 x 0.98 m for the complete structure (Figure 27). The walls of the spacecraft are 10 mm thick, and the central axis pole has a diameter of 40 mm. The mass of the frame alone is 499.188 kg. The dry mass of PERSEUS is 754.137 kg (Table 16). The fully-fueled mass of PERSEUS is 1803.275 kg (Table 16). Table 16 below provides a list of components alongside their mass, total volume, and part name. All components will be sufficiently tolerant to the radiation

environment. Figure 28 shows the layout of components (excluding the arm), and Figure 29 shows the components of the arm. The full PERSEUS system block diagram is shown in Figure 30.

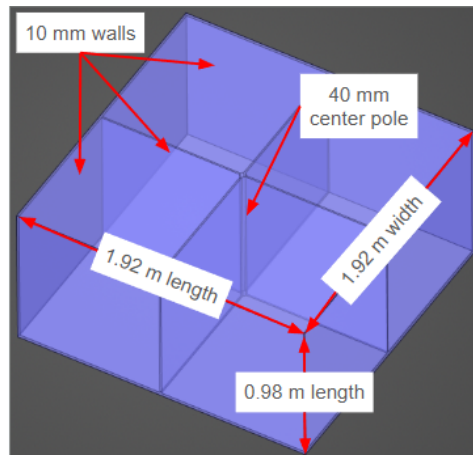


Figure 27. PERSEUS' Structural Layout and Dimensions

Table 16. Bill of Materials (BOM) including Component Names, Quantities, Volumes, and Masses

Part	Part name	Quantity	Vtot (m ³)	Mass (kg)
Structure	Custom Aluminum 4-Quadrant Frame	1	3.61E+00	499.188
REACCH and PFMA	KMI REACCH and PFMA	1	1.12E+00	90
Main Thruster	Redmond R-40B	1	9.00E-02	6.8
RCS thrusters	Aerojet Rocketdyne MR-106L	16	2.64E-03	32
Main prop tank (empty)	Custom Aluminum Sphere	1	4.64E-01	7.869
Main ox tank (empty)	Custom Aluminum Sphere	1	4.51E-01	7.718
Main Engine pressurant (empty)	Custom Aluminum Sphere	1	9.15E-02	28.654
RCS prop tank (empty)	Custom Aluminum Sphere	4	5.03E-02	0.256
RCS pressurant (empty)	Custom Aluminum Sphere	4	3.37E-05	0.148
Solar Panels	Lockheed Martin Small Satellite Solar Array	12	7.52E-03	38.1018
Primary Batteries	Ibeos 50V 1 kWhr Battery	3	1.78E-02	22.5
Secondary Batteries	Ibeos 8S4P 550Whr Battery	2	6.35E-03	8
Flash Lidar	ASC GSFL-16KS	1	5.07E-03	3.2
Optical camera	Infinity Avionics ORION12MP-550	1	2.50E-03	1.5
IR Camera	Teledyne BLACKFLY S BFS-U3-13Y3	1	2.52E-05	0.036
RWUs	Space Inventor Reaction Wheel WHL-500mNms	4	1.51E-03	3.2
IMU	HBK 3DM-CX5-IMU	1	9.60E-06	0.008
Star tracker	Terma T3 Star Tracker	1	3.60E-04	0.35
GPS reciever	L3Harris AS-48917 GPS BAND OMNIDIRECTIONAL ANTENNA	1	3.13E-03	0.454
X Radio	RocketLab Frontier-X	1	1.18E-03	0.92
S Radio	RocketLab Frontier-S	1	7.90E-04	0.59
Antenna	Flexitech 650MHz-10GHz Spiral Antenna	2	4.00E-03	2.566
Computer	FrontGrade Generation 8 (GR740 Based) SBC	1	3.20E-04	0.65

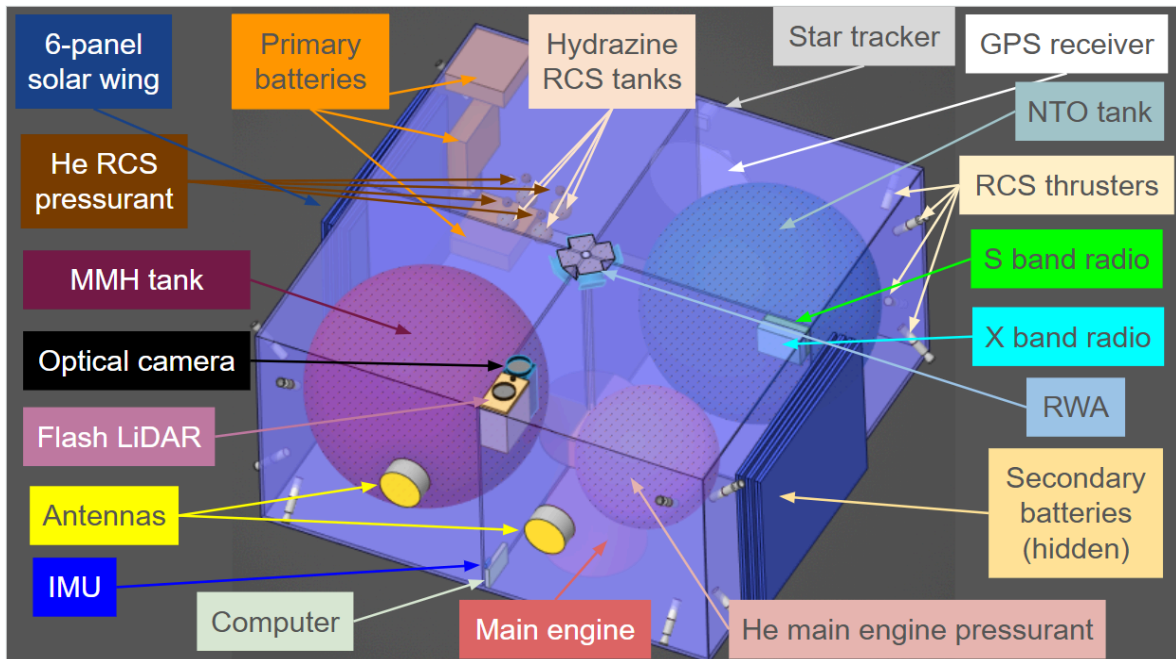


Figure 28. PERSEUS' Component Layout (Excluding the Capture Mechanism)

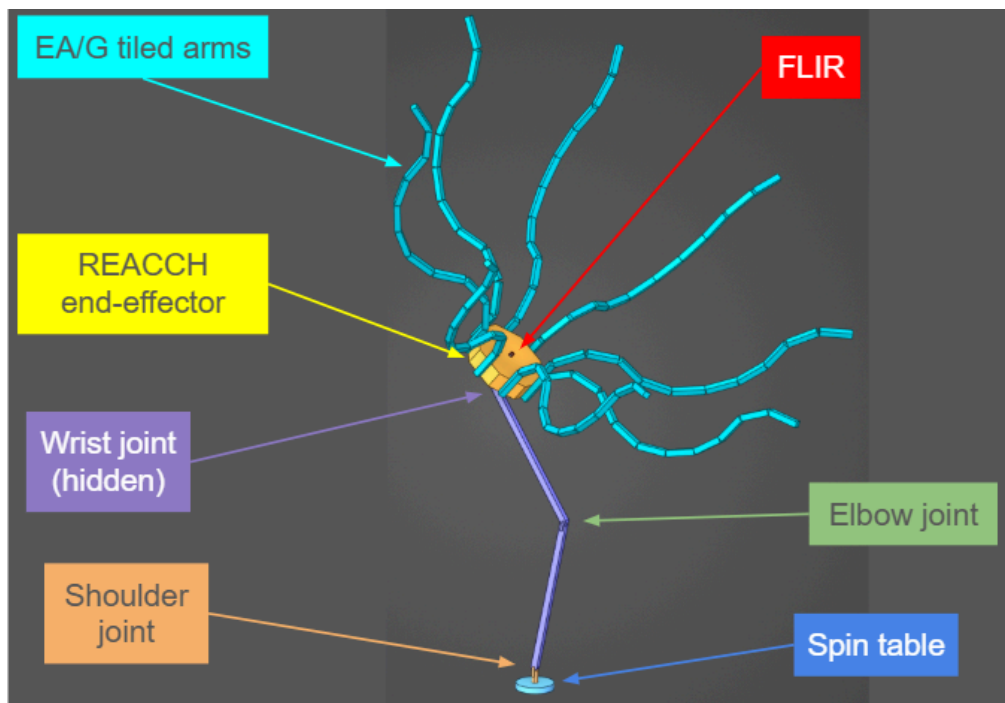


Figure 29. Capture Mechanism Layout

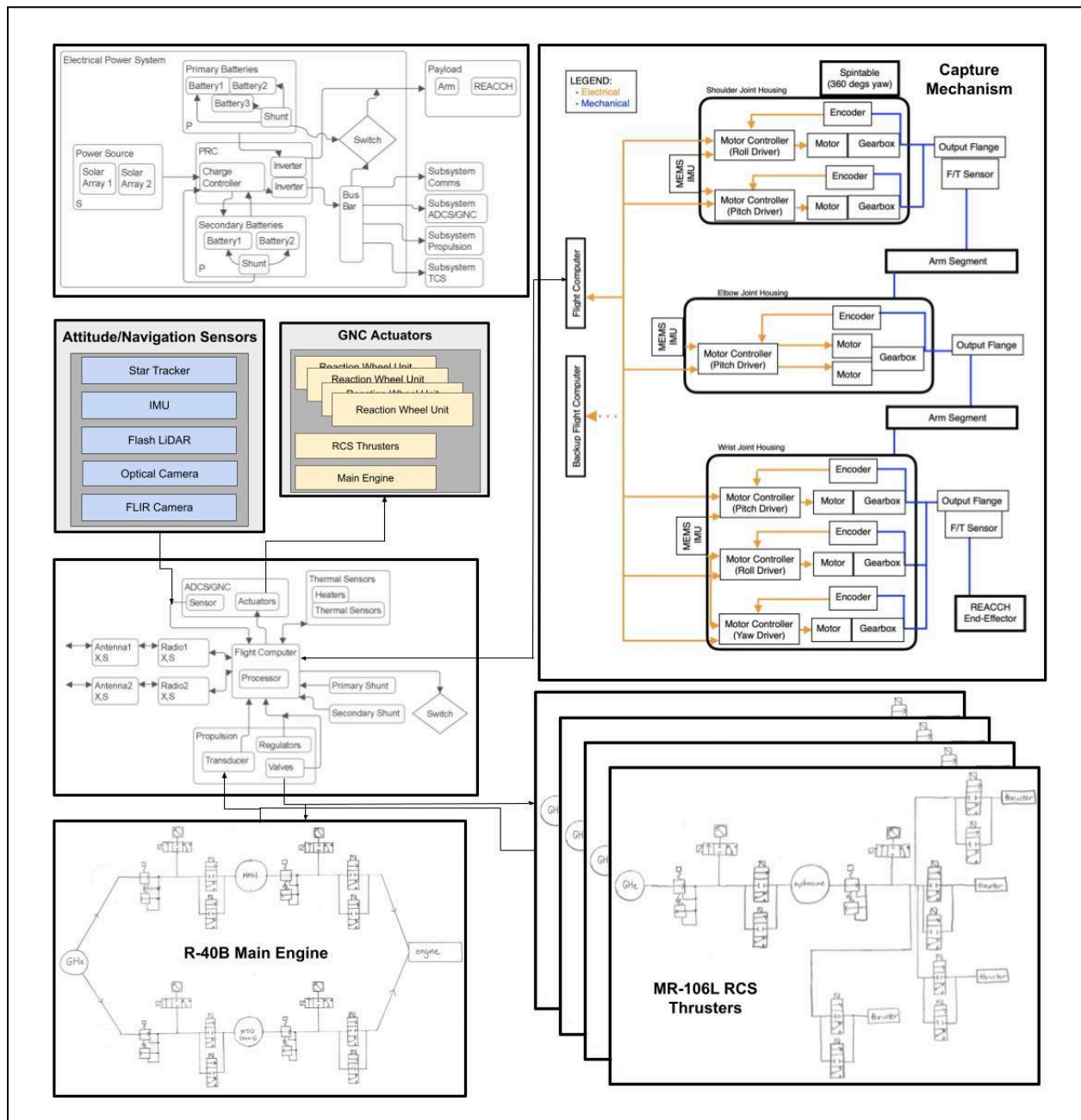


Figure 30. PERSEUS System Block Diagram

9.0 Spacecraft Mechanical

The capture mechanism subsystem is designed to enable the autonomous capture and stabilization of the uncooperative target object, as well as to secure the payload for a controlled deorbit. Major assemblies include a 2.4-meter proto-flight manipulator arm (P-FMA) and a modified eight-tentacle Reactive Electro-Adhesive Capture Cloth (REACCH) end-effector. This subsystem relies on an electromechanical and sensing architecture that ensures precise motion and high reliability during proximity operations and debris-capture maneuvers.

The capture mechanism is mounted to the spacecraft with a torsionally resistant isogrid interface ring, engineered to align the capture loads with the spacecraft's primary thrust structure to ensure that dynamic forces encountered during the capture phase are efficiently transferred and do not compromise the overall stability of the spacecraft (Meyer et al, 1973). To secure the capture mechanism during launch, an Ensign-Bickford Aerospace & Defense (EBAD) NEA Model 9011 non-pyrotechnic hold-down and release mechanism (HDRM) is incorporated. The HDRM provides structural reliability during launch vibrations and accelerations and a low-shock deployment of the robotic arm (EBAD, 2022).

9.1 Robotic Arm Manipulator

The robotic arm manipulator builds upon the P-FMA heritage design concept developed for the Space Shuttle program, incorporating modern actuators, sensors, and avionics. The robotic arm provides precise positioning and compliance of the REACCH end-effector during the capture phase by integrating a spin table at the shoulder joint, a revolute elbow joint, and a wrist joint, achieving 7 degrees of freedom (shoulder pitch and yaw, shoulder roll for indexing, elbow pitch, and wrist pitch, yaw, and roll) (Figure 31).

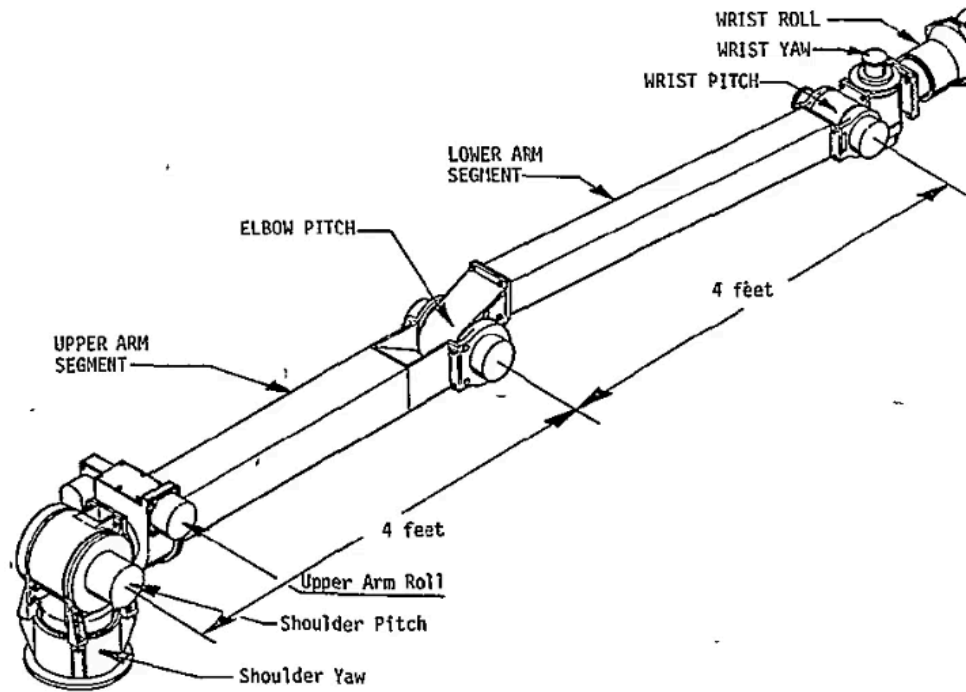


Figure 31. Proto-Flight Manipulator Arm (Britton, 1977)

The high-load shoulder and elbow joints are fabricated of titanium alloy (Ti-6Al-4V) to provide fatigue resistance, thermal stability, and an exceptional strength-to-weight ratio (Vasudevan & Balashanmugam, 2017). The arm segments and wrist housing are constructed from carbon-fiber-reinforced polymer (CFRP), providing maximum strength and stiffness to these secondary structures while minimizing overall mass (Connor, 2024). The arm's total mass (including the REACCH end-effector) is approximately 86 kg. Refer to the block diagram in Figure 32 for a visual overview of the robotic arm and Table 17 for an individual component breakdown by vendor/model.

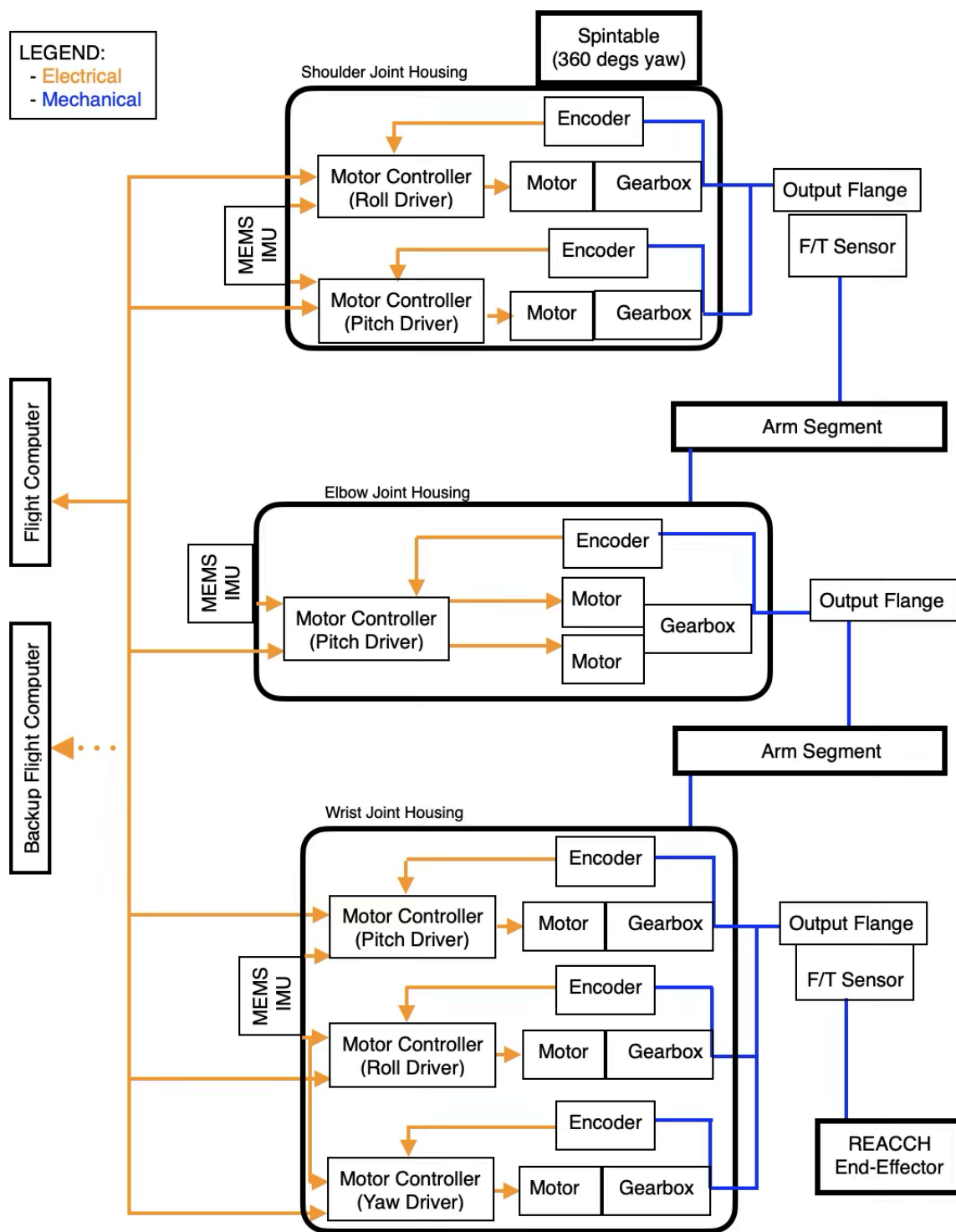


Figure 32. A Block Diagram of the Robotic Arm (Depicts electrical connections in orange and mechanical connections in blue)

Table 17. Mechanical Sub-System Component Breakdown by Vendor/Model/Quantity

Joint	Component	Vendor / Model	Quantity
Hold-Down & Release Mechanism	Ensign-Bickford Aerospace & Defense	NEA Model 9011 HDRM	1
Shoulder Base	Spintable	Moog UPT3 Rotary Gimbal	1
Shoulder Joint	Motor	Maxon EC-90 flat	2
*Note: 2 axes	Gearbox	Harmonic Drive SHG-32-100-2SO (with integrated mechanical brake)	2
	Motor Controller	Maxon EPOS4 70/15	2
	MEMS IMU	Honeywell HG4930	1
	Absolute Encoder	Renishaw Resolute UHV	2
	F/T Sensor	ATI Omega160	1
Elbow Joint	Motor	Maxon EC-60 flat	2 (in parallel)
*Note: 1 axis	Gearbox	Harmonic Drive SHG-25-100-2SO (with integrated mechanical brake)	1
	Motor Controller	Maxon EPOS4 70/15	1
	MEMS IMU	Honeywell HG4390	1
	Absolute Encoder	Renishaw Resolute UHV	1
Wrist Joint	Motor	Maxon EC-60 flat	3
*Note: 3 axes	Gearbox	Harmonic Drive SHG-20-50-2SO	3
	Motor Controller	Maxon EPOS4 70/15	3
	MEMS IMU	Honeywell HG4390	1
	Absolute Encoder	Renishaw Resolute UHV	3
	F/T Sensor	ATI Axia 80-M20	1
	Optical Camera	Teledyne FLIR Blackfly S	1
End-Effector	REACCH (8-tentacles)	KMI (custom model)	1
Avionics	Flight Computer	Frontgrade Generation 8 (GR740 Based) SBC	2 (1x active / 1x cold backup)
	High-Voltage DC-DC Converter	Horizon Power Systems SGRB12050S	1
	Thermal Heaters/ Sensors	Minco/ Thermofoil Heaters HK6907	17
Structural Materials	Ti-6Al-4V	Chalco Titanium	
	CFRP	Hexcel Corporation	
	MLI	QUEST 10-Layer IMLI	

Each joint utilizes a specialized actuator assembly that consists of a Maxon brushless DC (BLDC) motor and associated controller, a Harmonic Drive SHG-series strain-wave gearbox (with an integrated mechanical brake for the shoulder and elbow joints), an IMU, and force/torque sensors (for the shoulder and wrist joints only). The BLDC motors provide high-speed, reliable rotation, while the strain-wave gearboxes convert the fast, low-torque rotation into slow, high-torque motion, enabling precise arm movements (Harmonic Drive, 2021; Maxon, 2022). The joints will generate torques of approximately 125 Newton-meters at the shoulder, 54 Newton-meters at the elbow, and 13 Newton-meters at the wrist (Refer to Tables G1.1/G1.2 and Table G2 in Appendix G for joint torque calculations). Utilizing 12.9 m² of Quest 10-Layer Integrated Multi-Layer Insulation (IMLI) and 17 Minco HK6907 thermofoil heaters, the mechanical subsystem will have an operational temperature range of -35 degrees Celsius to 85 degrees Celsius (Minco, n.d.; Quest Thermal Group, n.d.) (Refer to Table G3 in Appendix G for the mechanical subsystem thermal calculations).

The BLDC motors are controlled by radiation-tolerant Maxon EPOS4 70/15 digital position controllers, which regulate joint movement by controlling motor speed, position, and torque (Maxon, n.d.). The controllers provide motor communications, current regulation, and telemetry, using high-resolution angular position data feedback from the Renishaw Resolute Ultra-High Vacuum (UHV) absolute optical encoders (Renishaw, 2023). The SpaceWire communication link provides low-latency, high-reliability communication between the joints and the Frontgrade Generation 8 (GR740-based) flight computer (Frontgrade Gaisler AB, n.d.; Parkes & Wilson, 2019).

In addition to joint force/torque sensors, which provide a measure of mechanical load, a Honeywell HG4930 Micro-Electro-Mechanical System (MEMS) IMU is also incorporated into

each joint assembly, providing 3-axis angular rate and linear acceleration data (Honeywell Aerospace, n.d.). To augment encoder data, the IMUs improve dynamic state estimation, including during transient events such as initial contact and load shifts, by enhancing disturbance rejection and increasing the accuracy of fine manipulation tasks. The combination of encoder/IMU feedback provides dual-sensor redundancy, and fault detection, fault isolation, and recovery (FDIR) (NASA, n.d.).

To account for the precession and nutation of the target object during the initial extension of the robotic arm, the elbow joint will adjust to match the angle of the target object. With a 2.4-meter arm, the precession angle will be limited to approximately 20 degrees. Once adjusted, the robotic arm can then be spun up via the Moog Ultra Performance Type 3 (UPT3) rotary gimbal until the relative velocity is less than 0.25 m/s, therefore aligning the REACCH mechanism with the target object's nozzle (Bylard et al, 2017; Moog Inc., 2020).

To effectively and securely grasp the target object, the wrist compliance of the robotic arm will enable final adjustments of the REACCH mechanism via translational and rotational springs and dampers (Bylard et al, 2017). The compliance and damping of the robotic arm is designed to provide limited amounts of off-nominal compliance to velocity or angle errors. It will enable angular momentum to be gradually transferred to PERSEUS within 30-35 seconds (Refer to Table G1.1 in Appendix G for time calculations). Once all relative rotation has been eliminated, the robotic arm will retract and its joints will become rigid to enable a controlled stabilization of the PERSEUS/target object pair.

9.2 Capture Mechanism

REACCH is a multi-armed articulated robotic system that is capable of grasping an unprepared object using geometric conformance and mechanical/electrical adhesion, which means no preexisting docking structure is required for capture (Morris et al, 2024). It consists of rigid-segmented tentacles equipped with low-power, high-grip Electro-Adhesive/Gecko (EA/G) tiles, designed for the non-destructive, geometric-compliant capture and manipulation of unprepared space debris ranging in diameter from 25 cm to greater than 6.5 m (Dremann et al, 2025; KMI, 2025). The REACCH base/body will be affixed to a robotic arm and will house the power and communication modules, relays, and control system. Each tentacle will contain an individual power and data line to connect the control system to the EA/G tiles and elastic stretch load sensors (Narayana et al, 2020).

Electro-Adhesive (EA) technology utilizes a voltage differential across a set of electrodes to create a polarized electric field on the surface material which generates a charge buildup and results in surface adhesion. While the adhesion strength is relatively weak, the advantages of EA tiles are effective and controllable attachment to a wide variety of surfaces ranging in composition and roughness, as well as being able to operate in a vacuum environment without the use of preloaded force or chemical bonds. In contrast, gecko, or dry adhesives, consist of micro-structures that deform under load to provide a large area of surface contact and create adhesion via Van der Waals forces. Non-directional dry adhesives typically generate high adhesion on smooth surfaces, but are found to lack controllability for rough surfaces (Bylard et al, 2017). On the contrary, gecko-like tiles, which utilize asymmetric micro-structure hairs to generate an adhesive force when loaded in a preferred direction, generate lower adhesion forces

but are more suitable for rough surfaces. By using a hybrid combination of EA and gecko-like tiles, the shear force applied to a surface is greatly increased (Narayanan et al, 2020).

The rigid-segmented spine of each tentacle serves as both an actuation and retraction mechanism to provide adequate load transfer from the EA/G tiles to the base/body structure. The mechanical link segments are connected by low-friction ball bearings, providing a highly compliant, rigid structure. The internal pulley design of the tentacle spine enables each tentacle to unroll and conform to the object's surface (Figure 33). The combination of mechanical grasp and EA/G grip reduces impact forces upon capture, preventing object damage and fragmentation (Morris et al, 2024).

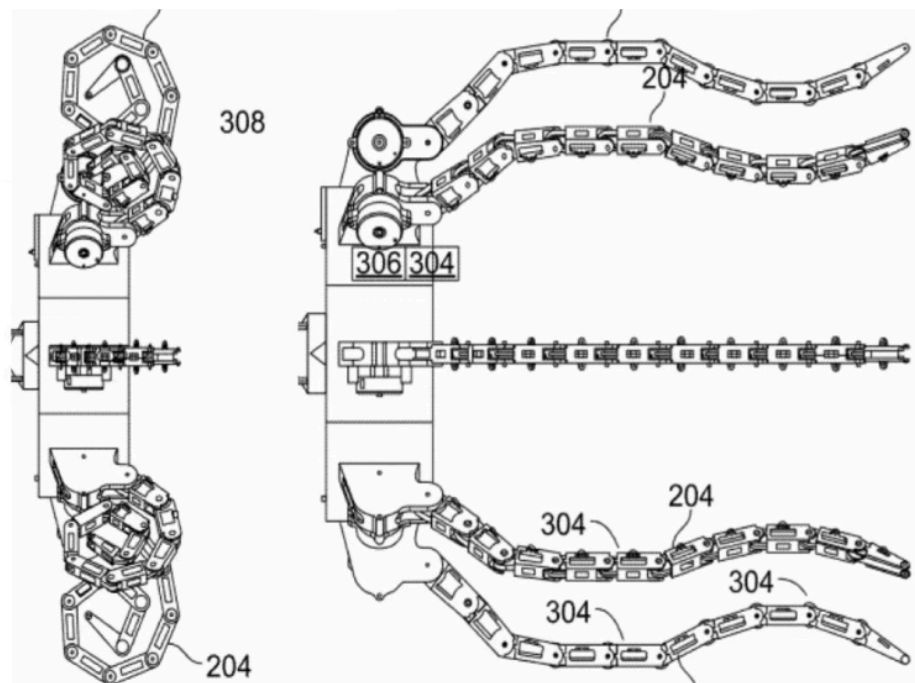


Figure 33. A Technical Illustration of the REACCH Rigid-Linked Robotic Tentacle Grasping Mechanism (Depicted is the stowed (left) and unstowed (right) positions)
US Patent #11,661,217 (KMI, 2025)

Originally developed by NASA JPL and the University of Southern California, REACCH is now exclusively licensed to Kall Morris Inc (KMI). Since 2022, KMI has completed over 1000 ground captures and 172 on-orbit captures with the REACCH robotic system. During testing, the EA/G tile technology was found to be effective on numerous materials and geometries, enabling a soft initial contact while maintaining a strong capture grip. A four-arm REACCH prototype demonstration was conducted and validated aboard the ISS in 2024 and currently has a technology readiness level of 7. KMI is working on the expansion to a functional 8-arm prototype for near-future operational missions (KMI, 2025). The modified REACCH end effector onboard PERSEUS will consist of eight 2-meter-long tentacles to enable the capture of an S/L-16 rocket booster with a diameter of 4 meters. Additionally, a Teledyne FLIR Blackfly S will be mounted in the center of the tentacle base to provide visual data that will be fused with the flash LiDAR data to provide accurate position data during the capture phase (Vicon, n.d.).

9.3 Power and Data Architecture

The power and data architecture of the mechanical subsystem has been scaled for peak torque and mass requirements. To regulate power from the nominal 28V DC spacecraft bus to the capture mechanism power is routed through a radiation-tolerant distribution unit containing a Horizon Power Systems SGRB12050S DC-DC converter (Horizon Power Systems, n.d.). Voltage rails in the distribution unit incorporate overcurrent protection and voltage sensing for operational safety. High-voltage isolation for EA/G tiles power distribution is incorporated to prevent arcing within the harsh vacuum and thermal space environment.

Idle, continuous, and peak power requirements for the REACCH end-effector are calculated to be 192 Watts, 645 Watts, and 1180 Watts respectively (See Table G4 and G5.1/5.2

in Appendix G for all mechanical subsystem power calculations). Accounting for the peak power of both the robotic arm and REACCH end-effector, a total power consumption during the capture of the 9000 kg target object is estimated to be 5475.3 Watts if all components were to operate simultaneously. To decrease the total power requirement of the mechanical subsystem, the capture phase has been broken up into individual subphases based on the capture timeline shown in Table 17.

Table 17. Capture Timeline of the Mechanical Subsystem with Associated Power Requirements

Sub-Phase	Time (min)	Time (s)	Spin Table	Robotic Arm	REACCH	EA/G Tiles	Total Power Per Sub-Phase (Watts)
Unstow - HDRM	0.00833333	0.5	0	30	0	OFF	30
Deploy Shoulder	3.75	225	0	2254	0	OFF	2254
Deploy Elbow	3.75	225	0	1400	0	OFF	1400
Deploy Wrist	7.5	450	0	3678	0	OFF	3678
Spin table spin-up	0.25	15	200	0	192	OFF	392
REACCH capture	15	900	150	0	1180	ON	1330
Spin table de-spin	0.58333333	35	200	0	645	ON	845
Shoulder retract	10	600	0	2254	645	ON	2899
Elbow retract	10	600	0	1400	645	ON	2045
Wrist retract	20	1200	0	3678	645	ON	4323
Total Time	70.8416666	4250.5					

Redundant SpaceWire networks are integrated to handle communication between the Frontgrade Generation 8 flight computer, the joint electronic modules, and the REACCH end-effector. The GR740 quad-core LEON4FT processor within the flight computer manages deterministic control loops, while the joint electronic modules fuse encoder and IMU data for local control. The Field-Programmable Gate Array (FPGA) assumed to be integrated into the REACCH end-effector is a reconfigurable microelectric device that allows custom digital circuits

to be incorporated. The FPGA would manage tentacle actuation and high-voltage control with sub-millisecond timing accuracy. Time-critical loops would operate at 500-1000 Hz, minimizing communication latency to less than 1 millisecond and ensuring responsive, real-time adaptation during the dynamic capture phase (Kim et al, 2015; Wu et al, 2018). The cold-backup flight computer and the redundant, fault-tolerant power and data paths, as well as the compliant mechanisms and dual-sensor FDIR, will enable the mechanical subsystem to minimize risk and maintain operational safety under partial system failures, including motor controller, absolute encoder, or high-voltage anomalies.

10.0 Spacecraft Thermal Control

10.1 Thermal Control System Overview

The thermal control system (TCS) of a spacecraft is responsible for maintaining all onboard equipment within allowable temperature limits throughout the mission. Since spacecraft are in a vacuum environment, the main mode of heat transfer is through radiation, where careful management of thermal energy is essential. The system ensures proper performance during all phases of the mission, including launch, parking orbit, Hohmann transfer, NMC, Rendezvous, capture, and deorbit.

Thermal control methods are typically classified into two categories: passive and active. Passive thermal control relies on design features and materials that regulate heat without consuming any power, such as multilayer insulations (MLI), surface coatings, thermal blankets, and heat pipes. These elements reduce heat loss or gain (Mell, R. 2000) and redistribute thermal energy across the spacecraft. Active thermal control, on the other hand, does require power to operate and includes components such as electric heaters, thermoelectric coolers, and fluid loops

with pumps. These systems allow fine-tune temperature regulation when passive methods alone aren't enough.

The primary objective of the TCS is to balance internal heat generated by electronic subsystems with external inputs from the Sun, Earth, and deep space, while minimizing power consumption and mass. This requires accounting for environmental variations, such as eclipse periods and attitude changes, as well as operational modes of onboard instruments. In some cases, pre-heating or controlled warm-up cycles are implemented to protect sensitive components from thermal shock or excessively low temperatures.

By integrating both passive and active strategies, the thermal control system ensures the spacecraft's instruments and subsystems remain within specified temperature ranges, preserving functionality, reliability, and mission longevity.

10.2 Thermal Control System Requirements

Thermal control systems must be defined in coordination with all other spacecraft subsystems, with requirements and constraints established early in the mission design process. Integrating thermal considerations from the start ensures a balanced and effective overall spacecraft design, rather than forcing a fixed configuration to operate within temperature limits after the fact. For PERSEUS, the thermal system must maintain the integrity and efficiency of all equipment (Juhasz, A., 1998). The Spacecraft internal temperature shall be 15°C with a tolerance of $\pm 15^\circ\text{C}$ to keep all components within their operating temperature (Table 18). At the optimal temperature of 15°C, the spacecraft hull is capable of radiating 5212.47 Watts of thermal energy into space. Thermal calculations were done with the blackout ratio and the spacecraft in three orientations relative to the Sun; ram direction, zenith direction, and 45 degrees angled. In the

Ram direction, the hull will radiate 770.77 Watts into space. In the zenith orientation, the hull radiates 1206.69 Watts into space more than is absorbed. In the angled orientation, the hull radiates 1447.35 Watts. If any two of the roll, pitch, or yaw are angled 45 degrees such that the corner of the craft is left indefinitely Sun facing, the craft will absorb more energy between the hull and solar panels than is emitted into space. Whenever possible, the craft will be oriented with the zenith towards the Sun for maximum efficiency with the solar panels and minimum heat absorption.

Table 18. Operating Temperatures of Each Component

Component	Min temp C	Max temp (C)	Min temp (K)	Max temp (K)
Robot arm	-35	85	238.15	358.15
Star Tracker	-40	30	233.15	303.15
IMU	-40	85	233.15	358.15
GPS reciever	-100	100	173.15	373.15
Optical Camera	-10	50	263.15	323.15
Flash LiDAR	0	30	273.15	303.15
FLIR	0	50	273.15	323.15
Reaction wheels	-20	60	253.15	333.15
Fuel tank	-50	88	223.15	361.15
Oxidizer tank	-30	50	243.15	323.15
Pressurant tanks	0	30	273.15	303.15
Secondary Battery	0	45	273.15	318.15
Primary Battery	0	45	273.15	318.15
Antenna Array	-20	85	253.15	358.15
X Radio	-25	55	248.15	328.15
S Radio	-30	60	243.15	333.15
Computer	-25	65	248.15	338.15

10.3 Thermal Control System Components

The PERSEUS spacecraft will be able to maintain its temperature within the operational range of the onboard components across all phases of the mission. This includes the launch and transfer to orbital operations. The system must ensure the spacecraft and its payload remain within operational and survivable temperature ranges, even when thermal energy is transferred

from the Sun, Earth, and deep space to the vehicle itself. PERSEUS utilizes a mixture of passive and active thermal control methods to accomplish this (Boeing, n.d).

10.3.1 Passive thermal control

Passive thermal control uses an Aeroglaze A276 thermal coating with a defined absorption and emissivity. Passive heat pipes (Figure 34) are integrated into the system to move excess heat away from electronics and high-dissipation components. Using ammonia as a phase-change fluid ($T_b = 240\text{K}$), these sealed tubes transfer thermal energy from hot spots to the aluminium frame, conduction through the frame transfers heat to the spacecraft surface where radiation then rejects excess heat to space.



Figure 34. Thermal Control Panel with Heat Pipes

10.3.2 Active Thermal Control

Active thermal elements, including the Minco HK6907 Thermofoil Heaters, are employed during eclipse periods, startup modes, or when components must be preconditioned before operation. By combining active and passive thermal control methods, the PERSEUS TCS provides a stable and controlled thermal environment, ensuring spacecraft reliability and extending mission life while minimizing power consumption.

10.3.3 Thermal Control Cooling System

The cooling system for the TCS entails multiple elements in operation. To start with, the hull acts as a large radiator with a total area of approximately 14.89 m². The surface rejects heat into space via radiation, and the radiating capacity is specified by the emissivity of the surface coating of Aeroglaze A276.

The hull must absorb all excess thermal energy from onboard equipment, which at maximum load is approximately 600 W, as well as the incoming heat from the surrounding space environment. Balancing these, the spacecraft must maintain the interior temperature of the craft at an ideal temperature of 15 °C (~285K).

The Stefan–Boltzmann law controls thermal performance. The emissivity of the coating is 0.895 to achieve maximum radiative efficiency with a low absorptivity of 0.23, which minimizes solar heating when in the sunlit state. Under steady-state operating conditions, the system can reject approximately 5.2 kW of cumulative heat. This margin covers both shaded and sunlit operations and is needed for safe operation even when worst-case thermal loading is encountered. To transfer heat from concentrated hot spots to the hull, aluminum heat straps and

embedded heat pipes cover the entire spacecraft structure. They serve as high-conductivity paths that collect excess heat from electronics and subsystems and transport it directly to the hull where it can be radiated into space.

10.3.4 Thermal Control Heating System

PERSEUS employs a distributed set of heaters to keep all avionics and payload equipment in their operating temperature regime when in eclipse or cold environments. Minco HK6907 Thermofoil Heaters (Figure 35) serve as the primary heating elements, due to their low-profile, lightweight form and high thermal stability.

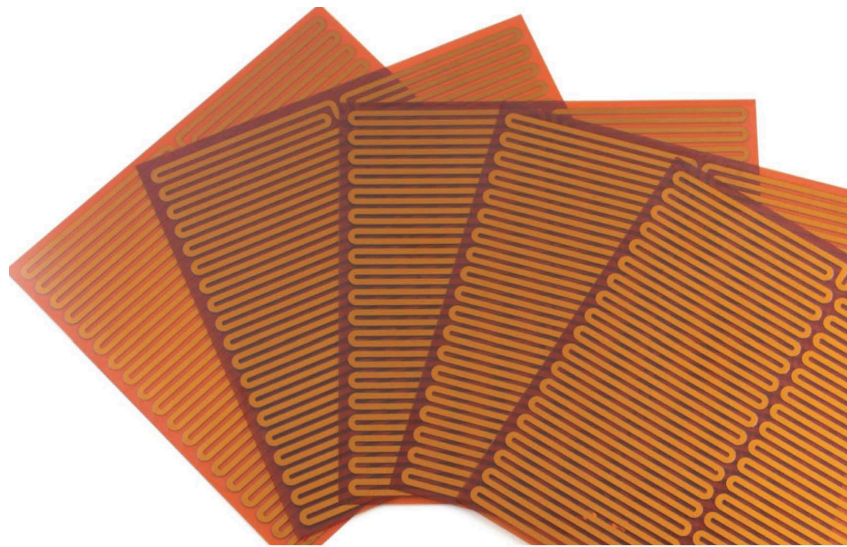


Figure 35. Space Rated Minco HK 6907 Thermofoil Heaters

Minco HK 6907 heaters are installed on key spacecraft surfaces, such as battery plates, avionics panels, and propellant tubing, and the adhesive-backed fastening provides effective thermal contact. They can be flexed to fit flat or curved surfaces, and their construction keeps

their mass low and their outgassing in check, suitable for use in the vacuum environment of space. Each is complemented with temperature sensors and is controlled by the thermal management system; power is provided only when required to preserve spacecraft power.

With this combination of patch heaters, heat pipes, and coating, PERSEUS maintains its internal environment at the preferred 15°C operating point, despite eclipse times and changes in orbital conditions.

11.0 Power Systems

11.1 Power Subsystem Overview

The power subsystem shall provide the required power through multiple modes including; 230.8 Watts for the low power mode, 809.36 Watts for the maneuver mode, and 1381.36 Watts for the deployment mode. Power will be generated by a Gallium Arsenide Multijunction small satellite solar array designed and built by Lockheed Martin. To meet the spacecraft requirements, the solar array will consist of 12 panels. Each panel has an effective area of ~0.592 square meters and can produce 175 watts at the Beginning Of Life (BOL). This provides ~295.712 watts per square meter. The spacecraft's orbital period is ~1.69666 hours with a blackout period of ~0.584622 hours. This leaves ~1.112028 hours or ~65.542% of the orbit as the effective charge time per orbit. During each orbit ~2.274 kW-hrs of power will be used if the spacecraft is in the deployment mode. This is split between ~1490.75 Whrs during the effective charge time and ~783.721 Whrs during the eclipse time. Two ibeos 28 volt modular secondary batteries are used to store the power during the eclipse time. Both are in the 8S4P

configuration providing a combined 1100 W-hrs. The use of two secondary batteries avoids a single point failure for the mission.

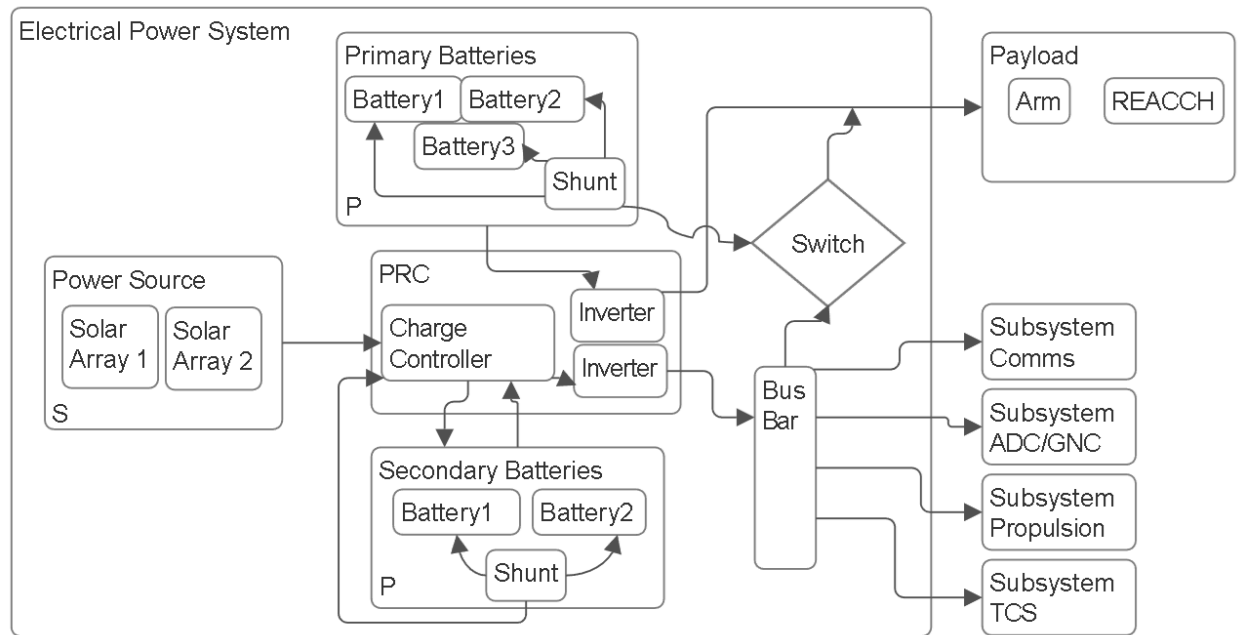


Figure 36. Power Block Diagram

Power is collected at the solar panels and transferred through the charge controller to the secondary batteries. There is a Shunt connected to each battery pack, as shown in Figure 36, which monitors the power levels and adjusts the power current flow between the hardware and charging batteries. Power flows from the batteries through inverters to a Bus Bar before moving onto the individual subsystems. The primary batteries push power out through a dedicated inverter to the payload subsystem. A dedicated shunt monitors the primary batteries and if low, controls a switch to allow power from the secondary batteries through to the payload.

11.2 Power Generation

The power generation options were a solar panel array, RTG, Fuel Cell, or Primary Battery. This mission duration calls for a power system that can last for three weeks. A duration of that length makes a sole primary battery configuration non-viable. The next consideration is cost for a single use spacecraft. The final consideration is minimizing mass. Fuel cells do not produce enough overall power to sustain the hardware for the mission duration and RTGs are too costly for a single use mission. Solar panels are light weight and will last the duration of the mission. Values for the selected solar panels are taken from the product data sheet.

Table 19. Solar Array

Power Required Altitude Charge Duration ratio Orbital Period Design Lifetime	1381.36 Watts 844 km .6554 1.69666 hr (101.7996 min) 1 year
Power Eclipse Power Daylight Eclipse Time Daylight Time Power Solar Array Panel Type	Pe = 1381.36 Watts Pd = 1381.36 Watts Te = 35.07732 min Td = 66.7223 min Psa = 2937.05 Watts Triple or quad junction
SiPo GaAsPo MultiJunctionPo Selected Panel Efficiency Selected Panel Type	248.6765 W/m2 310.8457 W/m2 470.4691 W/m2 21.72% GaAs MultiJunction
Given Power output of panels (data sheet) Mission Duration Duration decay performance P-EoL	175 Watt BOL <30 days Negligible 174.6047 Watt EOL
Area needed Based on EoL Mass (data sheet)	Asa = 9.448579 square meters Asa = 9.46997 square meters 38.1018 kg

11.3 Solar Panel Selection and Sizing

Solar Panels generate power while receiving incident solar radiation at a given distance from the Sun. The Solar radiance at Earth's orbital distance is ~1361 Watts per square meter. The panels have an effective collection area of ~0.591792 square meters and produce 175 Watts per panel with a general power collection of ~295.7118 watts per square meter. The highest energy mode of PERSEUS takes 1381.36 watts. The orbital period is ~1.6967 hours (101.7996 min) which comes out to a total of ~2343.7 Watt-hours per orbit split between a collection phase of ~1.112 hours (66.7223 min) and blackout phase of 0.574 hours (35.077 min). These values are also shown in Table 19.

11.4 Batteries

The battery selection goal is to provide enough power during the blackout period (Halpert, G. 1999) while keeping the mass to a minimum. The selected battery is the ibeos B28 28-Volt Modular Battery which comes in 4 configurations with varying Watt-hour capacity. The PERSEUS spacecraft uses two batteries in the B28-550 configurations (Table 20) which supply enough power during the eclipse portion and allow for redundancy. There are three 50V/1000Whr batteries used only to power the arm and REACCH end effector payload components. Should the primary batteries run out of power, the secondary batteries can transfer power to the payload.

Table 20. Battery Information

Solar panel table	
Mission Length	<30 days
Primary Power Storage	Ibeos B28-550 Configuration2
Secondary Power Storage	Ibeos 1000-Watt-hour
Orbital Parameters	
Eclipse Frequency	14.14543 eclipses per day
Eclipse Length	0.5846 Hours (35.077 min)
Power Use Profile	1381.36 Watts during eclipse
Voltage	28 Volts
Current	<80A Discharge, <20A Charge
Depth of Discharge	73.42%
Industry Standard DoD	80%
Battery Type	Lithium-Ion
Number of Batteries	Na=2 8S4P, Nb=3 1000 Whr
Efficiency lower end	n=0.95
Transmission efficiency	Cr=578.9141 W-hr 20.67669 Amp-hr

11.5 Power Summary

The power subsystem uses a solar array to generate all the needed power. The wattage is split between running the hardware and charging two lithium-ion batteries during the daylight portion of each orbit (McKissock, B. 2009). The two batteries are used to supply power during the eclipse portion of each orbit. The three primary batteries are used to supply power only to the arm and REACCH end effector. If the primary batteries are used up, the spacecraft will enter a phased operation mode, using the secondary batteries to toggle individual components of the arm and REACCH between recharges until splashdown.

12.0 Communications & Data Systems

12.1 Communications Subsystem Overview

The communications subsystem communicates with the Near Earth Network and consists of two Flexitech Aerospace 650MHz-10GHz Spiral Omnidirectional Antennas for redundancy, one AS-48917 GPS Band Omnidirectional Antenna, two RocketLab Frontier Radios configured for S and X bands, and a SpaceWire data processor. The uplink frequencies are centered at 2.07 and 7.19 GHz and downlink frequencies centered at 2.25 and 8.45 GHz. Two antennas are used for redundancy and for simultaneous uplink and downlink each having a pointing accuracy of ~ 34.64 degrees from nadir before a -3dB falloff of signal strength calculated with a 1dB acceptable loss (Shao et al, 2011). The ground swath length is 2923 km.

Data is received through either antenna from the ground stations and routed through the radio on QPSK modulation (Schaire, 2019) to the processor in the flight computer. The information is then passed to the relevant subsystem. Sensors, shunts, and transducers send signals to the processor which can then send the information to the surface or if in a comms blackout phase, process the data and send controls to relevant motors or heaters. Figure 37 shows the block diagram for the communications and data subsystem.

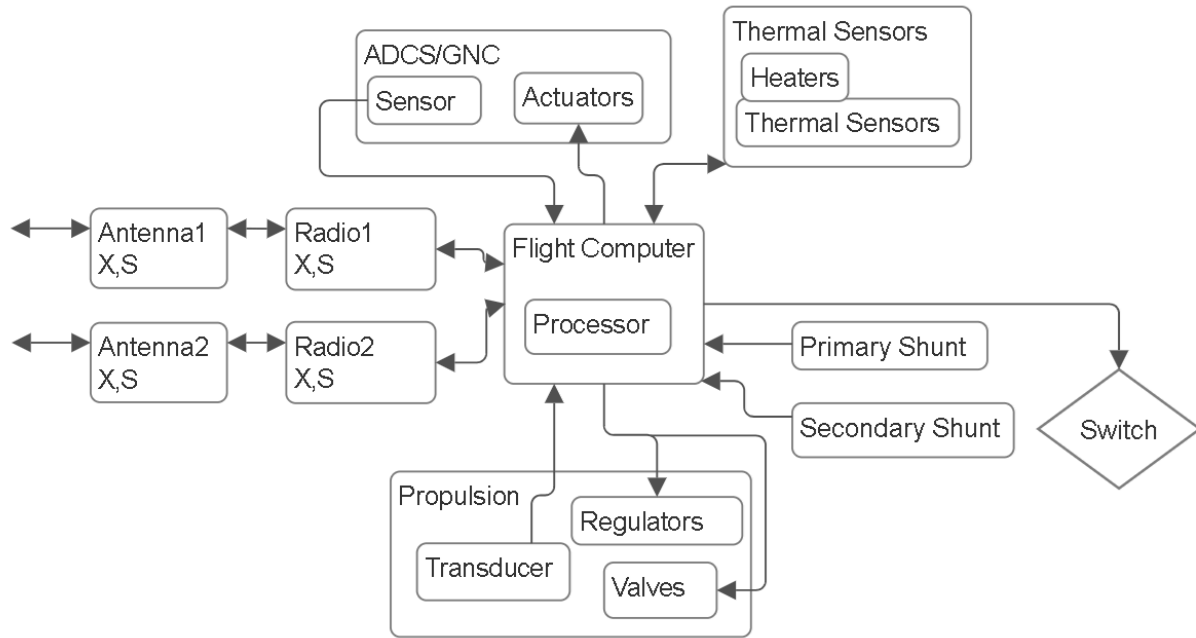


Figure 37. Communications and Data Block Diagram

Figure 38 and Figure 39 show the periods when the PERSEUS spacecraft will be in contact with one of the ground stations. The green periods are when telemetry and data will be able to be sent and received from the ground. During the black periods, PERSEUS will be operating autonomously based on previously sent instructions.

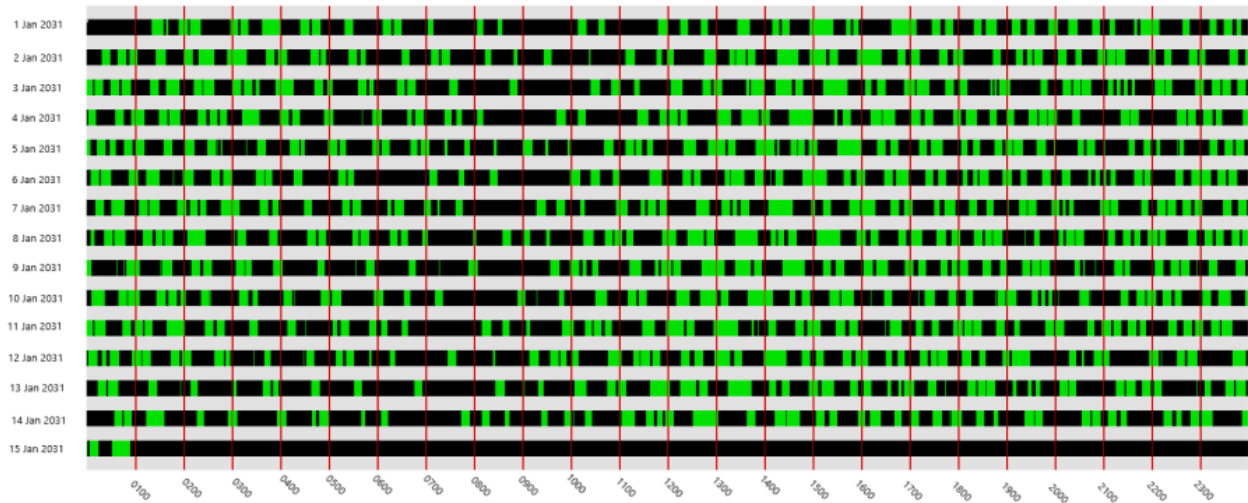


Figure 38. Communications Black Out Periods Before Capture

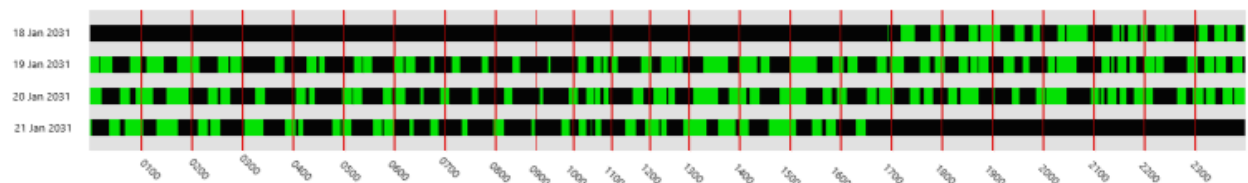


Figure 39. Communications Black Out Periods After Capture

12.2 Link Budget

The available uplink (Table 21) portion of the Link Budget allows for 247.3776 dB in the S Band and 253.1297 dB in the X Band. The Link budget shown is for the Fairbanks ground station. The station uses several ground antennas but our mission will only communicate with the 11.28 meter dish over the uplink frequencies shown in Table 21 and the downlink frequencies shown in Tables 21 and 22. With large amounts of communication blackouts, the large link margin becomes very important to sufficiently transmit telemetry and data.

Table 21. Uplink Budget

	Unit	S-Band	X-Band	Comments
Uplink Frequency	GHz	2.07	7.19	Central frequency for channels
Ground Station Type		Tracking Dish AS1		11.28M
Beamwidth	deg	0.85	0.23	
Gain	dBi	45.8	56.8	
EIRP	dBW	64.6	64.6	User's Guide
Propagation Range	km	844	844	
Space Loss	dB	-157.296	-168.111	
Atmospheric loss	dB	-10	-10	
Net Loss	dB	-167.296	-178.111	
Satellite Antenna, Type		Satellite Antenna, Type		
Coverage Area	deg ²	1156	1156	
Gain	dBi	3	3	
Received Carrier Power	dBW	-99.6963	-110.511	
G/T	dB/K	22	35.2	User's Guide
Receiver C/No	dB-Hz	314.6	327.8	k=228.6
Available Uplink	dB	241.4403	249.2327	

Table 22. Downlink Budget

	Unit	S-Band	X-Band	Comments
Download Frequency	GHz	2.25	8.45	
Antenna Type		Fixed Spiral Antenna		
Coverage Area	deg ²	1156	1156	
Gain	dBi	3	3	
Satellite Power	Watts	10	10	
EIRP		13	13	
Propagation Range	km	844	844	
Space Loss	dB	-158.02	-169.514	
Atmospheric Loss	dB	-10	-10	
Net Loss	dB	-168.02	-179.514	
Receiver Terminal Type		Tracking Dish AS1 11.28M		
Diameter	m	11.28	11.28	
Beamwidth	deg	0.85	0.23	
Gain	dBi	45.8	56.8	Producer provided
Receiver Carrier Power	dBW	-109.22	-109.714	
System Noise Temp	dBK	27	27	
G/T	dB/k	22	37.2	
Receiver C/No	dB-Hz	94.9795	98.68602	k=228.6
Available Downlink	dB	21.45768	19.41745	
End-End Eb/Co	dB	262.898	268.6502	
Required Eb/No	dB	15.2042	15.52042	
Link Margin	dB	247.3776	253.1297	

12.3 Communications Summary

The communications subsystem uses two omnidirectional antennas to receive and transmit between the PERSEUS and the 18 Near Earth Network Array and Commercial Array ground stations. Transmissions will occur in the S and X bands with higher data rates occurring in the higher frequency X Band region. Once received, data will be routed to the processor in the flight computer. The communications array is oversized for our overall low altitude of 844 km and provides a Link Margin of 247.3779 dB in S-Band and 253.1297 dB in the X-Band. Both antennas can also receive GPS signals should the GPS antenna malfunction.

13.0 Reliability & Risk

There are significant risks to the overall success of the PERSEUS mission. Careful design decisions have been made to ensure these risks aren't exacerbated by cheap design or skinny margins. During the design of each subsystem, component redundancy was considered so that even with failures in important spacecraft systems, the mission can still be accomplished using different failure modes. A qualitative analysis of these risks are laid out in Table 23 below. A fever chart expressing the risk level of each of the risks follows in Figure 40. A quantitative reliability analysis was not performed due to a lack of failure information in Commercial Off-The-Shelf (COTS) component data sheets.

Table 23. Qualitative Risk Analysis

	Source	Risk considerations	Risk Level
R-1	Power: Battery failure	With two medium capacity secondary batteries and three high capacity primary batteries, PERSEUS can operate with next to no interruption with single failures in either type of battery	Unlikely, low impact
R-2	Power: Solar Array deployment	Having two Solar Array wings allows PERSEUS to still generate power if deployment fails. The presence of primary batteries independent from the Solar Array drastically reduces the impact of a deployment failure. Solar Array deployment historically has a comparatively high failure rate	Unlikely, low impact
R-3	Communications: Pointing failure leads to loss of communications	There are two separate omni-directional antennas, neither are independently pointable	Unlikely, high impact
R-4	Data: Computer failure	Computer failure is unlikely, but would have a significant impact on mission success	Unlikely, medium impact
R-5	GNC: Sensor/reaction wheel failure	With a high heritage design and a short mission duration, failure is very unlikely. With multiple navigation sensors and two different pointing methods, impact would be somewhat low	Unlikely, somewhat low impact
R-6	Propulsion: Thruster/fuel failure	With built-in fuel margin and high-heritage thrusters, likelihood of failure is low, but the impact would be very high	Very unlikely, high impact
R-7	Environment: Single Event Effects (SEE)	Radiation tolerant design and a solar-minimum mission window mitigates the constant problems associated with radiation	Very likely, low impact
R-8	Capture: Breakoff of more debris	With an uncooperative docking scenario, even with careful capture mechanism operation, there is a possibility of overstressing the aged target. This may damage PERSEUS, and the goal of the mission is debris mitigation.	Unlikely, high impact
R-9	Astro/GNC: Can't characterize target motion	It isn't likely that tracking would fail to characterize the target's motion due to a high number of targeting sensors, but due to the delicate nature of the capture process, this could have a significant effect on mission success	Very unlikely, somewhat high impact
R-10	Astro: Collision risk	Once PERSEUS is added to the SCC catalog, mission operators will receive collision notifications, giving them opportunity to react	Very unlikely, high impact

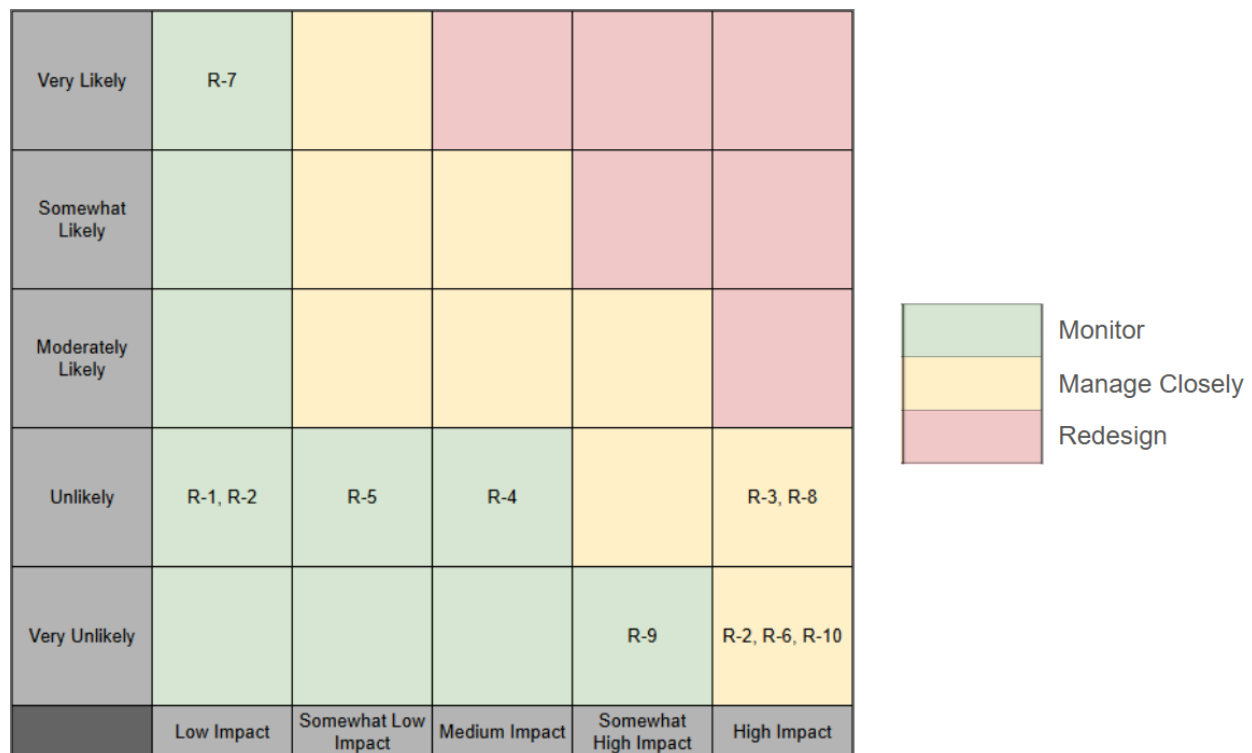


Figure 40. Fever Chart Showing Concerns and Their Risk Levels

13.1 Low Mission Impact Power System Failure

The R-1 and R-2 risks are shown to have a low impact despite the loss of significant power components, which would usually not be reasonable. After looking at the power needs of each subsystem, it was determined that while the capture mechanism requires the highest amount of power, it also needs it for the shortest duration. This meant that primary batteries would be a cheaper option for providing power during the capture arm deployment, so three primary batteries were selected. However, an arm deployment procedure was written for a triple primary battery failure-mode requiring only the secondary batteries using a significantly increased deployment time of the arm to use the Solar Array to recharge after each movement.

14.0 Launch & Mission Operations

The spacecraft will launch out of Vandenberg SFB, CA. Since Vandenberg SFB, CA launches from a latitude of 34.742 degrees, the launch azimuth will be 156.659 degrees to establish a 71 degree inclination for the parking orbit. The windows for launch are 0117GMT and 1126GMT. These two time windows for launch ensure the parking orbit will match the same plane as the target orbit.

The operators will operate out of NASA's Ames Research Center. The team will utilize NASA's Near Earth Network for TT&C with the spacecraft. Additionally, the operators will use several commercial sites for TT&C. This gives the team plenty of coverage in LEO so that they can utilize a handoff technique to ensure continuous communication during critical windows in the rendezvous phase. The antenna sites as well as their bands and location are listed in Appendix H.

The operators will use both S-band and X-band for sending commands and receiving data from the spacecraft. S-band will be utilized for sending commands and monitoring the state of health of the vehicle. This robust frequency band allows operators to be confident their commands will get to the vehicle even in adverse weather. Although the S-band is robust, it lacks data throughput. In order to overcome that limitation, the TT&C antenna on the vehicle as well as the ground antennas can communicate through X-band. X-band allows for much higher data throughput relative to S-band and will be used to collect timely data from the vehicle's on-board sensors during critical times throughout the mission. For example, the operators at Ames Research Center will rely on X-band to get information collected by the Flash LiDAR system while the vehicle is in an NMC around the target.

The mission crews will consist of a Mission Commander, Orbital Analysis and Satellite Vehicle Operator. These 3 person crews will have to operate 24/7 using 8 hour shifts during the week and 12 hour shifts during the weekend. Three crews will rotate through the weekday shifts, while 2 crews will rotate through the weekend shifts. This gives the mission a total of 5 crews with 1 reserve crew ready to take the place of another crew, or individual member, if an emergency happens. Therefore, the total mission operations personnel will consist of 18 operators.

15.0 Cost

15.1 Cost Estimation Method

To estimate the total cost of the PERSEUS mission, the QuickCost estimation tool was used (Shao et al, 2011). Values for each contributing variable were chosen based on the completed component selection and design of each subsystem. The standard error of the QuickCost estimation tool is 41%. The cost estimation variables are shown in Table 24 below. Additional adjustments shown in Table 25 are made to account for operational, facility, and launch costs.

Table 24. NASA QuickCost Estimation Variables for PERSEUS Mission Cost

Cost Driver	Value	Description	Rationale
Dry mass	758	Dry mass of spacecraft and instruments in kg	Current estimate for SC dry mass
Power	1360	LEO equivalent Beginning Of Life power (BOLP) in Watts	Current estimate for SC BOLP
Data%	0.25	Data Rate Percentile (fraction) relative to the state-of-the-art at Authority To Proceed (ATP) (0.5 for median data rate missions, <0.5 for lower data rates, >0.5 for higher data rates)	Data rate is expected to be low for this mission due to a lack of scientific objectives
Life	0.75	Advertised Design Life (in months) including extended operations	Current estimate for SC design life
New	0.3	Percent New (fraction): 0.2-0.3 = Simple Mod, 0.3-0.7 = Extensive Mod, 0.7-1.0 = New	Most of the spacecraft will be made up of COTS parts, but the capture mechanism makes use of a combination of TRL 7 components that are as of yet untested together
Planetary?	0	0 for Earth orbital, 1 for planetary	Earth orbital
Year	2027	ATP date in 4 digit calendar year minus 1960	Phase B is expected to begin in 2027 for our expected launch date
InstrComp%	0.5	Instrument complexity percentile (fraction) - relative to "average" instrument complexity (enter 0.5 for median complexity, <0.5 for lower complexity, >0.5 for higher complexity)	The capture mechanism has a medium level of complexity
Team	4	Team Experience: 1 = Unfamiliar, 2 = Mixed, 3 = Normal, 4 = Extensive	The team will be made up of members with an extensive level of experience

Table 25. Additional Adjustments to the QuickCost Estimate

Consideration	Percent increase (%)	Cost increase (2025 millions)
Ground station retrofit	9	7.428698184
Mission Operations and Data Analysis	10	8.254109094
Launch	N/A	70
Total	19	85.68280728

The estimated cost is 168.224 in 2025 millions. Considering the standard error of 41% in the QuickCost method, the total cost should fall in the range of 127.952 to 208.496 in 2025 millions.

16.0 Summary & Conclusion

With the mission design requirements provided in the proposal, the team has selected the hardware components for the spacecraft to carry out the mission. An analysis of the space environment has been conducted for which the spacecraft can expect to encounter. A conclusive breakdown of the astrodynamics has been completed and the launch vehicle has been selected. Guidance, Navigation, and Control hardware and algorithms have also been selected. For the Attitude Determination and Control System, trade studies have been conducted, and the moments of inertia found. Hardware and algorithm selections have therefore been verified. The main spacecraft engine and RCS thrusters have been selected. The schematics for the engine and thrusters has been created and the tank sizing has been completed. The robotic arm and capture mechanism to grab the booster have been successfully selected. The spacecraft structure and configuration have been designed and completed. The thermal control system components have been selected for both passive and active control. The Power subsystem has been decided to include solar arrays with two secondary batteries and three primary batteries to store power. The Communications and Data System subsystem has selected two omnidirectional antennas for ground communication, and a GPS antenna used in a GNC interface, and two radios to route data from the ground to a processor bus. It also includes a SpaceWire-compatible GR740 Quad-Core Space LEON4FT data processor. The Launch and Mission Operations have been thoroughly planned down to the shift of the employees. A qualitative analysis of risks was completed, and a risk fever chart has also been done. An estimation of cost brings the mission to approximately 168,224,000 million in 2025 dollars.

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Appendix B: Astrodynamics Calculations

B.1 Hohmann Transfer Rendezvous

A. $V_I = \sqrt{\mu \left(\frac{2}{R_I} - \frac{1}{a_I} \right)} = 7.48 \text{ km/s}$ velocity of PERSEUS parking orbit where

$R_I = 7121.95 \text{ km}$, $a_I = 7121.95 \text{ km}$, and $\mu = 398600.5 \frac{\text{km}^3}{\text{s}^2}$.

B. $V_F = \sqrt{\mu \left(\frac{2}{R_F} - \frac{1}{a_F} \right)} = 7.43 \text{ km/s}$ velocity of the target altitude for the Hohmann transfer

where $R_F = 7221.95 \text{ km}$, $a_F = 7221.95 \text{ km}$.

C. $a_H = \frac{R_I + R_F}{2} = 7171.95 \text{ km}$ semi major axis of the Hohmann transfer.

D. $t_H = \pi \sqrt{\frac{a_H^3}{\mu}} = 50.37 \text{ min}$ time of flight of the Hohmann transfer.

E. $V_{H1} = \sqrt{\mu \left(\frac{2}{R_I} - \frac{1}{a_H} \right)} = 7.51 \text{ km/s}$ velocity of the Hohmann transfer at the parking orbit altitude.

F. $V_{H2} = \sqrt{\mu \left(\frac{2}{R_F} - \frac{1}{a_H} \right)} = 7.40 \text{ km/s}$ velocity of the Hohmann transfer at the target altitude.

G. $\Delta V1 = V_{H1} - V_I = 26.03 \text{ m/s}$ change in velocity to enter the Hohmann transfer.

- H. $\Delta V2 = V_F - V_{H2} = 25.94m/s$ change in velocity to leave the Hohmann transfer and circularize the orbit at the target altitude.
- I. $\Delta V_{Total} = \Delta V1 + \Delta V2 = 51.97m/s$ total change in velocity to complete the Hohmann transfer.
- J. $\alpha_{Lead} = \omega_{tgt} * TOF = 178.1339$ degrees, the change in true anomaly of the target during the chase's Hohmann transfer where ω_{tgt} is 0.05894 deg/s and TOF is 3022.292273 seconds.
- K. $\phi_{final} = 180 - \alpha_{Lead} = 1.866063$ degrees, the lead angle of the target relative to the chase at the beginning of the Hohmann transfer to ensure the chase arrives at the desired relative position at the conclusion of the Hohmann transfer

B.2 Closing maneuver and deorbit

- A. $n = \sqrt{\frac{\mu}{a^3}} = 0.001028697 rad/s$ the mean motion of the chase vehicle
- B. $dV = a * e * n * \sqrt{1 + 3 * \cos^2(n * t)} = 10.72627288 m/s$ the delta V required to close the relative distance between the chase and the target where the time since perigee is $t=0$.
- C. $V1 = \sqrt{\mu * (\frac{2}{Ro} - \frac{1}{a})} = 7.43456km/s$ the velocity of the chase and target's orbit after the chase grapples the target where $Ro=7216.732381km$.

D. $a_{tran} = \frac{Ro + Rp}{2} = 6834.93469km$ the semi major axis of the deorbit transfer where

$Rp=6453.137km$ since the deorbit is targeting an altitude of 75km.

E. $V1_{tran} = \sqrt{\mu * (\frac{2}{Ro} - \frac{1}{a_{tran}})} = 7.221324771 km/s$ the velocity of the deorbit transfer
at the chase and target's altitude when they start the maneuver.

F. $dV1 = V1 - V1_{tran} = 213.2360946 m/s$ the total Delta-V of the deorbit maneuver.

B.3 Excel Calculations for Hohmann transfer, closing maneuver, and deorbit maneuver

SCC: 28353 TLE from SpaceTrack									
	1	28353U	04021B	25255.9664	-0.0000023	00000-0	-94855-4	0	9990
	2	28353	70.998	245.4497	0.0007219	297.003	63.0359	14.1456	97703
μ=	398600.5 km3/s2								
Re=	6378.137 km								
	843.8089								
	Chase				Target				
e=	0				e=	0.00072			
i=	70.998 deg				i=	70.998 deg		1.239149 rad	
Ω=	245.4497 deg				Ω=	245.45 deg		4.283905 rad	
ω=	297.0026 deg				ω=	297.003 deg		5.183673 rad	
n=	14.444576 rev/day				n=	14.1456 rev/day		0.001029 rad/s	
a=	7121.9459 km				a=	7221.95 km			
Rp=	7121.9459 km				Rp=	7216.73 km			
Ra=	7121.9459 km				Ra=	7227.16 km			
v=	0 deg				v=	1.86606 deg		0.032569 rad	
					Per=	6107.91 sec			
	Transfer								
R_c=	7121.9459 km				R_tgt=	7221.95 km			
v_c=	7.481171 km/s				v_tgt=	7.4292 km/s			
a_tran=	7171.9459 km								
TOF=	3022.2923 s								
v_tran_1=	7.5072036 km/s				v_tran_2=	7.40325 km/s			
dv1=	0.0260326 km/s				dv2=	0.02594 km/s		25.94202 m/s	
dv_tot=	0.0519746 km/s								
ω_c=	0.0601857 deg/s				ω_tgt=	0.05894 deg/s			
					α_lead=	178.134 deg			
					φ_final=	1.86606 deg			
Checkout (t_cc	1209600 s								
φ_initial=	-807.4889 deg								
n=	0.0010287 rad/sec								
t=	0 s				at target's perigee				
dV=	0.0107263 km/s				10.726273 m/s				

Deorbit

μ = 398600.5 km³/s²

Re=	6378.137 km		
we=	0.0000729 rad/s		
h_reentry=	120 km		
R_reentry=	6498.137 km		
hp=	75 km		
Rp=	6453.137 km		
Ro=	7216.7324 km		
vo=	7.4345609 km/s		
a_tran=	6834.9347 km		
v_tran_1=	7.2213248 km/s		
Δv1=	0.2132361 km/s	213.23609 m/s	
e_tran=	0.0558597		
Per_tran=	2811.7876 s	46.863126 min	
nu_reentry=	-0.517411 rad	-29.645475 deg	<-- must be negative for inbound
E_reentry=	0.490417 rad	28.098824 deg	
M_reentry=	0.4641074 rad	26.591396 deg	
n_reentry=	0.0011173 rad/s		
Δt=	2396.4023 s	39.940038 min	
Y=	-0.026345 rad	-1.5094366 deg	<-- flight path angle at reentry
Total Delta V=	265.21072 m/s	<-- Rendezvous and Deorbit	

B.4 Edelbaum Transfer

A. $dVt = v_c - v_{tgt} = 51.97525 \text{ m/s}$ the required change in velocity to achieve the desired target orbit where $v_c=7.481171\text{km/s}$ and $v_{tgt}=7.429196\text{km/s}$.

B. $TOF = \frac{acc_t}{dVt} = 3543.687s$ the time of flight for the Edelbaum transfer where the acceleration of the thruster is $acc_t=0.014667\text{m/s}^2$.

B.5 Excel Calculations for Edelbaum Transfer

SCC: 28353 TLE from SpaceTrack								
1	28353U	04021B	25255.97	-2.3E-06	00000-0	-94855-4	0	9990
2	28353	70.998	245.4497	0.000722	297.0026	63.0359	14.1456	97703
μ=	398600.5 km3/s2							
Re=	6378.137 km							
Chase					Target			
e=	0				e=	0.000722		
i=	70.998 deg		1.239149 rad		i=	70.998 deg		1.239149 rad
Ω=	245.4497 deg		4.283905 rad		Ω=	245.4497 deg		4.283905 rad
ω=	297.0026 deg		5.183673 rad		ω=	297.0026 deg		5.183673 rad
n=	14.44458 rev/day		0.00105 rad/s		n=	14.1456 rev/day		0.001029 rad/s
a=	7121.946 km				a=	7221.946 km		
Rp=	7121.946 km				Rp=	7216.732 km		
Ra=	7121.946 km				Ra=	7227.159 km		
v=	0 deg		0 rad		v=	-28.8649 deg		-0.50379 rad
					Per=	6107.905 sec		
Low Thrust Transfer								
R_c=	7121.946 km				R_tgt=	7221.946 km		
v_c=	7.481171 km/s				v_tgt=	7.429196 km/s		
ΔVt=	0.051975 km/s		51.97525 m/s					
at=	0.014667 m/s2							
TOF=	3543.687 s		59.06144 min		3.686688			
ω_tgt=	0.05894 deg/s							
α_lead=	208.8649 deg							
φ_final=	-28.8649 deg							

Appendix C: Navigation

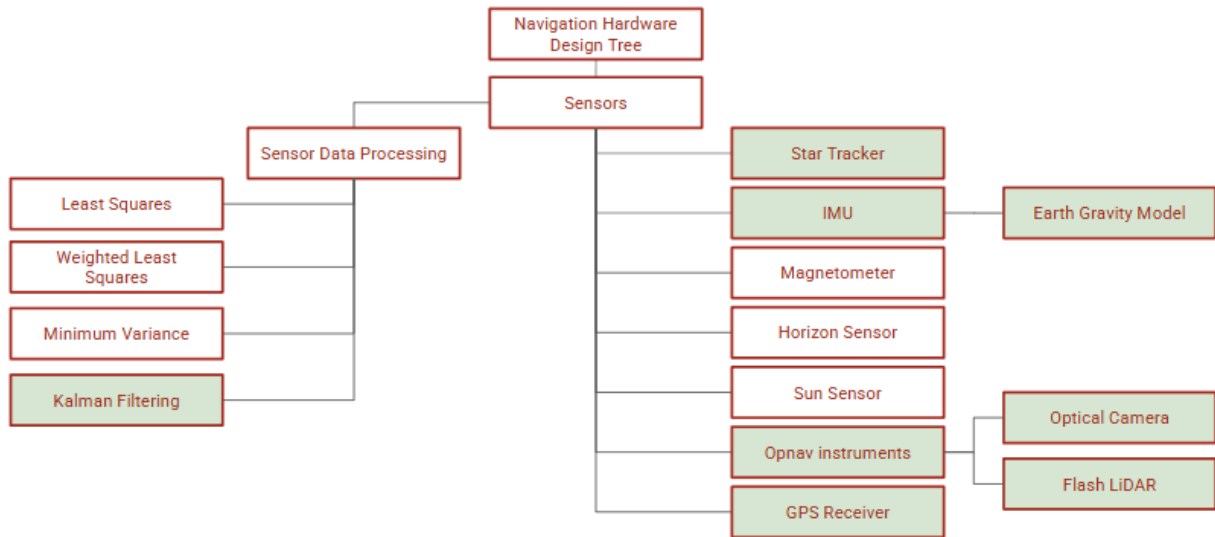


Figure C1. Navigation Hardware Design Tree

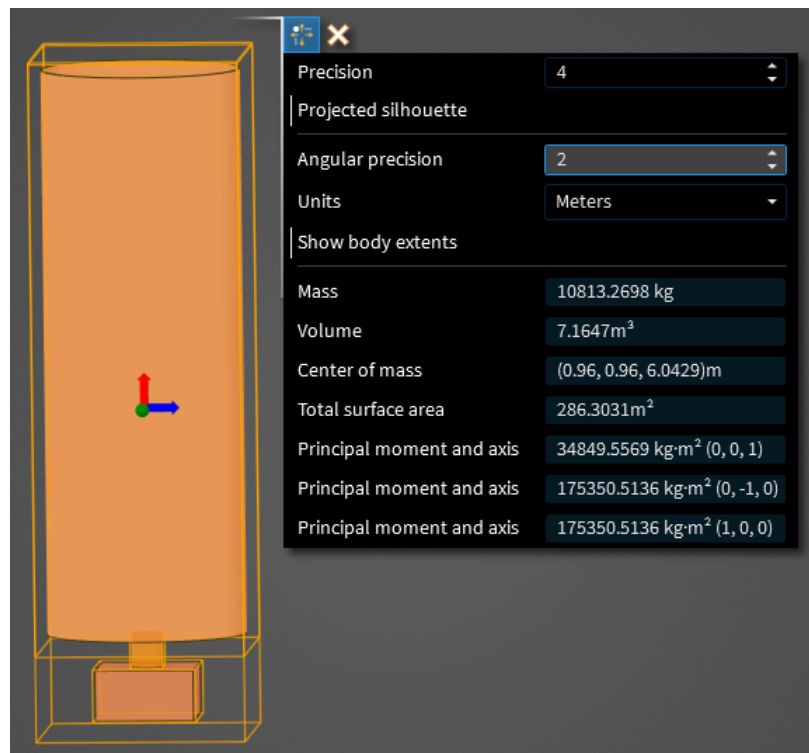
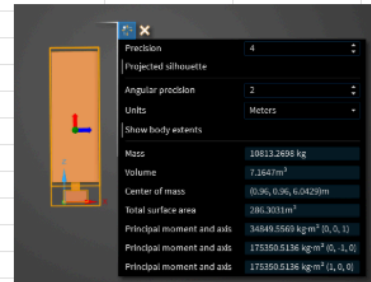


Figure C2. Calculation of PERSEUS/Target Object Configuration MOI

Table C1. Disturbance Torque Calculations

Disturbance Torque	PERSEUS	Combined Configuration						
Solar Radiation Pressure (Nm)	0	1.02E-03	SMAD					
Aerodynamic (Nm)	0	8.70E-04	SMAD					
Magnetic (Nm)	3.97E-05	7.93E-05	Elements of Spacecraft Design					
Gravity Gradient (Nm)	1.08E-04	3.89E-02	Elements of Spacecraft Design					
Center of Pressure	0.49	3.6385	m		5.5529			
Center of Mass	0.49	6.0429	m					
As	52.46290838	52.46290838	m2	Area of each simplified structure plus solar panel area				
q	0.77	0.77	unitless					
phi	1367	1367	W/m2					
rho	0	0	deg					
c	300000000	300000000						
rho	0	0	kg/m3					
Cd	2.5	2.5	unitless					
Ar	52.46290838	52.46290838	m2					
V	7429.1952	7429.1952	m/s					
Center of Pressure	0.49	3.6385	m					
Center of Mass	0.49	6.0429	m					
M	1	2	Am2					
R	7221946	7221946	m					
rearth	6378000	6378000	m					
B0	0.00003	0.00003	T					
L	71	71	deg					
B	0.00003965107192	0.00003965107192	T					
theta	0.0872664626	0.0872664626	rad					
Iz	1,058.77	34,849.56						
Iy	667.30	175,350.51						
mu	398600441800000	398600441800000						
R	7221946	7221946						



Appendix D: Propulsion Calculations

The following equations were used to find propellant mass and burn time to conduct trade studies on the main spacecraft engine and the RCS thrusters:

$$m_p = m_{dry,0} (1 - \exp(\frac{-\Delta V}{I_{sp} g_0}))$$

$$t_b = \frac{m_{dry,0} \Delta V}{T}$$

The burn time equation is an approximation. More concise calculations will be conducted later.

The equations shown below were used to complete the tank sizing:

$$\dot{m}_f = \frac{\dot{m}_p}{1+OF}$$

$$\dot{m}_o = \dot{m}_p - \dot{m}_f$$

$$m_f = \frac{m_p}{(1+OF)}$$

$$m_o = m_p - m_f$$

$$V_f = \frac{m_f}{\rho_f}$$

$$V_o = \frac{m_o}{\rho_o}$$

$$V_{gf1} = \frac{(\frac{P_{TF}}{P_{T1}}) V_f}{(1 - \frac{P_{TF}}{P_{T1}})}$$

$$V_{go1} = \frac{(\frac{P_{TF}}{P_{T1}}) V_o}{(1 - \frac{P_{TF}}{P_{T1}})}$$

$$r_f = \left(\frac{3*V_f}{4*\pi} \right)^{1/3}$$

$$r_o = \left(\frac{3*V_o}{4*\pi} \right)^{1/3}$$

$$t_f = \frac{P_{TF} * V_f * FS}{2 * \sigma_y}$$

$$t_o = \frac{P_{TF} * V_o * FS}{2 * \sigma_y}$$

$$m_{He} = \frac{P_{T1} * 1000000 * V_{gtot}}{\rho_{He} * 300}$$

$$V_{He} = \frac{m_{He} * \rho_{He} * 300}{P_{b0} * 1000000}$$

$$r_{He} = \left(\frac{3*V_{He}}{4*\pi} \right)^{1/3}$$

$$t_{He} = \frac{P_{TF} * V_{He} * FS}{2 * \sigma_y}$$

$$m_{He_{Tank}} = 4 * \pi * R_{He}^2 * t_{He} * \rho_{Al}$$

$$m_{f_{tank}} = 4 * \pi * R_f^2 * t_f * \rho_{Al}$$

$$m_{o_{tank}} = 4 * \pi * R_o^2 * t_o * \rho_{Al}$$

The schematics are shown below:

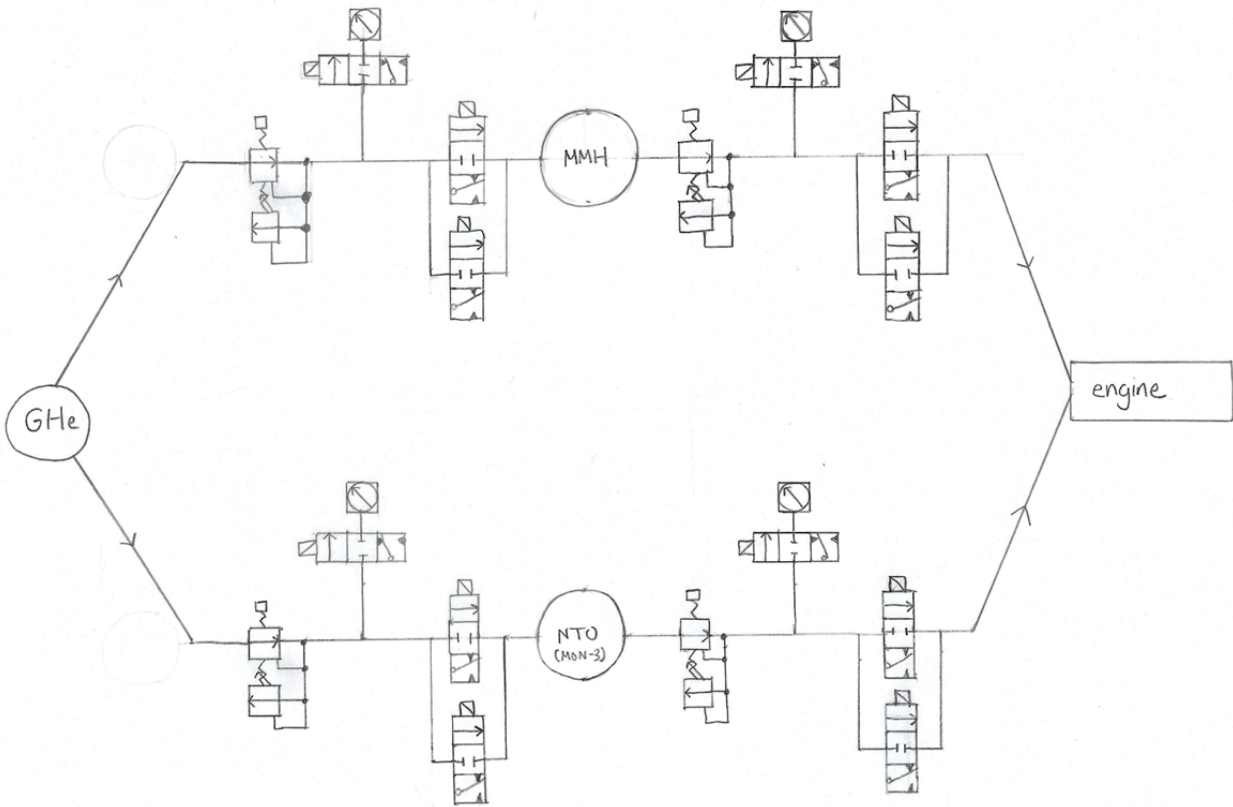


Figure D1. Main Spacecraft Engine Schematic

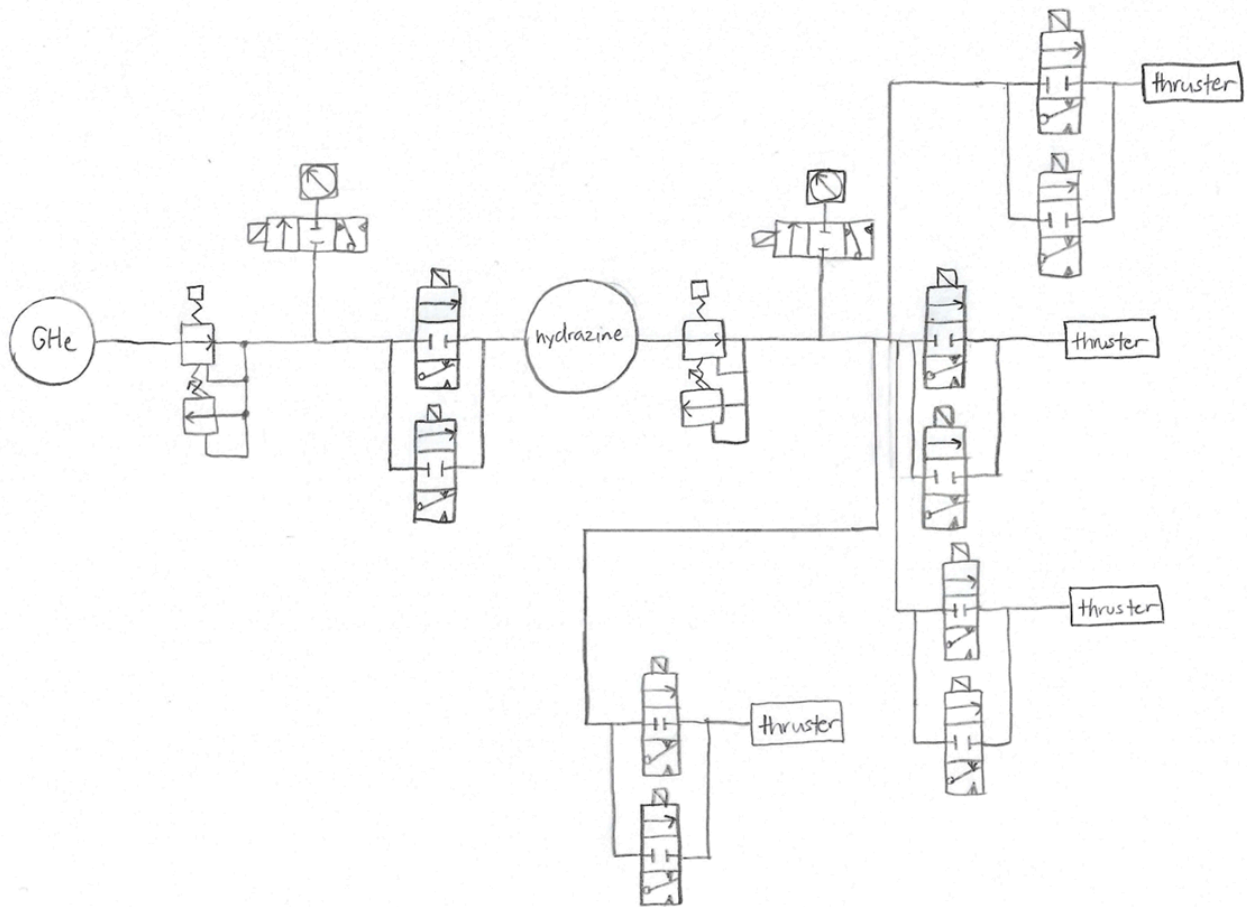


Figure D2. RCS Thrusters Schematic

Appendix E: Mechanical Subsystem Calculations

Table E1.1. Required Joint Torque Calculations for the Robotic Arm Assembly

	Target Mass	m_tgt (kg)	9000			
	Arm Length	L_arm (m)	2.4			
	Max relative velocity for capture	v_rel (m/s)	0.25	*Bylard et al, 2017: dry adhesion is effective for velocities up to 0.15-0.25 m/s		
	Dist fm s/c center of mass to shoulder joint	r_shoulder (m)	2.6			
	Radius of r/b	r_rb (m)	2			
Shoulder Torque						
		T = I * alpha (Nm)				
	Moment of Inertia	I_cyl = 1/2 * m*r_rb^2 (kg/m^2)	18000			
		Angular Vel Comparison	Calculated based on dry adhesion effectiveness values	Given based on Bylard et al, 2017		
			Radian/s	Degrees/s	Radians/s	
	Angular velocity	w = v/r_cg	0.096153846153846	6	0.10471975511966	*Bylard et al, 2017: "for conservatism we set 6 deg/s as the initial axial spin of the R/B"
	time	t (sec)	30	33	*note necessary time increase by 2s to calc same torque values	
	Angular deceleration	Alpha = dw/dt (rad/s^2)	0.003205128205128	0.003173325912716		
	Nominal Inertial Torque @ Target	T = I * alpha (Nm)	57.6923076923078	57.1198664289055		
	Safety Factor	SF	2	2		
		T*Sf_accel (N/m)	115.384615384616	114.239732857811		

Table E1.2. Required Joint Torque Calculations for the Robotic Arm Assembly Continued

Elbow Torque		$T_{\text{elbow}} = I * \alpha * r_{\text{elbow}} / r_{\text{shoulder}} \text{ (Nm)}$	
		$r_{\text{elbow}} \text{ (m)}$	1.15
		$r_{\text{shoulder}} \text{ (m)}$	2.6
		$T_{\text{elbow}} = I * \alpha * r_{\text{elbow}} / r_{\text{shoulder}} \text{ (Nm)}$	25.51775147929
	Safety Factor	SF	2
		$T_{\text{elbow}} * \text{SF} \text{ (Nm)}$	51.03550295858
Wrist Torque		$T_{\text{wrist}} = I * \alpha * r_{\text{wrist}} / r_{\text{shoulder}} \text{ (Nm)}$	
		$r_{\text{wrist}} \text{ (m)}$	0.15
		$r_{\text{shoulder}} \text{ (m)}$	2.55
		$T_{\text{wrist}} = I * \alpha * r_{\text{wrist}} / r_{\text{shoulder}} \text{ (Nm)}$	3.39366515837105
	Safety Factor	SF	2
		$T_{\text{wrist}} * \text{SF} \text{ (Nm)}$	6.7873303167421

Table E2. Joint Torque Output Calculations for the Robotic Arm Assembly

		Shoulder	Elbow	Wrist	
	Torque Requirements (Nm)	~116	~53	~7	
	Motor	Maxon EC-90 flat	2 x Maxon EC-60 Flat motors in parallel	Maxon EC-60 Flat	
	Gearbox	Harmonic Drive SHG-32-100-2SO	Harmonic Drive SHG-25-100-2SO	Harmonic Drive SHG-20-50-2SO	
	$T_{out} = T_{motor} * G * N$				
Motor Torque	T_{motor} (Nm)	1.49	0.638	0.319	
Gearbox Ratio	G (x:1)	100	100	50	
Gearbox Efficiency	Assume N = 84%. *Garcia et al, 2020 - Harmonic Drive Strainwave GB. (NOTE: Harmonic Drive SHG series is an advancement over the CSG series)	0.84	0.84	0.84	
	$T_{out} = T_{motor} * G * N$ (Nm)	125.16	53.592	13.398	

Table E3. Thermal Heater Requirement Calculations for the Robotic Arm Assembly

STEADY HEATER POWER REQ'D (assuming worst-case)		Without MLI	With QUEST 10-Layer IMLI	
Desired Control Temperature	T_set (K)	273.15	273.15	
Environmental Temperature	T_env (K)	3	3	
Total Mass	m (kg)	85.638	86.112	
Thermal Emissivity	Epsilon (E)	0.85	0.01	*Conservative emissivity est - Datasheet lists 0.0025 for best case scenario
Stefan-Boltzmann Constant	Sigma (W/m ² K ⁴)	0.00000005670374	0.00000005670374	
Radiative Flux	q_rad = E*sigma*(T_set ⁴ - T_env ⁴)(W/m ²)	268.30912522551	3.15657794382953	
Area of Surfaces to be Heated (assuming exposed area is 0.15 m ² /kg)	A_surf (m ²)	12.8457	12.9168	
Total Radiative Heat Loss	Q_rad = A_surf*E*sigma*(T_set ⁴ - T_env ⁴)(W)	3446.61852990933	40.7728859848572	
*assume 30% design margin	Q_req,mar = Q_rad * 1.3 (W)	4480.60408888213	53.0047517803144	
HEATER POWER DENSITY				
Required	q_req = Q_req,mar / A_surf (W/m ²)	348.801862793163	4.10355132697838	
Minco Thermofoil Heaters Watt Density	q_Minco (W/in ²) *fm datasheet	20	20	
	Conversion from W/in ² to W/m ²	1550.0031	1550.0031	
	q_Minco (W/m ²)	31000.062	31000.062	
Heater Area Required	A_heat = Q_req,mar / q_Minco (m ²)	0.144535326699738	0.001709827282936	
	A_heat (in ²)	224.029756384594	2.65023228855114	
Heater Power Required	P_max = q_minco * A_heat	4480.60408888213	53.0047517803144	
Minco Thermofoil Heaters HK6907	P_minco (W)	32.8		
		136.603783197626	1.61599852988763	
	**WITH MLI	Shoulder	Elbow	Wrist
	Heater Area/ Assembly	16	8	6
	# of Heaters/ Assembly	8	4	3

Table E4. Power Consumption Calculations for the Shoulder, Elbow, and Wrist Joints of the Robotic Arm Assembly

			Shoulder	Elbow	Wrist
		Maxon BLDC Model	EC-90 Flat	EC-60 Flat	EC-60 Flat
*Values obtained from Maxon Motor Datasheets	Max Cont Torque	T (Nm)	125	319	319
	Motor Voltage	V (V)	48	48	48
	Motor Efficiency	N	0.87	0.88	0.88
	Max Cont Current	I (A)	5.99	2.3	2.3
	Winding Resistance	R (K/W)	0.89	3.5	3.5
	Speed Constant	K_v (rpm/V)	41.3	83.4	83.4
	Operating Angular Speed	$w = (V - I \cdot R) / K_v$ (rad/s)	1.03314527845036	0.479016786570743	0.479016786570743
		$P_{elec} = (T \cdot w) / N$ (W)	148.440413570454	173.643585131894	173.643585131894
		Number of motors/joint	2	2	3
		P_joint (W)	296.880827140908	347.287170263788	520.930755395682
		P_joint total (W)	1165.09875280038		

Table E5.1. Power Consumption Calculations for the REACCH End-Effector

REACCH		# of 2-meter Tentacles	8	
ASSUMPTIONS:		Segments per Tentacle	24	
	Segment Length	l_{seg} (m)	0.0833333333333333	
	Estimated Contact Force	F_{contact} (N)	20	
	Mechanical Torque (per segment)	$T_{\text{seg}} = F_{\text{contact}} * l_{\text{seg}}$ (Nm)	1.66666666666667	
	Mechanical Torque (per segment) (rounding up for SF)	T_{seg} (Nm)	2	
	Mechanical Angular Speed (per segment)	w_{seg} (rad/s)	1	
	Motor Efficiency	N	0.85	
	EA/G Tile Area (per segment)	A_{tile} (m²)	0.0004	
Dielectric	Thickness	d (m)	0.00005	
	Relative Permittivity	E_r (F/m)	4.5	*typical acrylic elastomer permittivity value range: 3.0-4.7
	Permittivity of Free Space	E_o (F/m)	0.00000000008854	
	EA/G Tile Capacitance (per segment)	$C_{\text{seg}} = E_o * E_r * (A_{\text{tile}}/d)$ (F)	0.000000000318744	
	EA/G Tile Capacitance (per tentacle)	$C_{\text{ten}} = C_{\text{seg}} * \#$ of segments	0.000000007649856	
	EA/G Drive Voltage	V (Volts)	5000	
	EA/G Switching Frequency	f (Hz)	50	
	Control Electronics (per tentacle)	$P_{\text{elec,ten}}$ (Watts)	24	

Table E5.2. Power Consumption Calculations for the REACCH End-Effector Continued

TOTAL CONTINUOUS POWER	*Assume all 8 tentacles active, EA at 5 kV, motors holding		
	Mechanical Power (per segment)	$P_{\text{mech,seg}} = (T \cdot w) / N$ (Watts)	1.96078431372549
	Mechanical Power (per tentacle)	$P_{\text{mech,ten}} = P_{\text{seg}} \cdot \# \text{ of segments}$ (Watts)	47.0588235294118
	EA/G Switching Power (per segment)	$P_{\text{EA,seg}} = C \cdot (V^2) \cdot f$ (Watts)	0.39843
	EA/G Switching Power (per tentacle)	$P_{\text{EA,ten}} = P_{\text{EA,seg}} \cdot \# \text{ of segments}$ (Watts)	9.56232
	Total Power per Tentacle	$P_{\text{tot,ten}} = P_{\text{mech,ten}} + P_{\text{EA,ten}} + P_{\text{elec,ten}}$ (Watts)	80.6211435294118
	Total REACCH Power	$P_{\text{tot,REACCH}} = P_{\text{tot,ten}} \cdot \# \text{ of tentacles}$ (Watts)	644.969148235294
TOTAL PEAK POWER	*Assume higher EA voltage at 10 kV		
		EA power scale	4
	EA per segment at 10 kV	$P_{\text{peak EA,seg}} = P_{\text{EA,seg}} \cdot \# \text{ of segments}$ (Watts)	9.56232
	EA per tentacle at 10 kV	$P_{\text{peak EA,ten}} = P_{\text{peak EA,seg}} \cdot \# \text{ of tentacles}$ (Watts)	76.49856
		$P_{\text{peak,ten}} = P_{\text{mech,ten}} + P_{\text{peak EA,ten}} + P_{\text{elec,ten}}$ (Watts)	147.557383529412
		$P_{\text{peak,REACCH}} = P_{\text{peak,ten}} \cdot \# \text{ of tentacles}$ (Watts)	1180.4590682353
TOTAL IDLE POWER	*Assume motors off, EA off, electronics only		
		$P_{\text{idle}} = P_{\text{elec,ten}} \cdot \# \text{ of tentacles}$ (Watts)	192

Table E6. A Detailed Breakdown of the Mass/Cost/Power Budget for the Mechanical Subsystem

Joint	Component	Vendor / Model	Quantity	Total Mass (kg)	Estimated Total Cost (USD)	Total Power (Watts)
Hold-Down & Release Mechanism	Ensign-Bickford Aerospace & Defense	NEA Model 9011 HDRM	1	0.7	\$10000.00	30
Shoulder Base	Spintable	Moog UPT3 Rotary Gimbal	1	5.67	\$150000.00	150
Shoulder Joint	Motor	Maxon EC-90 flat	2	1.976	\$780.00	298
*Note: 2 axes	Gearbox	Harmonic Drive SHG-32-100-2SO (with integrated mechanical brake)	2	5.88	\$9000.00	
	Motor Controller	Maxon EPOS4 70/15	2	0.744	\$1610.00	1470
	MEMS IMU	Honeywell HG4930	1	0.14	\$10000.00	3
	Absolute Encoder	Renishaw Resolute UHV	2	0.38	\$4000.00	2.5
	F/T Sensor	ATI Omega160	1	2.72	\$10500.00	
Elbow Joint						
*Note: 1 axis	Motor	Maxon EC-60 flat	2 (in parallel)	0.94	\$540.00	348
	Gearbox	Harmonic Drive SHG-25-100-2SO (with integrated mechanical brake)	1	1.31	\$5000.00	
	Motor Controller	Maxon EPOS4 70/15	1	0.372	\$805.00	735
	MEMS IMU	Honeywell HG4390	1	0.14	\$10000.00	3
	Absolute Encoder	Renishaw Resolute UHV	1	0.19	\$2000.00	1.25
Wrist Joint						
*Note: 3 axes	Motor	Maxon EC-60 flat	3	1.41	\$810.00	522
	Gearbox	Harmonic Drive SHG-20-50-2SO	3	2.43	\$6000.00	
	Motor Controller	Maxon EPOS4 70/15	3	1.116	\$2415.00	2205
	MEMS IMU	Honeywell HG4390	1	0.14	\$10000.00	3
	Absolute Encoder	Renishaw Resolute UHV	3	0.57	\$6000.00	3.75
	F/T Sensor	ATI Axia 80-M20	1	0.3	\$4500.00	
	Optical Camera	Teledyne FLIR Blackfly S	1	0.36	\$880.00	3

Table E6. A Detailed Breakdown of the Mass/Cost/Power Budget for the Mechanical Subsystem Continued

Joint	Component	Vendor / Model	Quantity	Total Mass (kg)	Estimated Total Cost (USD)	Total Power (Watts)
Avionics						
	Flight Computer	Frontgrade Generation 8 (GR740 Based) SBC	2 (1x active / 1x cold backup)	1.3	\$4400.00	12.8
	High-Voltage DC-DC Converter	Horizon Power Systems SGRB12050S	1	0.85	\$12000.00	400
	Thermal Heaters/ Sensors	Minco/ Thermofoil Heaters HK6907	17	0.0442	\$2006.00	53
Structural Materials	Ti-6Al-4V	Chalco Titanium		24	\$672.00	
	CFRP	Hexcel Corporation		19	\$2090.00	
	MLI	QUEST 10-Layer IMLI		0.43	\$2500.00	
End-Effector	REACCH (8-tentacles)	KMI (custom model)	1	13	\$75000.00	1180
TOTALS				86.1122	\$343508.00	5475.3

Appendix F: Thermal Calculations

This is the calculation performed to find the radiated thermal wattage in from the sun and out through the thermal coated surface via radiation. This was accomplished by calculating the total surface area of the spacecraft (14.89 m²) and finding the value of Q_{out} to the surrounding space. This is the equation for Q_{out}:

$$Q_{out} = e\sigma AT^4$$

The next step was to find the range of possible values of Q_{in}, the radiated heat to the sun-facing side of the spacecraft. This was calculated using three possible sun-facing areas (ram orientation, zenith orientation, or 45 degree angled orientation) in order to capture the highest, lowest, and most likely orientations faced towards the sun. This is the equation for Q_{in}:

$$Q_{in} = IAa \cdot \frac{t_{sun}}{t_{orbit}}$$

Subtracting Q_{out} from Q_{in}, we get:

$$Q_{in} - Q_{out} = Q_{net}$$

$$Q_{net,ram} = - 770 W$$

$$Q_{net,zenith} = - 1206 W$$

$$Q_{net,angled} = 1447 W$$

Appendix G: Power and Communication

Table G1. Uplink Budget Communications Calculations

FairBanks AS1 11.28M

Forward Link Cases	Units	S-Band	X-Band	
Uplink Frequencies	GHz	2.07	7.19	
Diameter	m	11.28	11.28	Users Guide
Beamwidth	deg	0.85	0.23	Users Guide
Gain	dBi	45.8	56.8	Users Guide
Transmit Power	W			
Backoff and Line loss	dBi			
EIRP,gateway	dBW	64.6	64.6	Users Guide
EIRP per user				
Propagation Range	km	844	844	
Space Loss	dB	-157.296	-168.111	
Atmospheric Loss	dB	-10	-10	
Net Path loss	dB	-167.296	-178.111	
Ssatellite Antenna type				
Coverage Area	deg2	1156	1156	
Antenna Efficiency	%	0.95	0.95	
Gain	dBi	3	3	
Line Loss on Satellite	dB			
Received carrier power, C	dBW	-99.6963	-110.511	
sys noise temp	dB-K			
G/T	dB/K	22	35.2	Users Guide
Receiver C/No	dB-Hz	314.6	327.8	228 k
Data Rate per User	dB-Hz	73.1597	78.56729	
Available Eb/No Uplink	dB	241.4403	249.2327	

Table G2. Downlink Budget Communications Calculations

Downlink Frequency	GHz	2.25	8.45	
Satellite Antenna, type				
Coverage area	deg ²	1156	1156	
Antenna Efficiency	%	0.95	0.95	
Antenna Gain	dBi	3	3	
Satellite TX power	W	10	10	
Backoff and link loss	dB			
EIRP per channel	dBW	13	13	
EIRP per user	dBW			
Propagation Range	km	844	844	
Space Loss	dB	-158.02	-169.514	
Atmospheric Losses	dB	-10	-10	
Net Path loss	dB	-168.02	-179.514	
User Terminal Type				
Diameter	m	11.28	11.28	
Beamwidth	deg	0.85	0.23	
Gain	dBi	45.8	56.8	
Line Loss	dB			
Receive Carrier Power C	dBW	-109.22	-109.714	
System noise temperature	dBK	27	27	
G/T	dB/K	22	37.2	Users Guide
Receiver C/No	dB-Hz	94.9795	98.68602	228 k
Data Rate per User	dB-Hz	73.52183	79.26857	
Available Eb/No Downlink	dB	21.45768	19.41745	
End-to-End Eb/No	dB	262.898	268.6502	
Modem Implementation loss	dB	-1.2	-1.2	QPSK
Required Eb/No	dB	15.52042	15.52042	
Link Margin	dB	247.3776	253.1297	
Channel Bandwidth				
Number of Channels		1	1	
No of users/ Channel				
Single User Data Rate	Mbps	40	350	Users Guide
Code Rate				
Single User Bandwidth	MHz			
Bandwidth user/channel	MHz			
Total Capacity	Mbps	40	350	

Table G3. Initial Solar Array and Battery Sizing Calculations

Solar Panels				Blackout			
dim	inch	cm	m				
H	27.3	69.342	0.69342	Target Alt	844 km		
L	33.6	85.344	0.85344	Rearth	6378 km		
W	0.5	1.27	0.0127	MuE	398600 km ³ /s ²		
Area			0.591792 m ²	Vcirc	7.429163 km/s		
Watt Bol			175 W	circumfram	45377.16 km		
	0.655428						
power			295.7118 W/m ²	time	6107.978 sec		
					101.7996 min		
eff			21.72754 %		1.69666 hr		
		Daylight time		arc angle	124.046 deg		
		1.112038	66.7223		2.165011		
	2343.699			arclength	15635.71 km		
				arc time	2104.639 sec		
No Payload power			52 W	blacjout	35.07732 min		
					0.584622 hr		
power/orbit		1381.36	W	Battery Sizing			
Area Needed		4.671305	m ²	Blackout Power needed			
panels Needed		7.893486		2907265 Total Wattage			
Rounded UP		8 Panels		2907.265 kW	807.5736 Whrs		
				Battery			
Total Power Per Orbit				Whr	275	550	825
		8437316 W		W	470.3894	940.7787	1411.168
Watt Needed with BO				#batts	2.936631	1.468316	0.978877
		2107.57 W		Rounded	3	2	1
area needed		7.127108		Mass (kg)	6.3	8	5.9
Panels Needed		12.04326					7.8
Rounded UP		13 Panels					
				BO Battery No Payload			
	807.5736			109441.3 W	63981.77		DoD
				109.4413 kW	63.98177		0.734158
	1.112038	66.7223		807.5736			
				Whr	275	550	825
1536.125384	2343.699			W	470.3894	940.7787	1411.168
				#batts	0.110547	0.055273	0.036849
	65.54278			Rounded			0.027637
				mass(kg)	0.232148	0.221093	0.217409
					0.215566		
		24					
frequency	14.14543						
				807.5736			

Appendix H: Ground Operations Antennas

Table H.1. Ground Antenna Names, Bands, and Location

	Antenna	Bands	Lat	Long
NASA's Near Earth Network	NASA Alaska Satellite Facility, Fairbanks	S/X	64.85	-147.5
	Fairbanks Command and Data Acquisition Station, AK	S/X	64.98	-147.51
	Blossom Point, MD	S/X	38.41	-77.1
	Guam Remote Ground Terminal, Guam	S/X	13.59	144.84
	Remote Ground Relay Terminal, Australia	S/X	-31.39	136.88
	NASA Kennedy Uplink Station	S/X	28.59	-80.64
	NASA Ponce de Leon Station	S/X	29.07	-80.91
	NASA Wallops Ground Station, Virginia	VHF/S/X	37.94	-75.47
	NASA White Sands Ground Station, New Mexico	VHF/S/Ka	32.83	-106.25
	NASA White Sand Complex, New Mexico	VHF/S/Ka	32.4	-106.6
	NASA McMurdo Ground Station, Antarctica	S/X	-77.85	166.68
Commercial	KSAT Singapore	S/C/Ku	1.37	103.99
	KSAT Svalbard, Norway	S/X	78.23	15.39
	KSAT TrollSat, Antarctica	S/X	-72.01	2.32
	SANSA Hartebeesthoek, South Africa	S/X	-25.89	27.71
	SSC Kiruna, Sweden	S/X	67.88	21.06
	SSC Santiago, Chile	S	-33.08	70.4
	SSC Space US North Pole, Alaska	S/X	64.48	-147.39
	SSC Space US Dongara, Australia	S/X	-29.05	115.35
	SSC Space US South Point, Hawaii	S/X	19.01	-155.4

Table H.1 depicts the ground antennas which will be used by the operators. These antennas are a combination of NASA Near Earth Network antennas and commercial antennas. Nearly all the antennas being utilized are capable of S-band and X-band. The above table also shows the latitude and longitude of the antennas.



Figure H.1. Ground Antenna Location Map

Figure H.1 shows the location of the ground antennas on a map. This is also highlighted in Figure H.2 which shows the antennas on a map overlaid with the spacecraft's ground track.

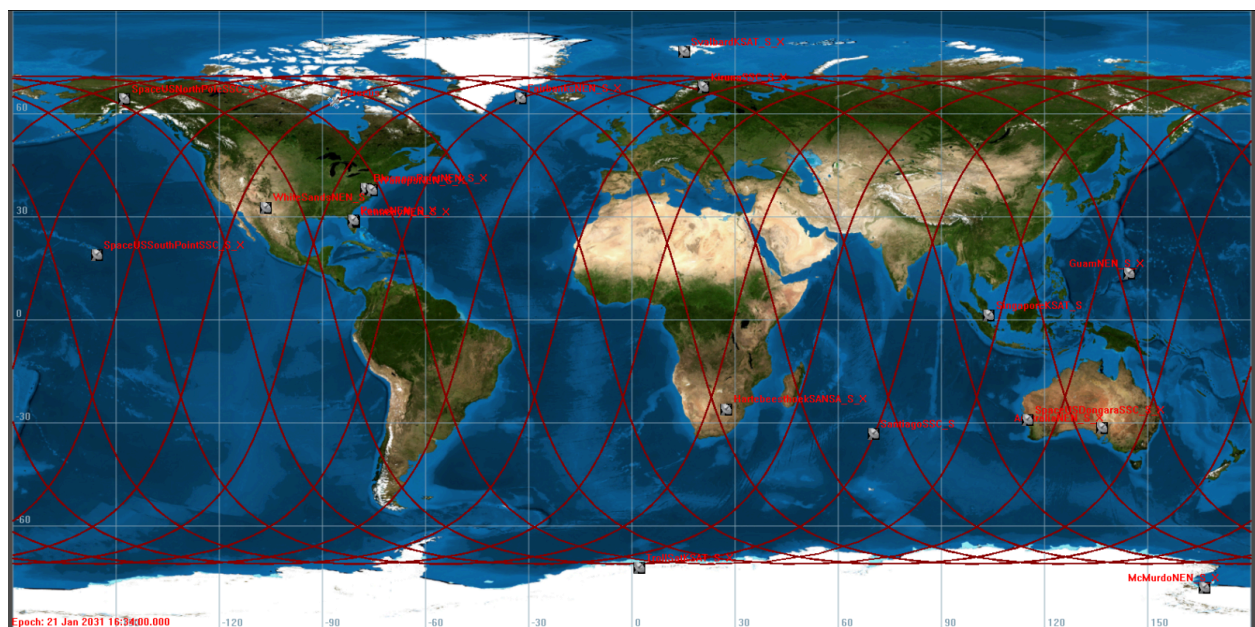


Figure H.2. Spacecraft Ground Track and Ground Antenna Access